2026

BlackRock Fixed Income Risk Model

HIGHLIGHTS

This paper describes a suite of fundamental fixed income risk models estimated and maintained by BlackRock. These are collectively known as BlackRock Fixed Income Risk Models, or BFIRM. The purpose of these models is to describe the structure and sources of fixed income portfolio risk globally.

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1 Introduction

1.1 Executive Summary

This document provides an overview of BlackRock fixed income risk models (BFIRM). These models identify and quantify portfolio risks, serving as essential tools to understand the contributions and interrelationships of different portfolio components. While portfolio managers often intuitively grasp their portfolios' positioning and potential risk exposures, risk models offer a systematic perspective on the drivers of risk and return in individual portfolios.

The BFIRM suite employs a fundamental factor-based approach to describe the risk characteristics of fixed income portfolios using a limited set of risk factors, which are economically intuitive and statistically significant. Key features of the models include:

- Key rate factors for sovereign yield curve risks covering 40+ currencies across global bond markets, including developed and emerging market government curves.
- Key spread basis factors for swap/ARR curves, inflation curves, and the muni AAA curve.
- Implied volatility factors for callable bonds, MBS, and interest rate derivatives.
- Granular coverage of asset-specific systematic factors, calibrated from daily cross-sectional regressions, to capture unique behavior of individual asset classes.
- DxS-based exposures to spread factors for most credit products, including corporate bonds, emerging markets, leveraged loans, and asset-backed (ABS and CMBS) securities.[1]
- Return based factors for distressed securities and non-Agency mortgages, with beta multipliers to account for security level risk differentiation.
- Spread duration-based exposures for tight spread sectors such as AAA government related securities and Agency mortgages.
- Broad coverage of currency factors for FX derivatives, as well as portfolio level currency risks.

1.2 Model Objective

Multi-factor Risk Models are essential to the portfolio management process, allowing portfolio managers to quantify expected volatility of performance deviation ("tracking error volatility") between a portfolio and a benchmark, as well as the severity of extreme events ("tail risk"). Multi-factor Risk Models provide risk managers with insight into the common drivers of risk in a portfolio so they can control their exposures and understand the contributions of different portfolio components to total risk.

The models are designed to:

- Accurately forecast the return and risk behavior of fixed income and multi-asset portfolios using a small set of intuitive factors. These factors capture broad market movements and specific nuances of individual asset classes, simplifying the model estimation process. They enable portfolio managers to distinguish significant sources of return volatility across portfolios and identify offsetting risks within portfolios/sub-portfolios.
- Use an approach that incorporates structural and historical data-driven elements, producing risk estimates with a bias towards recent data while reflecting economic intuition. Extensive empirical analysis and back-testing support the model, balancing financial market theory with statistical estimation. This research enables the model to produce valid risk estimates in historical market conditions and adjust dynamically to significant regime shifts.

[1] Duration Times Spread (DxS) methodology is explained in section 2.2.2

- Provide an intuitive and detailed interpretation of risk characteristics for various fixed income portfolios to facilitate investment decision-making. BlackRock's model prefers a larger set of economic factors over a smaller set of statistical factors, which lack interpretability and relevance in differing market regimes. The model structure and factor selection are determined by statistical evidence and user input, capturing granular data and economic intuition. This approach helps identify basis across different traded assets and adjusts to market stress, ensuring accurate risk interpretation and hedge alignment.

Typical use cases of the model include:

- Generation of *ex ante* risk forecasts;
- Structuring an active or passive portfolio that is "efficient" from the perspective of balancing forecast volatility against expected return;
- Evaluation of the impact of a potential trade on portfolio risk before placing the trade;
- Informing discussions with or increasing transparency to asset owners or investors;
- *Ex post* performance attribution, permitting a user to decompose common factor sources of excess return;
- Optimized rebalancing of a portfolio allowing for minimization of ex ante risk whilst respecting complex constraints on decision criteria such as portfolio weights, transaction size, rebalancing costs or taxes;
- Scenario analysis, facilitating the specification of scenarios by modifying certain factors and imputing changes consistent with the model for the remaining factors.
- Asset allocation;
- Risk management, risk budgeting and firmwide oversight of multiple portfolios or strategies.

A multi-factor risk model supports investment decisions by providing clarity into common factor exposures across assets and relationships across multiple horizons and different market regimes, generating a granular but comprehensive picture of portfolio risk characteristics.

1.3 Model Scope

This document outlines the model description for the BlackRock Fixed Income Risk Models (BFIRM), a suite of 43 factor risk models, which encompasses four model categories (economic, credit, securitized and idiosyncratic), and more than 3000 risk factors, designed to cover multiple fixed income asset classes.

- Economic factors are observed risk components that capture systematic influences across fixed income asset classes, such as curve movements, convexity, and volatility. This includes fixed income asset classes such as Rates, FX, and swaps.
- Credit spread factors are estimated factors, derived from spread returns to reflect risks in credit instruments like government-related and covered bonds, corporates, municipals, emerging market bonds, CDS and CDS indices, and leveraged loans.
- Securitized spread factors are estimated factors which are tailored to structured products such as mortgages, CMBS, ABS and CLOs.
- Idiosyncratic factors represent asset-specific risks not explained by systematic factors, capturing issuer-level nuances and concentration risks.

2 Model Methodology

2.1 Model Overview

Fixed-income securities are exposed to a range of market and issuer-specific risks. To measure these risks in a consistent and interpretable manner, the BlackRock Fixed Income Risk Model (BFIRM) employs a multi-factor framework that decomposes security-level returns into components attributable to well-defined systematic risk factors and a residual idiosyncratic component. This structure allows the model to quantify how different economic drivers influence observed asset returns.

In this framework, the return on security i at time t is represented as:

$$r_i(t) = \sum_{j=1}^{K} x_{i,j}(t) f_j(t) + \varepsilon_i(t) \tag{2.1}$$

Here $f_j(t)$ denotes the j-th systematic factors, while $x_{i,j}$ represents the exposure of security i to factor j, ε_i captures idiosyncratic risk, defined as the portion of return not explained by the set of K systematic fixed-income risk factors. This decomposition enables the model to attribute security-level risk to coherent and economically interpretable categories.

Key risk drivers in Fixed Income

The risk profile of fixed-income instruments is anchored in the valuation of contractual cash flows and the sensitivity of those cash flows to changes in the macroeconomic environment. For government bonds and interest-rate–dominated instruments, the primary drivers are yield curve movements, which reflect market expectations of future interest rates. For corporate and other credit-sensitive securities, issuer creditworthiness, including default and recovery expectations, becomes an additional core source of risk. The model further incorporates market structure effects, such as supply and demand imbalances through the use of spread curves.

Importantly, there is substantial alignment between the construction of credit-spread-related factors and methodologies employed in equity fundamental factor models: in both settings, returns or spreads are grouped into economically meaningful cohorts based on shared issuer or obligor characteristics.

Return Decomposition and Systematic Factor Mapping

At the heart of the fixed-income risk model is a return-splitting procedure, which linearly decomposes a bond's total return into standardized components aligned with distinct risk drivers. Following the removal of time return, which is the deterministic return attributable to the passage of time, the model separates the remaining return into four categories shared across all asset classes within a currency: Curve Return, Convexity Return, Swap Return,Volatility Return.[2]

These components are modeled using systematic factors tied to underlying economic variables derived from the valuation engine. Factor exposures are taken from standard risk analytics such as key rate durations and volatility durations, ensuring consistency between pricing analytics and the risk model. Because these components exist across currencies and asset classes, the decomposition supports an integrated and uniform modeling framework.

The portion of return not explained by the above components is classified as spread return, or excess return. Spread returns represent the primary focus of cross-sectional modeling, as they capture credit- and sector-specific risks that vary meaningfully by asset class. Accordingly, each fixed-income asset class is equipped with a dedicated spread risk model and tailored set of spread risk factors. For more details on return splitting and risk decomposition, please refer Section 2.2.1.

End-to-end workflow

[2]Here, certain less common components (e.g. mortgage basis, OIS-swap basis) are omitted for illustration purpose

A comprehensive workflow underpins the generation of risk model estimates and outputs. The process begins with the collection of security returns, issuer and security attributes, economic data, upstream risk analytics, and benchmark weights. These data feed the return-splitting engine, which produces the time, curve, convexity, swap, volatility, and spread return components.

The spread return component then serves as the input to systematic and idiosyncratic factor calibration routines. Calibrated factors and exposures undergo model-quality controls before being published to downstream databases. A final model overlay is applied to incorporate governance-driven adjustments, after which the system produces the full suite of risk model outputs and reports used across analytics, portfolio management, and governance functions. A brief illustration of the production workflow is provided in Figure 1.

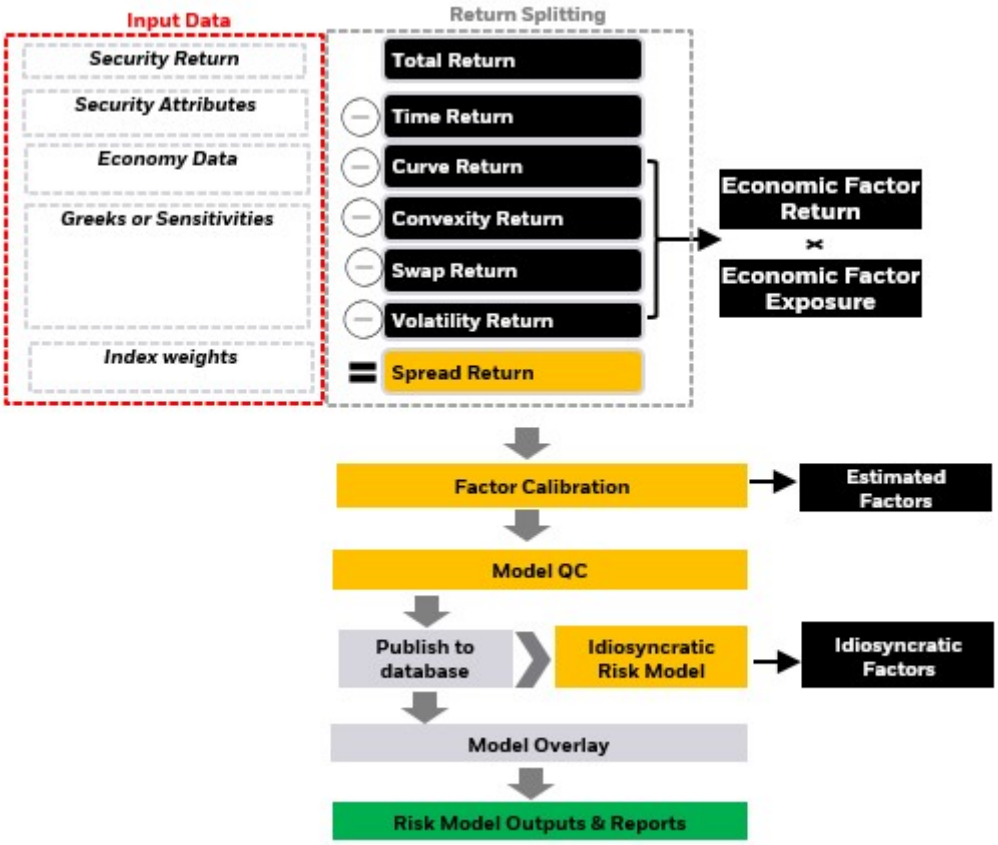


Figure 1: Fixed income risk model - workflow

2.2 Model Theory

An overview of the model construction is provided in this section, focusing on the key components that establish the general modeling framework for fixed income risk factor models. The model specification and calibration process for each asset class is then detailed, highlighting risk factors specific to individual asset classes to provide a comprehensive understanding of the model structure and calibration procedure.
Several sources of risk are bundled together in fixed income securities. A multi-factor modeling framework is employed to capture these risks by identifying common sources across different dimensions. These common sources are referred to as systematic fixed income risk factors. Risk not captured by these systematic factors is considered idiosyncratic and depends on the choice of systematic risk factors.

Fixed-income securities are structured assets, with theoretical prices estimated by discounting future cash flows. Risk arises from various sources that impact the number, timing, and size of these cash flows:

- The issuer may fail to meet one or many of the future payments described in the term-sheet of the security.
- The security may afford the issuer or the holder the option to cancel the contract through some repayment of the principal.
- The amount of future payments may vary depending on variables or indices specified in the contract.
- The security or certain cash flows may be denominated in a currency different from the investor's benchmark currency, known as the numeraire in financial economics. This currency unit is used to compare the relative traded prices of all other assets.
- The coupons payable on other assets during the security's maturity period may vary, or the initial auction price of those securities may change. This affects an investor's preference for holding or purchasing more units of the security.

The goal is to decompose the total return of each fixed income security in the model universe during the period $[t, t + \Delta T)$ into components related to the highlighted sources of risk. This process, known as return splitting, uses a theoretical pricing model for each security with input variables aligned with the conceptual risks. The pricing model links the non-deterministic elements of the return to time-varying factors indicative of the risks and changes in the security's return. Consequently, the decomposition depends on the form of the theoretical pricing model for the security.

As highlighted in the section *Model Objectives* (see section 1.2), a goal of the model is to assist an investor in reasoning about the common sources of risk across their portfolio. This motivates the use of a risk model with a simple structure with a linear mapping between changes in the factors, which are aligned with the conceptual risks highlighted above, and their influence on the non-deterministic component of the asset return.

2.2.1 Return Splitting and Risk Decomposition

In general, the total return of a fixed income security over a period may be decomposed into two components:

- Coupon payments which are defined in issuance documents;
- Variation in the price of the asset in the secondary market.

The coupon payment may be fixed or variable and linked to an index.

$$R^{Total} = R^{Coupon} + R^{Price} \tag{2.2}$$

Generically, the theoretical price of a fixed income asset may be expressed as a function of a number of variables and can be written as:

$$p_t = P(t,\ YC_t, Vol_t, ARR_t,\ OAS_t) \tag{2.3}$$

The full price of a security at time t is related to the current time t, benchmark yield curve YC_t, implied volatility surface Vol_t, Alternative Reference Rate ARR_t and option-adjusted spread OAS_t.[3]

Based on the pricing function 2.4, the security's period total return is decomposed into different stochastic and deterministic components:

$$R^{Total}_{t+1} = R^{Time}_{t+1} + R^{Curve}_{t+1} + R^{Convexity}_{t+1} + R^{Swap}_{t+1} + R^{Vol}_{t+1} + R^{spread}_{t+1} \tag{2.4}$$

[3]This section outlines the return splitting process and establishes the foundation of risk modeling. Asset-specific pricing functions will be detailed in later sections.

The subsequent sections provide a comprehensive description of the return components and their associated risk factors.

Time Return

Time return (also known as carry) is the deterministic return of the security minus the risk-free funding cost when market conditions do not change. Carry is determined by a security's term-sheet, assuming other variables remain unchanged. For most fixed income securities, this is simply the coupon return plus any return from the passage of time:

$$R_{t+1}^{Time} = R_{t+1}^{Coupon} + \frac{P(t+1, YC_t, Vol_t, OAS_t)}{P(t, YC_t, Vol_t, OAS_t)} - 1 \tag{2.5}$$

Note that carry is earned even when market is not open – ergo, carry is accumulated over the weekend/holidays to the next business date. In addition, as it is deterministic, time return is only included in parametric return calculation and does not contribute to risk calculation.

Curve and convexity return

Curve and convexity returns are the portion of the total return due to variations in benchmark yield curve used to discount cashflows of the security. Curve return is defined as,

$$R_{t+1}^{curve} = \frac{P(t, YC_{t+1}, Vol_t, OAS_t)}{P(t, YC_t, Vol_t, OAS_t)} - 1 \tag{2.6}$$

For most vanilla fixed income securities, yield curve fluctuations affect prices through changes in discount factors. More complex securities, like MBS, floating-rate securities, and interest rate derivatives, have cash flows dependent on interest rates. This return component captures variations in theoretical price due to changes in projected cash flows and discounting. Typically, this component is modeled using duration-convexity approximation as follows:

$$R^{curve} \cong \underbrace{-\sum_i krd_i \cdot \Delta kr_i}_{\text{first-order component}} + \underbrace{\sum\sum_{i,j} \frac{krbc_{i,j} \Delta kr_i \Delta kr_j}{2}}_{\text{second-order convexity return}} \tag{2.7}$$

where kr_i's are selected rates along the yield curve, known as key rates, krd_i's are key rate durations that measure the first-order price sensitivity to the (isolated) change of corresponding key rate, and $krbc_{i,j}$ are key-rate-bucket convexities that measure the price sensitivity to second-order key rate movements. This key-rate approach[4] captures yield curve dynamics with high accuracy using 11 key rate points (3m, 1yr, 2yr, 3yr, 5yr, 7yr, 10yr, 15yr, 20yr, 25yr, and 30yr) for most markets. Some European countries include 40yr/50yr points. Second-order movements are computed using key rate bucket convexities[5] for four key rates (2yr, 5yr, 10yr, 30yr), resulting in 4 diagonal and 6 cross terms for major currencies (USD, EUR, CAD, GBP, JPY). A 10yrx10yr convexity is used for the rest of the universe.

The local treasury curve is typically chosen as the base curve for each currency. Certain asset classes use other curves as benchmark discount curves[6], such as the LIBOR/Alternative Reference Rates (ARR) curve for credit default swaps and Muni curves for USD Municipal bonds. Consequently, curve return is split into base (treasury) curve components and benchmark-to-base-curve spreads, like swap spread and muni-to-treasury spread. Details on curve risk factor coverage and market nuances will be discussed in later sections.

Swap Return

[4]Money market securities, which are mostly short-term and non-coupon-bearing fixed income instruments, require a different risk decomposition approach. The typical 11 key rate points do not apply because the short-end of the curve is not granular enough, and the long-end is mostly irrelevant. Therefore, dedicated interest rate and spread risk factors are used to capture the short-end yield curve movement more accurately. These risk factors are described in the money market sub-section (refer 4.1.13).

[5]4 key rates instead of 11 are chosen here to balance out computation intensity and accuracy. For details of key rate bucket convexity calculation, refer to "Key Rate Bucket Convexities"

[6]Refer to "Security Valuation Model Matrix" for details on the discount curve of each asset class

Swap return is attributed to changes in swap spreads and is modeled similarly to curve return, with key swap spreads replacing key rates. The term "swap spread" will be used throughout this document. Technically, curves are constructed from Alternative Reference Rates instead of Libor rates. Unlike curve risk factors, swap spread risk factors are included only in a subset of fixed income securities where swap returns are more relevant or applicable.

- Fixed income derivatives, such as swaps, options, swaptions, futures etc.
- Instruments that are quoted in swap spread space, including most securitized products and floating-rated instruments such as leveraged loans.
- Instruments with no alternative spread risk model coverage. Swap spread factors serve as a basic proxy for spread risk in this situation.

With the recent developments in the risk model due to Libor retirement in major markets like USD, GBP, JPY, CHF, the new sets of replacement swap spread factors (ARR factors) have been introduced which leverage the Alternative reference rate in these markets to capture the basis with the local market Treasury rates.

While structurally similar, there are some differences in the swap risk models between fixed- and floating-rate products. For floating-rate products and securities using swap curve as discount curve, swap return is modelled as

$$R^{swapspread} = -\sum_i krd_i \Delta key_{swapspread_i} \tag{2.8}$$

where krd_i is the key rate duration calculated by shocking the base curve of the security, $key_{swapspread,i}$ is the key swap spread factor constructed as the difference of the corresponding key rate point of the swap curve and the yield curve. For fixed rated securities,

$$R^{swapspread} = -\sum_i krd_i \cdot \frac{spd_dur_i}{\sum_i krd_i} \cdot \Delta key_{swapspread_i} \tag{2.9}$$

The difference in treatment arises because the swap spread is included in both the cash flow (numerator) and discount (denominator) parts of the pricing equation for floating-rate securities. Conceptually, the economic meaning of swap return varies between floating and fixed-rate securities. For floating-rate securities, swap return captures a component of interest rate risk, making it part of the benchmark risk. In contrast, for fixed-rate securities, swap return captures a small component of spread risk.

Volatility Return

The volatility return captures the change in a fixed income security's price due to changes in interest rate volatility. This component is only included for interest rate derivatives (interest rate caps/floors, swaptions, etc.) and instruments with embedded options (callable bonds, MBS, etc.). Volatility return is defined as

$$R^{Vol}_{t+1} = \frac{P(t, YC_{t+1}, Vol_{t+1}, OAS_t)}{P(t, YC_t, Vol_t, OAS_t)} - 1 \tag{2.10}$$

Generally, this component is linearly approximated as

$$R^{vol} = -vol_{dur} \cdot F^{vol} \tag{2.11}$$

where vol_{dur} is the volatility duration which measures the first-order sensitivity of security price to the implied volatility surface, and F^{vol} is the single volatility factor mapped to at-the-money swaption implied volatilities. For swaptions, a comprehensive set of volatility factors is employed based on option term/underlying swap term pair.[7]

Interest rate implied volatility quotations directly influence the pricing of interest rate derivatives like caps and floors, as well as other products whose cashflows depend on future interest rates, such as mortgage-backed securities and callable or puttable bonds. Modeling the entire market of interest rate derivatives to ensure pricing of complex assets

[7]See list of volatility factors in section 3.4 on Volatility Factors

in line with simpler derivatives is a complex and extensively researched area. Aladdin analytics use such models to obtain theoretical prices and exposure measures like partial durations used in the fixed income model. However, from the perspective of the risk model, a reduced set of ATM IR implied volatility factors are used. Swaption-based quotes are preferred to caps and floors and are the sole quotes used in the model. For securities with embedded options, the volatility is proxied using the 5-year x 10-year swaption implied volatility of the same currency.

With the retirement of Libor quotations in major markets like USD, GBP, and JPY, the source of volatility factors has shifted from Libor (CMS) quotes derived volatility surfaces to ARR (Alternative Reference Rate) derived volatility surfaces. This update means that the volatility factors, which previously represented interest rate volatility in the market derived from Swaptions/Options quotes with Libor-based underlying, have now moved to Swaptions/Options quotes with ARR-based underlying (e.g., SOFR for USD).

Spread Return

The models for the mentioned components are straightforward, as the factors can be easily linked to market observables, and the factor loadings (exposures) are standard risk analytics for fixed income securities. These factors follow a similar paradigm across different currencies and are shared by various asset classes in the same currency market.

The last component of return is spread return, which is the portion of the total return unexplained by previous terms (carry, changes in yield curve, and changes in volatility). In the risk model, spread return is defined as the remaining component.

$$R_{t+1}^{spread} = R_{t+1}^{total} - R_{t+1}^{time} - R_{t+1}^{curve} - R_{t+1}^{convexity} - R_{t+1}^{swap} - R_{t+1}^{vol} \tag{2.12}$$

Hence R_{t+1}^{spread} contains not only the portion of return due to OAS movement but also all cross-effects. Broadly speaking, R^{spread} is further decomposed into a systematic piece, modeled as a linear combination of spread risk factors with factor loadings, and an idiosyncratic piece, which captures whatever is left unexplained.

$$R^{spread} = \sum_j exposure^j \cdot F^j + \epsilon \tag{2.13}$$

Systematic risk factors (F^j) capture common components of spread returns for securities in the same asset class. The selection of risk factors depends on the characteristics of each asset class. Every major asset class within a currency market has its own spread risk model and associated factors. For instance, the USD Corporate spread risk model primarily uses industry as its key factor, while the USD MBS model focuses on moneyness. These distinctions reflect the unique drivers of spread returns across asset classes. Model specifications are tailored to align with portfolio managers' investment perspectives and are customized to account for local market conditions and investor behavior. Factor structures are determined through market input, empirical analysis, and historical data patterns, with performance validated through in-sample and out-of-sample tests across different universes and time periods.

Idiosyncratic risk represents residual risk not explained by systematic factors. It is independent of systematic risk and can be diversified away. While idiosyncratic risk often constitutes a significant portion of total risk at the security level, it becomes negligible in large portfolios due to diversification. Consequently, this component is particularly relevant for portfolios with limited holdings and serves as a key measure of concentration risk.

Idiosyncratic risk is modeled in two parts. First, the level of idiosyncratic risk for individual securities is estimated using residuals from the systematic model, capturing variations across asset classes over time. Second, an intra-issuer correlation structure accounts for residual correlations among bonds from the same issuer. For example, a Ford bond's idiosyncratic risk is independent of a GMAC bond but correlated with another Ford bond, even if issued in a different currency. Detailed model structures and estimation procedures will be discussed in Section 6.

2.2.2 Measuring Spread Sensitivity

Credit spreads represent the most significant non-interest-rate risk in fixed income markets and serve as a forward-looking indicator of an issuer's creditworthiness. Historically, credit spread sensitivity was measured using spread duration, analogous to interest rate duration. Spread duration estimates the price impact of a bond for a given change

in its credit spread. Portfolio allocations were often expressed in terms of contributions to spread duration, calculated as:

$$\text{Spread Duration Contribution} = (\%\ \text{Portfolio Market Value}) \times (\text{Average Spread Duration of Securities}) \tag{2.14}$$

While spread duration provides useful insights, it has limitations. Bonds with higher spreads are inherently riskier and exhibit sharper responses to market shocks. Two portfolios with identical spread duration can have very different risk profiles if one contains higher-spread bonds.
The Duration-times-Spread (DxS) model offers a more robust approach for managing spread risk under varying market conditions. DxS incorporates both spread level and duration, providing a more accurate measure of risk. BFIRM applies DxS across many spread products, reflecting portfolio sensitivity to parallel shifts in spreads within a market cell.
Therefore, spread return is modeled in two ways:

- **Absolute Space** : $R_{spread} = D \cdot \Delta s$
- **Relative Space**: $R_{spread} = D \cdot s \cdot \frac{\Delta s}{s}$

Here, D is the duration, $D \cdot s$ is the DxS sensitivity, and s is the spread level.

Spread volatility is approximated as:

$$\sigma_{spreadreturn} = D \cdot \sigma_{absolutespreadchange} \tag{2.15}$$

$$\sigma_{spreadreturn} = D \cdot s \cdot \sigma_{relativespreadchange} \tag{2.16}$$

The relative approach is preferred because it offers:

- Stability: Delivers more consistent volatility estimates compared to absolute spread changes.
- Enhanced Risk Prediction: Incorporates both spread and duration, enabling more accurate forecasts of excess return volatility.
- Standardized Framework: Provides a standardized measure applicable across asset classes, including investment-grade and high-yield bonds.

While the relative approach is preferred, DxS assumptions may break down under certain conditions:

- Low Spreads: Illiquidity or mispricing can obscure credit components; absolute spread duration may be more appropriate.
- Distressed Securities: Distressed securities trade on default/recovery expectations, making spread less meaningful; modeled in total return space.
- Non-Credit Spread Movements: For example, mortgage pass-throughs where spread changes reflect prepayment risk rather than credit risk.

In summary, DxS provides a more accurate and stable measure of credit risk by accounting for both spread and duration.

2.2.3 Calibration Methodology

In general, spread risk factors are estimated daily in the following regression:

$$\frac{R_{i,t}^{spread}}{exposure_{i,t-1}} = \sum_j I_{i,t-1}^j f_t^j + \sum_k I_{i,t-1}^k f_t^k + \cdots + \sum_l I_{i,t-1}^l f_t^l + \sum_m X_{i,t-1}^m f_t^m + \epsilon_{i,t} \tag{2.17}$$

where $i \in \{estimation_universe\}$, $R_{i,t}^{spread}$ is the excess return after taking out all other return components from total return, $exposure_{i,t-1}$ is the exposure type ($D{\times}S, spd_dur$, or 1) depending on the exposure choice.

The regression includes two types of explanatory variables on the right-hand side of the equation:

- $I_{i,t-1}^j, I_{i,t-1}^k, \cdots, I_{i,t-1}^l$ are dummy variables indicating membership of security to different factor cohorts, $j \in J, k \in K, l \in L$ are different types of factors with dummy exposures to be calibrated (e.g. industry, maturity, subordination, country factors in US corporate spread risk model). These are classification variables, meaning each security is mapped to a factor based on its membership in a factor cohort.
- M is the set of factors with non-dummy exposures (e.g. liquidity factors in GBP corporate spread risk model)

Finally, $\epsilon_{i,t}$ is the residual return.

A single factor from a classification group, typically the most populated cohort (e.g., 5-year maturity bucket among maturity factor group in the USD Corporate risk model), is intentionally excluded to avoid the dummy variable trap[8]. This ensures the regression model is properly identified. The influence of the excluded cohort is implicitly captured through the intercept or is embedded in the sector-based factors, while the remaining factors represent the relative effect compared to the dropped category, controlling for other variables.
For a number of asset classes that are relatively simpler or smaller, or when there exists only one dominant set of characteristics, for example USD Agency bonds, EUR Covered bonds, and Pan Euro Securitized bonds etc., there only exists the first-order set of factors in the spread risk models, and equation 2.17 essentially reduces to

$$\frac{R_{i,t}^{spread}}{exposure_{i,t-1}} = \sum_j I_{i,t-1}^j f_t^j + \epsilon_{i,t} \tag{2.18}$$

Factor returns in this case are the "Market" or "Intercept" factors that capture the weighted average cohort-level spread movements (in absolute or relative space, depending on exposure choice).

2.2.4 Calibration Variations

Largely speaking, the previous equation captures the regression specification for all spread risk models, with a few exceptions:

- European Corporate Spread Risk Model: A two-stage regression approach is employed for European spread risk model calibration process, to more cleanly extract the industry effects and country effects respectively.
- CDS spread risk model and USD EM Sovereign: In these models, a step-wise regression approach is taken, where first-order factors are first estimated (as Equation 2.17), then model residuals from the last step is used to estimate the next set of factors, and this process is repeated until all factors are estimated.

[8]The dummy variable trap is a common problem in statistical modeling, especially in regression analysis, when using categorical variables that are converted into multiple binary (dummy) variables. The dummy variable trap occurs when all categories of a categorical variable are included as separate dummy variables in a regression model. This leads to perfect multicollinearity, meaning one variable can be exactly predicted from the others, making it impossible for the model to estimate unique coefficients for each variable. To prevent the dummy variable trap, one category is always omitted (usually the most common or a reference group). The omitted category becomes the baseline, and the coefficients of the included dummies represent the effect of each category relative to this baseline.

- Non-Asia EM Corporate Spread Risk Models: Restricted regression approach is employed. In other models with multiple types of cohort-based factors, the most prevalent cohort type is dropped for all the non-first-order factor types (or else there is an identification issue as there is linear dependency among the regressors). In the three non-Asia (Latin America, Central & Eastern Europe, and Mid-East and Africa) EM corporate spread risk models, additional restrictions are imposed on the second-order variables to solve this issue instead.
- USD MBS Spread Risk Model: Exposures to moneyness factors are continuous, measuring the (normalized) distance between the security's WAC (weighted average coupon) to the WAC of the two closest moneyness factors, with the sum of the exposures to be 1.
- Maturity Factors in Non-Asia EM Corporate Spread Risk Models and USD ABS spread risk model: In other models, exposures to maturity factors are dummy variables, indicating if the security maturity falls into the range of maturity bucket of the factor. Continuous (numerical) exposures are used instead, measuring the proportional distance between the security maturity and the midpoint of the factor maturity bucket, to provide a much smoother exposure profile over time and to avoid jumps in volatility forecasts because of switching maturity buckets.
- Non-Agency Mortgage Risk Model: Due to lack of liquidity , factor return is directly estimated from third-party subindices' (cohort-level) price time series, instead of estimated cross-sectionally from security level returns and exposures.
- CRT model: Due to lack of reliable security-level calibration universe, third-party subindices' price data is used to estimate the CRT model. Systematic factor returns are estimated using daily cross-sectional regressions in price return space, with each security's weight determined by its current balance. In order to account for differential pay-down characteristics of CRT securities, factor exposures are further refined using a credit beta adjustment. More details can be found in Section 5.1.11.

2.3 Factor Selection Methodology

Research Proposal Model development begins with a comprehensive review of the asset class and any existing risk model framework. This initial stage seeks to address key questions:

- **Asset Class Characteristics**:
  Identify the unique features of the asset class and the fundamental attributes considered by market participants during investment and risk management.
- **Data Sources**:
  Evaluate the most reliable data sources, considering historical depth, evidence of structural changes, and expectations of future shifts. Assess the frequency of structural changes relative to other asset classes.
- **Pricing and Analytics Framework**:
  Review pricing and analytics models historically used for the asset class, noting any changes that may influence research or implementation.
- **Current Model Limitations**:
  Examine constraints within the existing risk model, including assumption changes that may have impacted performance and the market segments most affected.

Following this review and discussions with model users, a preliminary view of the future model structure and candidate factors is formed through empirical analysis, with emphasis on addressing known limitations.

Data Collection A research plan is drafted, and historical data is gathered to support factor analysis. Typically, three core datasets are required:

- **Representative universe (monthly):**
  An index, either third-party or internal (created by BlackRock's dedicated indexing division, known as BlackRock Index Services) is selected to represent the asset class. Its constituents form the basis for research and calibration, with monthly rebalancing introducing minor changes.
- **Security Returns and Prices:**
  Prices or returns are sourced from vendors, while risk analytics such as time return, key rate durations (KRDs), spread duration, and option-adjusted spread (OAS) are derived from internal analytics models. These inputs support excess return generation and factor exposure estimation.
- **Fundamental attributes (static)**
  Features such as industry, country of issuance, issue size, and rating are used to segment the universe and analyze cross-sectional behavior. Selected attributes are incorporated into the final model for ongoing calibration.

Additional characteristics may be considered where relevant. For example, liquidity, though not directly observable, can be proxied using bid-ask spreads or trading volumes to construct liquidity factors.

Exposure Measurement Historical data is analyzed to understand statistical properties and identify unusual patterns with modeling implications. For spread risk modeling, exposure measurement is critical. The statistical behavior of spreads is assessed to determine whether DxS, spread duration, or %NAV (price based) is most appropriate. While DxS is the standard measure, alternatives may perform better in tight-spread sectors or distressed securities. Adjustments such as floors or caps may be applied to extreme spread values to enhance robustness.
Two key analyses are typically performed:

- **Normalized returns over time:** Realized returns are standardized using ex-ante volatility forecasts based on relative or absolute spread movements. The resulting time series is reviewed to assess whether DxS provides stable volatility estimates and stationarity.
- **Cross-sectional volatility and spread analysis:** Spread volatility is plotted against average spread levels across quantiles to visualize relationships. For example, analysis of U.S. Corporate Investment Grade bonds shows a clear linear relationship between spread volatility and spread level, which weakens in tight-spread regions.

Conclusions on exposure measurement are drawn from these analyses. Where evidence is inconclusive, further testing is conducted. DxS generally performs better in high-risk environments, while its advantages may diminish during low-risk periods.

Factor Selection Factor selection defines the model structure. Techniques such as principal component analysis or clustering identify dominant effects and co-movement patterns. Stepwise cross-sectional regressions are then applied:

- Begin with the first-order factors, which represent the primary sources of cross-sectional variation, structured in a nested hierarchy that progresses from the broadest to the most granular factors. [9]
- Regress residuals sequentially across levels, reviewing marginal t-statistics at each stage (10% threshold as a guideline).
- Apply economic rationale to ensure factor groupings reflect actual market segmentation.

The final model structure is determined based on explanatory power, consistency across market regimes, and economic interpretability. Comparative analysis is performed at:

- **Aggregate Level (In-Sample):** Cohort performance across time periods.
- **Security Level (In-Sample):** Ability to capture cross-sectional variation and incremental explanatory power.

[9] These are considered the most important effects driving cross-sectional movements of the asset class. Every spread risk model has a first-order set of factors that each security is exposed to. Some models only have first-order effect factors, e.g., country factors for emerging market sovereign spread risk models.

- **Index Level (Out-of-Sample):** Effectiveness in explaining portfolio returns relative to the prior model.

Empirical findings are validated through discussions with market participants and model users before finalization.

2.4 Factor Details

Typically, fixed income risk factors are classified into the following four categories:

- **Economic factors**: These factors, like interest rates and swap spreads, drive returns across various fixed income asset classes rather than being specific to one. Table 1 outlines risk model coverage for such factors. The model covers all major currencies and many emerging market currencies. The coverage is continuously reviewed and expanded as new markets emerge or new data becomes available. Economic factors capture the risk derived from curve, convexity, swap, and volatility return, which loosely correspond to the deterministic return components of a fixed income security with few exceptions:
  - **Time Return:** Time return does not have an associated risk factor due to its deterministic nature. It is calculated at the security level and incorporated only into parametric return-related risk analytics, such as stress profit and loss (PnL) and historical Value-at-Risk (VaR).
  - **Foreign Exchange (FX) factors:** FX factors measure risk arising from currency movements across all fixed income asset classes. Other risk factors are generally expressed in local currency; therefore, FX risk is assessed at the portfolio level when the portfolio's base currency differs from that of the underlying securities or when FX instruments are present.
- **Credit Spread factors**: These are estimated risk factors which are designed to capture asset class specific risk characteristics derived from spread return for the following asset classes: Government-related bonds, covered bonds, corporates, municipals, emerging market bonds, CDS and CDS indices, and leveraged loans.
- **Securitized Spread factors**: These are estimated risk factors which are designed to capture asset class specific risk characteristics derived from spread return for the following asset classes: mortgages, CMBS, ABS and CLOs.
- **Idiosyncratic factors**: These factors capture idiosyncratic risk which can be correlated among bonds from the same issuer but have zero correlation with systematic risk factors and idiosyncratic risk from other issuers. The table's last column shows the current coverage of the idiosyncratic risk model, with plans to expand coverage so that concentration risk can be measured more accurately.

Table 1: Model coverage – economic factors

Asset Class	Factor Type	Coverage	#Factor	Exposure
ALL* Fixed Income Securities	Interest Rates	42 currencies, 11/13 key rates	466	KRDs
IR derivatives, Swaps Securitized	ARR Spreads	24 currencies, 11/13 key rates	270	Swap KRDs
IR derivatives, Swap (ARR Basis factors)	ARR Spreads	USD/GBP/JPY/CHF/CAD, 11/13 key rates	61	KRDs, Basis KRDs
Swaps & IR Derivatives (ARR)	OIS-Swap Basis	EUR–13 key rates	13	Basis KRDs
European Sovereigns	Sovereign Spreads	16 countries, 11/13 key rates	180	KRDs
TIPs Inflation swaps	Inflation Spreads	17 currencies, 11/13 key rates	187	Inflation KRDs
IR derivatives Bonds with embedded options	Interest Rate Volatility	USD/EUR/GBP/JPY/AUD	274	Volatility Duration
FX derivatives**	FX	136 currencies	136	%NAV
Muni Tax-exempt	Muni Spreads	USD	11	KRDs
USD Mortgages	Mortgage Basis	USD	4	Mortgage Basis Duration

* Except price-based securities
** In addition, FX is included in portfolios with multiple currency-denominated securities
***ARR factors, at the time when the document is written, are there in only 6 currencies.

Table 2: Model Coverage - Estimated factors - By region and asset class

Region	Asset Class	Subclass	Link	Systematic Factor #	Description[10]	Idio
US	Credit	US Agency	4.1.1	4	Maturity	N
		Government Related	4.1.2	4	Sector/Rating	Y
		Corporate IG	4.2.1	55	Industry, Maturity, Style, Capsec, Yankee, Seniority	Y
		Corporate HY	4.2.1	41	Industry, Maturity, Style, Capsec, Yankee, Seniority	Y
		Preferred Equity	4.2.2	2	Rating	Y
	Loans	Leveraged Loans	4.5.1	25	Industry, Subordination	Y
		CLO	5.7	1	Market	N
	EM	EM Sovereign	4.3	86	Country, Maturity	Y
		EM Corporate	4.3.5	68	Country, Industry, Maturity, Rating	Y
	Securitized	Agency Mortgages[11]	5.1	17	Moneyness, Agency, Tenor	N
		Agency IOPO	5.1.6	30	Moneyness/Structure	N
		Agency CMOs/ARMs	5.1.8	14	Structure/Maturity	N
		Agency CRTs	5.1.11	4	Tranche Type	N
		Non-Agency	5.2	7	Collateral Type/Coupon Type	N
		CMBS	5.3	21	Tranche Type	Y
		ABS	5.4	15	Collateral Type, Maturity	Y
	Municipals	Muni IG	4.6.1	31	Industry, Maturity	Y
		Muni HY	4.6.2	32	Industry, Maturity	Y
		Muni Taxables	4.6.3	3	Maturity	Y
	Money Market	Money Market	4.1.13	28	Sector/Maturity	N
Pan Euro	Credit	Government Related	4.1.3, 4.1.5	8	Sector/Rating	Y
		EUR Corporate IG	4.2.3	40	Industry,Maturity, Seniority,Liquidity, Country, Capsec	Y
		EUR Corporate HY	4.2.3	40	Industry,Maturity, Seniority,Liquidity, Country, Capsec	Y
		Sterling Corporate IG	4.2.4	22	Industry, Maturity, Subordination, Liquidity	Y
		EUR EM Sovereign	4.3	37	Country	Y

Continued on next page

Region	Asset Class	Subclass	Link	Systematic Factor #	Description	Idio
Pan Euro	Securitized	Covered Bonds	4.1.6	9	Country	Y
		ABS, RMBS, CLO	5.6	15	Country/Collateral Type	N
		DKK MBS	5.5	3	Moneyness	N
	Money Market	EUR Money Market	4.1.13	27	Sector/Maturity	N
		Sterling Money Market	4.1.13	27	Sector/Maturity	N
Global	Credit	CAD Corporate	4.2.5	17	Industry, Maturity	Y
		CAD Provincials	4.1.7	3	Maturity	Y
		AUD Corporates	4.2.6	5	Industry, Maturity	Y
		AUD Govt-Related	4.1.8	3	Sector	Y
		JPY Credit	4.2.8	16	Industry/Rating, Maturity	Y
		CNY Policy Bank	4.1.11	4	Maturity	N
		CNY Corporate	4.2.7	17	Industry, Maturity, Rating	N
		INR Local	4.1.12	4	Maturity	N
	CDS	Single-name CDS	4.4.1	91	Industry, Maturity	Y
		CDS Indices	4.4.2	63	CDS Indices	N
	Money Market	CAD	4.1.13	15	Maturity	N
		AUD	4.1.13	15	Maturity	N

2.5 Model Assumptions & Limitations

The BFIRM suite is designed to provide robust and consistent risk estimates; however, users should be aware of the following assumptions and considerations:

Core Methodological Assumptions

- All standard assumptions of BlackRock's Analytical Value-at-Risk (AVaR) methodology apply.
- BFIRM suite relies on BlackRock's asset pricing and curve-generation analytics as key inputs. Updates to these upstream models may necessitate revisions to BFIRM to maintain consistency. Such updates are reviewed to determine whether structural changes or historical factor return backfills are required.

Data and Universe Consideration

- Model calibration is based on a representative universe using the most reliable data available at the time of development. While efforts are made to ensure robustness for out-of-universe securities, results may be less precise when applying the model to portfolios containing assets not well represented in the estimation universe.

[10] Each model has at least one set of factors. Some asset classes have multiple classes of factors to cover different risk drivers. E.g. US Corporate IG/HY models. There is typically one "first-order" set of factors in those models and we highlight them in bold when applicable. In addition, note that, "/" means one set of factors representing both attributes while "," separates different sets of factors.

[11] Mortgage related factors such as Agency Mortgages, Agency IOPO and Agency CMOs/ARMs have version-based factors to be consistent with the versions of prepayment models. The # factors shown refer to one version.

- Accurate factor returns depend on liquid and reliable price data. Fixed income instruments, however, often exhibit limited liquidity, and some securities trade infrequently. This may introduce staleness in return data or deviations from true market conditions. Alternative data sources are reviewed where appropriate to help address these limitations.

Factor Representation

- Systematic factors are generally represented as categorical indicators (e.g., sector, country, maturity bucket). Depending on the modeling approach, exposures may be expressed in spread-risk or price space. Certain factors, such as maturity or style, use continuous measures (e.g., fractional weights across adjacent buckets or standardized z-scores) rather than binary indicators.
- Each model includes at least one set of first-order factors, which represent primary risk drivers. Additional factor sets may be introduced for specific asset classes (e.g., U.S. Corporate IG/HY) or mortgage-related instruments to align with prepayment model versions.[12]

Forecast Risk Horizon

- The model's forecast horizon is typically one month. Performance assessments prioritize monthly results, although other horizons are reviewed.
- Factor returns are assumed to have zero correlation with asset-specific returns, and idiosyncratic returns across issuers are assumed to be uncorrelated.
- Idiosyncratic risk estimation assumes that the systematic risk model captures all relevant risk drivers. Missing factors may inflate residual risk estimates. Principal Component Analysis (PCA) is employed to detect residual correlation, and updates to the systematic model may trigger corresponding adjustments to idiosyncratic risk specifications.
- Serial correlation is incorporated into volatility forecasts to improve monthly idiosyncratic risk estimates.

Operational Framework

- Models are calibrated daily and made available to Aladdin clients on a T-1 basis (U.S. time), subject to upstream data availability and exception handling. Idiosyncratic models are calibrated on a weekly basis.
- The risk analytics and sensitivities used for spread factor calibration are typically derived from a standardized Aladdin centralized environment. When these calibrated factors are subsequently combined with risk analytics and sensitivities generated within individual client environments, discrepancies may arise. This integration can occasionally lead to unexpected or inconsistent behaviors, reflecting inherent limitations in aligning centralized calibration with client-specific configurations.
- Efforts are made to apply the latest upstream analytical model for calibrating exposures and factors. In most cases, only a single version is available; where alternatives exist, selecting the same version or setting is recommended to maintain consistency as per performance evaluation. There are a few exceptions, such as mortgages, multiple versions may be developed to accommodate different modeling choices.

  While introducing multiple versions can reduce discrepancies, complete consistency cannot be guaranteed. This is generally not an issue for vanilla instruments, but for complex securities with uncertain payoffs, variations may result in significant numerical differences.

Periodic updates may be required to align with market developments and sustain accuracy.

[12] See section 5.1.1 for more information.

2.6 Model Data Inputs

BlackRock's Fixed Income Risk Model integrates a diverse set of data sources to deliver robust and comprehensive analytics. The model leverages security-level data, market pricing,etc., in multiple ways, with inputs sourced both internally and from vendors. Key components include:

- **Calibration Index:** Sourced from external vendors or from BlackRock Index Services (BIS), a dedicated function within BlackRock responsible for the design, governance, and administration of proprietary indices across asset classes for its clients
- **Risk Analytics:** Generated by BlackRock's proprietary analytics and risk calculation engine, which provides valuation and risk metrics across asset classes
- **Returns:** Derived either directly from vendor-provided return data or calculated using vendor pricing inputs
- **Bond Attributes:** Characteristics such as industry, country, and currency are vendor-provided and maintained within the Aladdin platform
- **Exposures:** Produced by BlackRock at security level which feed into the broader risk modeling framework

2.7 Data Handling Framework

In the previous section, we introduced the data sources and fields that underpin BlackRock's Fixed Income Risk Model. Before these inputs are applied to model calibration, they undergo a series of preprocessing steps to ensure consistency and reliability.
While sourced from trusted vendors and internal systems, raw data can still exhibit gaps, minor inconsistencies, or extreme values. To address these issues, BFIRM employs a structured data handling process that focuses on four key areas- Completeness and Quality Checks, Outlier Management, Data Transformation for Model Readiness and Industry Representation and Factor Stability.

Completeness and quality checks Securities lacking essential risk analytics or exposures are excluded, while minor gaps, such as missing issuer identifiers, are resolved using standardized methods. This ensures the final universe includes securities with valid exposures, returns, and index weights.

Outlier Management Extreme observations can distort factor estimates. To mitigate this, statistical techniques such as winsorization are applied on quantiles.
A quantile threshold (q_t) is determined based on group size:

$$q_t = \begin{cases} 0.001, \text{ if } n_t \geq 1000 \\ \text{round}\left(\frac{1}{n_t}, 3\right), \text{ if } 100 < n_t < 1000 \\ 0.01, \text{ if } n_t \leq 100 \end{cases} \tag{2.19}$$

Values in the top and bottom q_t-quantiles are capped and floored accordingly. This helps maintain stability without compromising the integrity of underlying trends

Data Transformation for Model Readiness Raw inputs like spread returns and option-adjusted spreads (OAS) are converted into derived measures such as levels and deltas, which serve as the foundation for risk modeling. Factors are grouped into categories (Duration (D), Duration × Spread (DxS), and Net Asset Value (NAV)) with additional safeguards like floors and caps applied to exposures to prevent distortions from illiquid or distressed securities.
For factor type D or DxS:

$$level = \frac{\text{oas}_{\ t-1}}{100} \tag{2.20}$$

$$delta = \frac{-\text{ eret }_t}{\text{exposure }_{t-1}} \tag{2.21}$$

For factor type NAV:

$$level = \frac{price_{t-1}}{100} \tag{2.22}$$

$$delta = eret_t \tag{2.23}$$

Note that *eret* is defined as the excess return or spread return, computed from the return splitting equation as explained in section 2.2.1.
Typically, DxS-type exposures are floored and capped at 20 bps and 10,000 bps, respectively. Exceptions include models such as USD Corporate HY and Municipal HY, where the DxS floor is set at 300 bps. These floors apply exclusively to regression weight calculations.
Deltas and levels are winsorized according to Equation 2.19, and an additional outlier detection layer is applied to eliminate noisy analytics, typically originating from upstream risk processes

Industry Representation and Factor Stability To avoid skewed results from thin or underrepresented sectors, the model applies thin industry correction. This ensures factor calibration reflects broader market dynamics rather than isolated data points. Thinness is assessed at both bond and issuer levels using the effective number of assets, calculated as the reciprocal of the Herfindahl Index.[13]

$$n_eff_j^L = \frac{\left(\sum_{i=1}^{n_j^L} w_{i,j}^L\right)}{\sum_{i=1}^{n_j^L} \left(w_{i,j}^L\right)^2} \tag{2.24}$$

In cases where representation is insufficient, proxy or synthetic exposures may be introduced into the factor calibration to preserve robustness.

2.8 Model Outputs and Downstream Uses

BFIRM suite generates risk factor time series sourced from a combination of inputs, as outlined in Section 2.6.
These outputs serve as the foundation for a wide range of applications across portfolio management and risk analysis.

Below is a high-level overview of the most common uses. Detailed methodologies will be covered in dedicated white papers.

- Analytical VaR

  Factor return time series are leveraged to construct a factor covariance matrix, which is then combined with factor exposures to estimate portfolio volatility and Value-at-Risk (VaR). This approach also supports related analytics such as factor risk contributions, standalone factor risks, and beta calculations.

- Historical VaR

  Historical factor returns and exposures are used to generate parametric return distributions. These empirical distributions form the basis for calculating VaR, expected shortfall, and other risk metrics.

- Monte Carlo VaR

  BFIRM factor returns are fed to the Covariance Matrix estimation and then are used to simulate joint risk-factor scenarios, which are passed through security pricing to generate portfolio P&L distributions, supporting VaR, expected shortfall, and risk contribution analytics for nonlinear exposures.

[13]The Herfindahl Index, also known as the Herfindahl–Hirschman Index (HHI), is a widely used measure of market concentration and diversification. It quantifies how concentrated or competitive a market (or portfolio) is by summing the squares of the market shares (or weights) of all entities in that set.

- Stress Testing and other scenario-based analysis

  Historical and hypothetical shocks are applied to factor return time series to model portfolio and security-level P&L under various stress scenarios.

- Portfolio Optimization

  Factor covariance matrices and exposures serve as key inputs to optimization engines, enabling portfolio construction under specified constraints.

- Private Credit Factors

  Private credit factors are mapped to BFIRM's factor structure, adopting spread components from BFRE and BFIRM. Factor returns are integrated into covariance calculations through these mapped relationships.

3 Economic Factors

3.1 Rates

3.1.1 Treasury/Government Curve

The sovereign yield curve forms the foundation of all fixed income risk models. It embodies the economic principle of the time value of money, which determines the relative importance an investor assigns to receiving cash flows today versus in the future. This concept is critical for pricing across asset classes. In most developed markets, the government yield curve represents the borrowing cost for the entity with the lowest credit risk in that jurisdiction, making it the benchmark against which other issuers are evaluated. The Eurozone introduces additional complexity to this framework, and the approach for constructing sovereign discount curves within the Single Currency Zone is discussed separately in the section on European Sovereigns.

A brief description of the Rates -treasury/government curve model is provided in Table 3.

Table 3: Rates – treasury/government curve

Factor Type	Key Rate Duration Factor	Key Rate Bucket Convexity Factor
Exposure Universe	All fixed income (except price-based and money market) securities	
Parametric Return Equation	$\underbrace{-\sum_i krd_i \cdot \Delta kr_i + \sum_i \sum_j \frac{krbc_{i,j} \cdot \Delta kr_i \cdot \Delta kr_j}{2}}_{R^{Curve}}$	
Exposure Type	Key rate durations (KRDs) are calculated by shocking individual currency's Treasury/government curve[14]	Key Rate Bucket Convexities (KRBCs) are calculated by shocking individual currency's Treasury/government curve
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y (all currencies) 40Y, 50Y (GBP, EUR, JPY only)	2Y^2, 5Y^2, 10Y^2, 30Y^2, 2Y×5Y, 2Y×10Y, 2Y×30Y, 5Y×10Y, 5Y×30Y, 10Y×30Y
Currency Coverage	G10 (10): AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, SEK, USD Other developed markets (4): DKK, HKD, ILS, SGD Emerging markets (29): ARS, BRL, CLF, CLP, CNH, CNY, COP, CZK, EGP, HUF, IDR, INR, KRW, LKR, MXN, MYR, NGN, PEN, PHP, PLN, RON, RUB, SAR, THB, TRY, TWD, UYU*, VND, ZAR Currencies not listed above: ALL (mapped to EUR)	
Factor Estimation Methodology	The systematic rates risk factors are estimated using calibrated curves. Monotone spline fitted curve construction methodology is used for the following curves: USD, GBP, EUR, AUD, JPY, CHF, DKK, NOK, PLN, SEK and SGD. For currencies not listed above, we use third-party yield curve data to estimate the factors.	
Factor Volatility Calculation	Volatility of absolute factor changes	

[14] Hereinafter, we use krd to refer to key rate durations calculated against local treasury curve and kr to local treasury rates, unless otherwise noted.

3.1.2 BRL Onshore Treasury/Government Curve

A brief description of the BRL Onshore Treasury/Government curve model is provided in Table 4

Table 4: BRL Onshore Government Curve

Factor Type	USD onshore rate Factor(USB)
Exposure Universe	Brazilian market instruments like DDI futures, USD BRL FX futures
Parametric Return Equation	$\underbrace{-\sum_i krd_i \cdot \Delta kr_i}_{R^{Curve}}$
Exposure Type	Key rate durations (KRDs) are calculated by shocking individual currency's Treasury/government curve[15]
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y
Currency Coverage	Brazil(BRL)
Factor Estimation Methodology	Estimated as level of USB coupon curve built using DDI futures
Factor Volatility Calculation	Volatility of absolute factor changes

3.1.3 European Sovereigns

The BlackRock Fixed Income model decomposes risk in Eurozone sovereign bonds by using German government issuance as the base curve. Relative credit risk across participating sovereigns is represented as credit spreads over Germany, the lowest-yielding member. Key Rate Durations (KRDs) for Euro-denominated fixed income securities are mapped to German government rates, adjusted by each sovereign's curve spread versus Germany. This approach is consistent with the Aladdin analytical engine, which calculates duration using the German benchmark curve. The bond universe of ICE BofA All Maturity Euro Government Index is used to calibrate the following Euro-zone treasury curves: Austrian, Belgian, Spanish, Finnish, French, Irish, Italian, Netherlands, and Portuguese.

[15] We use krd to refer to key rate durations calculated against local treasury curve and kr to local treasury rates, unless otherwise noted.

Table 5: Rates – European sovereign spread factor

Factor Type	European Sovereign Spread Factor
Exposure Universe	Eurozone (ex. Germany) government bonds and interest rate derivatives
Parametric Return Equation	$R^{Curve} - \sum_i \Delta krd_i \cdot \left(\Delta kr_i^{\text{Sov}} - \Delta kr_i^{DEM}\right)$
Exposure Type	Key rate durations (KRDs) calculated by shocking the German government curve
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y, 40Y, 50Y for each sovereign covered
Sovereign Coverage	Austria, Belgium, Finland, France, Greece, Ireland, Italy, Netherlands, Portugal, Spain, Slovenia, Slovakia, Cyprus, Latvia, and Lithuania
Factor Estimation Methodology	Spread between individual country's government curve and the German government curve. We use the monotone spline fitted curve construction methodology for all those curves.
Factor Volatility Calculation	Volatility of absolute factor changes

3.1.4 Muni AAA

Non-callable Muni AAA curve is the discounting curve for US tax-exempt Muni securities and is used to obtain KRDs of these securities. On the risk factor side, we decompose the total Muni AAA curve into (1) Treasury and (2) spread to Treasury components, both exposing to the same set of KRDs. Effectively, the interest rate return of these securities is the product of Muni AAA key rates and Muni KRDs across all key rate points.
A brief description of Rates – Muni AAA spread factor is provided in Table 6

Table 6: Rates – Muni AAA spread factor

Factor Type	Muni AAA Spread Factor
Exposure Universe	Muni tax-exempt securities
Parametric Return Equation	$R^{Curve} - \Sigma_i Munikrd_i \cdot (\Delta USMuniAAAkr_i - \Delta USTreasurykr_i)$
Exposure Type	Key rate durations (KRDs) calculated by shocking the Muni AAA curve
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y
Currency Coverage	USD
Factor Estimation Methodology	Spread between Muni AAA curve [16] and the US treasury curve
Factor Volatility Calculation	Volatility of absolute factor changes

3.1.5 Inflation

Inflation reflects the rise in the price of a representative basket of goods and services, reducing the purchasing power of a currency unit. Because basket composition and measurement differ across countries, BFIRM simplifies this complexity by using risk factors derived from break-even inflation (BEI) curves. BEI is inferred from the yield spread between nominal and inflation-linked bonds of similar maturity, where inflation-linked bonds adjust cash flows based on a published price index. This spread serves as a proxy for market-implied inflation expectations.

Issuance of inflation-linked bonds is driven by investor demand, sovereign market-development goals, and policy considerations. Most inflation-linked bonds follow a capital-indexed structure; interest-indexed structures are less common.

[16] see `https://spc.blackrock.com/acs/literature/aladdin-product-overview/municipal-bonds-brs.pdf`

BFIRM abstracts from market-specific conventions by applying BEI curves consistently across assets. Because BEI estimates from treasury and swap markets reflect differing credit and liquidity characteristics, two curves are maintained: one from government inflation-protected bonds and one from inflation swaps.

Within BFIRM, inflation-linked securities carry exposure to both nominal curve factors (through Key Rate Durations) and inflation-specific factors (through BEI spreads). Consistent with the Fisher relationship, which states that real interest rate equals nominal interest rate minus expected inflation, Aladdin decomposes risk into nominal rate and inflation components. Thus, reporting sensitivities to both nominal interest rate and BEI factors. Further details on BEI curve calibration are provided in [1].

A brief description of Rates - inflation rate factor is provided in Table 7.

Table 7: Rates – inflation rate factor

Factor Type	Inflation Rate Factor
Exposure Universe	Inflation-linked government bonds
Parametric Return Equation	$R^{Curve} - \sum_i infl_krd_i \cdot \Delta inf_kr_i$
Exposure Type	Inflation KRDs. In the absence of inflation KRDs, scaled Treasury KRDs $Treasury_krd_i \cdot \frac{infl_dur}{\sum_i Treasury_krd_i}$ are used instead.
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y (all currencies) 40Y, 50Y (GBP only)
Currency Coverage	USD, EUR, GBP, CAD, SEK, JPY, AUD, KRW, ILS MXN, BRL, CLP, PLN, THB, FRF, ZAR TRY
Factor Estimation Methodology	Estimated as the spread of nominal curve and real curve built internally using government inflation-protected bonds[1]
Factor Volatility Calculation	Volatility of absolute factor changes

3.1.6 Real Rate Factors

In several Latin American jurisdictions with prolonged histories of elevated inflation, a distinctive market convention has developed for the issuance of financial instruments. Certain sovereign and retail products, such as government bonds and mortgage loans,are denominated in a synthetic, inflation indexed reference unit published by the corresponding monetary authority (e.g., the Central Bank). This reference unit does not circulate as legal tender; instead, all cashflows settle in the domestic currency, adjusted using a backward looking consumer price index to preserve purchasing power.

Under these conventions, market participants effectively transact directly in real interest rates. Because the principal is continuously adjusted for realized inflation, the quoted yield on these instruments represents a real yield; inflation is not modeled or traded as an independent risk factor. To remain consistent with this market structure, Aladdin introduces risk factors based on real rates, rather than decomposing nominal yields into real rates and breakeven inflation (BEI) as is standard in markets without inflation indexed reference units. This approach is particularly relevant in markets such as Mexico, Chile and Colombia, where government securities are commonly quoted in real terms and investors seek to measure their portfolio's sensitivity to real rate dynamics rather than to nominal rates and implied inflation separately.
Table 8 summarizes the inflation indexed reference units employed by selected Latin American sovereigns.

Table 8: Inflation-adjusted Currencies in LatAm

Country	Currency
Chile	CLF
Colombia	COU
Mexico	MXV

Table 9 provides additional detail on the inflation adjustment mechanisms that underpin these units.

Table 9: Real rates factor

Factor Type	Real Rate Factor
Exposure Universe	Securities issued in CLF, COU or MXV; Government bonds, Corporate bonds and derivatives are examples.
Parametric Return Equation	$\underbrace{-\sum_i krd_i \cdot \Delta kr_i}_{R^{Curve}}$
Exposure Type	Key rate durations (KRDs) are calculated by shocking individual currency's Treasury/government curve[1]
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y
Currency Coverage	CLF, COU,MXV.
Factor Estimation Methodology	We use third-party yield curve data to estimate the factors
Factor Volatility Calculation	Volatility of absolute factor changes

3.1.7 Brazil Real-Rate & Onshore Market Factors

Similar to the practice developed in Chile, Colombia and Mexico, Brazil has experienced periods of high inflation, but maintain their inflation-linked securities in the nominal currency (BRL). Nonetheless, it is a convention and common practice to assess exposure to real rates as risk factors for these securities, and to model the real rate curve as well as a function of the price of these securities (NTN-Bs).

Table 10: Real rates brazilian factor

Factor Type	Real Rate Factor
Exposure Universe	Real rate quoted securities like Government Inflation Bonds (NTN-B Bonds), Inflation Corporate Bonds (IPCA), Interest Rate futures (DAP futures)
Parametric Return Equation	$\underbrace{-\sum_i krd_i \cdot \Delta kr_i}_{R^{Curve}}$
Exposure Type	Key rate durations (KRDs) are calculated by shocking individual currency's Treasury/government curve[1]
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y
Currency Coverage	BRL.
Factor Estimation Methodology	Estimated from the calibrated real rate curve from NTN-Bs prices
Factor Volatility Calculation	Volatility of absolute factor changes

Brazil has developed a large and sophisticated onshore derivatives market centered on B3, where FX and interest rate contracts are actively traded. Because Brazilian FX futures and options are non-deliverable and settle in BRL, they have become essential tools for hedging in an economy marked historically by high inflation, volatile interest rates, and tight capital controls. Episodes of economic instability, particularly in the late 1980s and early 1990s, fueled demand for derivatives to manage inflation and interest rate risks. Later, following the Real Plan and greater macroeconomic stability, increased foreign currency borrowing by the private sector further boosted demand for FX futures. Regulatory and tax constraints also helped shift activity toward exchange-traded products, as exchanges allow more efficient netting of profits and losses compared to OTC markets. Additionally, strict controls on the FX spot market and the non-convertibility of the Brazilian real reinforced the use of derivatives as substitutes for cash transactions.

Price discovery in Brazil's FX market is driven largely by the FX futures market, whose depth contrasts with the relatively underdeveloped spot market. The two markets are linked through "casado" or differential transactions, which synthetically align futures and spot pricing. The central bank's active use of derivatives through instruments such as non-deliverable currency swaps and FX repos has further supported market liquidity without immediately impacting FX reserves. Beyond FX, B3 also lists the DDI (Dollar DI) interest rate future, which effectively trades the spread between Brazil's overnight CDI rate and the USD/BRL exchange rate, creating a proxy for an onshore USD interest rate. Within Aladdin, the most efficient way to represent this structure is by introducing a dedicated currency for the USD onshore market, designated as "USB."

A brief description of brazil onshore interest rate factors and Onshore FX factor is provided in Table 11.

Table 11: Brazilian market-specific factors

Factor Type	Onshore interest rate factor	Onshore FX factor
Exposure Universe	Securities on shore with exposure to DI and exchange rate risk	FX factors
Parametric Return Equation	$\underbrace{-\sum_i krd_i \cdot \Delta kr_i}_{R^{Curve}}$	
Exposure Type	KRDs	NAV
List of factors	3M,1Y,2Y,3Y,5Y,7Y,10Y,15Y,20Y,25Y,30Y	USB_FX_USD
Currency / Curve Coverage	BRL/USB	BRLUSB
Factor Estimation Methodology	Onshore : Based on Onshore rates calculated using local market nuances	Ratio of USB & USD
Factor Volatility Calculation	Volatility of absolute factor changes	

3.2 Basis Factors

The swap curve has traditionally played a key role in valuing financial assets, serving as a reference for coupon indexation in floating-rate bonds and for discounting liabilities. Prior to the 2008 Financial Crisis, it was also widely used as a base curve for benchmarking corporate bond market credit risk. However, evolving market insights have challenged this approach, highlighting the influence of interest rate hedging demand on swap curve movements. Reflecting these developments, the BFIRM model computes the corporate credit curve relative to the sovereign base curve corresponding to the bond's currency denomination.

3.2.1 Swap Spreads

Following the migration of swap quotes to Alternative Reference Rates (ARRs), swap spreads have been replaced with the corresponding newly elected ARR spread. A brief description of ARR Spread factors is provided in Table 12.

Table 12: ARR Spread Factors

Factor Type	ARR Spread Factor
Exposure Universe	1. Fixed income derivatives, such as swaps, options, swaptions, futures etc. 2. Instruments that are quoted in swap spread space, including most securitized products and leveraged loans. 3. Instruments that are discounted off ARRs.
Parametric Return Equation	$R^{Curve} - \underbrace{\sum_i \tilde{krd}_i \cdot (\Delta(ARR)kr_i - \Delta Treasury\, kr_i)}_{R^{ARR\ Spread}}$ $\tilde{krd}_i$ bears different meanings per security type, see "Exposure Type" below for details
Exposure Type	For floating-rated securities: key rate durations (KRDs) calculated by shocking the security's base curve (per equation 2.1.8) For fixed-rated securities: spread duration scaled by base curve KRDs (per equation 2.1.9)
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y (all currencies) 40Y, 50Y (EUR, JPY, GBP, ALL only)
Currency Coverage	G10 (9) - AUD, CAD, CHF, EUR, GBP, JPY, NZD, SEK, USD Other developed markets (3) - HKD, KRW, SGD, TWD Emerging markets (11) - BRL, CLP, CNH, CNY, CZK, HUF, INR, MXN, PLN, THB, TRY, ZAR, IDR, MYR, PHP Currencies not listed above - ALL (mapped to EUR)
Factor Estimation Methodology	Spread between individual currency's LIBOR or ARR curve and Treasury/government curves [2]
Factor Volatility Calculation	Volatility of absolute factor changes
Note	Brazilian Real (BRL): In lieu of a LIBOR curve, the BRL swap spread factors measure the spread between OIS and Treasury curves.

3.2.2 Muni Swap Spreads

The construction methodology for Muni Spreads is shown in Table 13.

Table 13: Muni Swap Spread Factor

Factor Type	Muni Swap Spread Factor
Exposure Universe	USD municipal swaps
Parametric Return Equation	$R^{Curve} = \sum_i krd_i \cdot (\Delta Muni_swap_kr_i - \Delta Treasury_kr_i)$
Exposure Type	Key rate durations (KRDs) calculated by shocking the US treasury curve
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y
Currency Coverage	USD
Factor Estimation Methodology	Spread between BMA Muni swap curve and US Treasury curve
Factor Volatility Calculation	Volatility of absolute factor changes

3.2.3 Other Swap Spreads

The methodology used to compute other forms of spread is shown in table 14.

Table 14: Other Swap Spread Factors

Factor Type	OIS – LIBOR Spread Factor
Exposure Universe	1. OIS swap 2. Other swap securities using OIS discounting (e.g. inflation swap)
Parametric Return Equation	$R^{Curve} + R^{ARR_Spread} - \underbrace{\sum_i OIS\ kr_i \cdot (\Delta OIS\ kr_i - \Delta ARRkr_i)}_{R^{OIS\ ARR\ Spread}}$
Exposure Type	OIS KRDs: Key rate durations calculated by shocking the OIS LIBOR basis curve
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y
Currency Coverage	EUR
Factor Estimation Methodology	Spread between ESTR curve and EURIBOR curve
Factor Volatility Calculation	Volatility of absolute factor changes

3.2.4 Inflation Swap

Table 15 shows the methodology used for construction of the inflation rate factor used in risk calculations of inflation swaps.

Table 15: Swaps - Inflation Rate Factor

Factor Type	Inflation Rate Factor
Exposure Universe	Inflation swaps and other inflation-linked interest rate derivatives
Parametric Return Equation	$R^{Curve} + R^{ARR\ Spread} + R^{OIS\ Spread} - \sum_i infl_krd_i \cdot \Delta infl_swap\ kr_i$
Exposure Type	Inflation KRDs. In the absence of inflation KRDs, scaled Treasury KRDs ($Treasurykrd_i \cdot \frac{infl_duration}{\sum_i Treasury\ krd_i}$) are used instead.
List of factors	3M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 15Y, 20Y, 25Y, 30Y (all currencies) 40Y, 50Y (GBP only)
Currency Coverage	USD, EUR, GBP. USD, EUR, GBP now use the SOFR, ESTR, SONIA factors respectively instead of the Libor factors adopted in the past.
Factor Estimation Methodology	Estimated using internally built inflation curves using government inflation swaps
Factor Volatility Calculation	Volatility of absolute factor changes

3.2.5 Mortgage Basis

The mortgage basis represents the average option-adjusted spread between mortgage-backed securities (MBS) and the underlying yield curve, now the ARR. It indicates the compensation an investor in an MBS receives for the risk that a mortgage holder may prepay. Typically, there is a smile in moneyness space, with the lowest spreads when the

prepayment option is at-the-money (ATM) (rate is close to the current mortgage rate) and increases as the option moves into or out of the money. The mortgage basis factor in the model includes mortgages across the moneyness axis.

To understand the differential spread across the smile, the model contains additional moneyness factors (see section on US Agency Mortgages).

Table 16: Mortgage Basis

Factor Type	Mortgage Basis Factors
Exposure Universe	All agency MBS including agency pass-through, TBA, IOPO, and MSR
Parametric Return Equation	$R^{Curve} - mtg_basis_{dur} \cdot \Delta mtg_basis$
Exposure type	Mortgage Basis duration
List of factors	15Y, 30Y
Currency Coverage	USD
Factor Estimation Methodology	Estimated daily using par coupon TBA duration and parallel interest rate shock. Par coupon is calculated daily by interpolation and extrapolation along the TBA stack. Mortgage beta captures the correlation between par coupon and yield curve. Par coupon movements are regressed individually with movements in the 2-year, 5-year and 10-year yield curve and then normalized to calculate coefficients. Unversioned- $mtb_{15Y} = 15Y ParCouponRate - (0.17 \times r_{2Y} + 0.4 \times r_{5Y} + 0.36 \times r_{10Y})\beta_{15Y} - 0.31 \times 1 \times 10 ATM\ Vol$ $mtb_{30Y} = 30Y ParCouponRate - (0.08 \times r_{2Y} + 0.22 \times r_{5Y} + 0.55 \times r_{10Y})\beta_{30Y} - 0.31 \times 1 \times 10 ATM\ Vol$ Versioned- $mtb_{15Y} = 15Y ParCouponRate - (coeff_1 \times r_{2Y} + coeff_2 \times r_{5Y} + coeff_3 \times r_{10Y})\beta_{15Y} - 0.31 \times 1 \times 10 ATM\ Vol$ $mtb_{30Y} = 30Y ParCouponRate - (coeff_1 \times r_{2Y} + coeff_2 \times r_{5Y} + coeff_3 \times r_{10Y} + coeff_4 \times r_{30Y})\beta_{30Y} - 0.47 \times 1 \times 10 ATM\ Vol$ In the versioned case, coefficients are calibrated daily as part of mortgage rate model.
Factor Volatility Calculation	Volatility of absolute factor changes
Note	Bonds are mapped to these factors depending on the underlying collateral's original term. Original term 15 get mapped to the 15-year factor. Original term between 15-year and 30-year get mapped to both the factors, while those with original term 30year get mapped to the 30-year factor. These factors are versioned based from prepayment model 6.0 onwards. The ATM volatility used in the above equation are now based off SOFR volatility surface, which was earlier based off Libor volatility surface.

3.3 Foreign Exchange (FX)

FX factors are used to capture risk associated with foreign exchange rate movement across all fixed income asset classes. All risk factors are expressed with respect to local currency numeraire of the market from which the factor is drawn. Portfolio risk estimates are computed with respect to a numeraire which is a unit of the base currency of the portfolio.

A brief description of the foreign exchange spot rate factor is provided in Table 17.

Table 17: Foreign exchange – foreign exchange spot rate factor

Factor Type	Foreign Exchange Spot Rate Factor
Exposure Universe	FX derivatives and all non-USD denominated fixed income securities
Parametric Return Equation	Any rates components as described above when applicable $+\%NAV \cdot \frac{\Delta fx}{fx}$
Exposure Type	%NAV (Delta-adjusted for FX derivatives)
List of factors	One spot FX rate factor for each currency covered, 135 in total G10 ex. US (9): AUD, CAD, CHF, EUR, GBP, JPY, NOK, NZD, SEK Other developed markets (7): ANG, DKK, HKD, ILS, KRW, LUF, SGD Pre-EUR legacy currencies, mapped to EUR (4): NLG, DEM, FRF, ITL Emerging markets (65): AED, ARS, BDT, BGN, BHD, BRL, BWP, CLF, CLP, CNH, CNY, COP, COU, CYP, CZK, ECS, EEK, EGP, GHS, HRK, HUF, IDR, INR, ISK, JMD, JOD, KES, KWD, KZT, LBP, LKR, LTL, LVL, MAD, MUR, MXN, MXV, MYR, NAD, NGN, OMR, PEN, PHP, PKR, PLN, QAR, RON, RSD, RUB, SAR, SIT, SKK, THB, TND, TRL, TRY, TTD, TWD, UAH, VEB, VEF, VND, ZAR, ZMK, ZMW Other currencies (50): AFN, AMD, AOA, AWG, AZN, BAM, BBD, BIF, BMD, BYR, BYN, BZD, CRC, DOP, DZD, GEL, GTQ, GYD, HNL, IRR, KHR, KMF, KYD, LRD, MDL, MGA, MMK, MNT, MOP, MWK, MZN, NPR, PAB, PGK, PYG, RWF, SSP, SVC, SZL, TZS, UGX, UYU, XAF, XAU, XCD, XDR, XOF, YER, ZWD, ZWR One spot FX rate factor for BRL (onshore BRL) – USB Commodities(3): XAG (Silver), XPD (Palladium), XPT (Platinum)
Currency Coverage	135 currencies as listed above
Factor Estimation Methodology	Third-party market data
Factor Volatility Calculation	Volatility of relative factor changes

3.4 Volatility Factors

3.4.1 IR Volatility

Interest rate implied volatility quotations directly influence the pricing of interest rate derivatives such as caps and floors, as well as other instruments with cash flows dependent on future interest rate distributions. This includes

mortgage-backed securities, where borrower prepayment behavior is sensitive to future rates, and callable or puttable bonds.

Modeling the entire market of interest rate derivatives to ensure complex assets are priced consistently with simpler instruments is a sophisticated and extensively researched area. Aladdin analytics leverage such models to derive theoretical prices and exposure measures, including partial durations used in the fixed income model. From a risk modeling perspective, a reduced set of at-the-money (ATM) interest rate implied volatility factors is applied. Swaption-based quotes are preferred over caps and floors and serve as the sole input for the model.[17]

Table 18: ARR interest rate volatility factor

Factor Type	Interest Rate Volatility Factor
Exposure Universe	Interest rate derivatives and bonds with embedded options
Parametric Return Equation	$R^{curve} + R^{ARR_spread} - vol_dur \cdot \Delta vol$(IR derivatives) $R^{curve} - vol_dur \cdot \Delta vol$(bonds with embedded options)
Exposure Type	Volatility duration
List of factors	USD (56 factors): Normal SOFR volatility (53 factors) {1M,3M,6M} * {1Y,2Y,3Y,5Y,7Y,10Y,15Y} {1Y,2Y,3Y,5Y} * {1Y,2Y,3Y,5Y,7Y,10Y,15Y,20Y} Log-normal SOFR volatility (1 factor) Mortgage implied SOFR volatility (2 factors) {15Y, 30Y} (mapped to 5Y*10Y normal volatility) GBP,JPY (84 factors each): Normal volatility (46 factors) {1M,3M,6M} * {1Y,2Y,3Y,4Y,5Y,7Y,10Y,12Y,15Y,20Y,25Y,30Y} {1Y,2Y,3Y,5Y} * {1Y,2Y,3Y,4Y,5Y,7Y,10Y,12Y,15Y,20Y,25Y,30Y}
Currency Coverage	USD, EUR, GBP, JPY
Factor Estimation Methodology	Estimated using at-the-money swaptions
Factor Volatility Calculation	Volatility of absolute factor changes
Note	While interest rate derivatives are mapped to the risk factor that corresponds to the security term, bonds are always mapped to the 5Y*10Y factor regardless of their maturity date.

[17] Following the retirement of LIBOR, interest rate implied volatility quotes transitioned from contracts referencing LIBOR to those based on Alternative Reference Rates (ARRs).

3.4.2 FX Volatility

Table 19: Foreign exchange – foreign exchange volatility factor

Factor Type	Foreign Exchange Volatility Factor
Exposure Universe	FX option
Parametric Return Equation	$\%NAV \cdot OptionDelta \cdot \frac{\Delta fx}{fx} + \%NAV \cdot OptionGamma \cdot (\frac{\Delta fx}{fx})^2 + \%NAV \cdot optiontheta \cdot \Delta time + \%NAV \cdot Optionvega \cdot \Delta vol$
Exposure Type	Vega-adjusted %NAV
List of factors	AUD/CAD, EUR/GBP, GBP/NZD, USD/ILS, AUD/CHF, EUR/HKD, GBP/PLN, USD/INR, AUD/JPY, EUR/HUF, GBP/SEK, USD/JPY, AUD/MXN, EUR/ILS, GBP/USD, USD/KRW, AUD/NOK, EUR/JPY, NOK/SEK, USD/MXN, AUD/NZD, EUR/MXN, NZD/CAD, USD/NOK, AUD/SGD, EUR/NOK, NZD/CHF, USD/PHP, AUD/USD, EUR/NZD, NZD/JPY, USD/PLN, BRL/JPY, EUR/PLN, NZD/USD, USD/RUB, CAD/CHF, EUR/SEK, TRY/JPY, USD/SEK, CAD/JPY, EUR/TRY, USD/BRL, USD/SGD, CHF/JPY, EUR/USD, USD/CAD, USD/THB, CHF/SEK, EUR/ZAR, USD/CHF, USD/TRY, EUR/AUD, GBP/AUD, USD/CLP, USD/TWD, EUR/BRL, GBP/CAD, USD/CNH, USD/ZAR, EUR/CAD, GBP/CHF, USD/CZK, XAU/USD, EUR/CHF, GBP/HKD, USD/HKD, ZAR/JPY, EUR/CNH, GBP/JPY, USD/HUF, EUR/CZK, GBP/NOK, USD/IDR.
Currency Pair Coverage	74 currency pairs as listed above
Factor Estimation Methodology	Estimated using 3M FX options
Factor Volatility Calculation	Volatility of absolute factor changes
Note	The gamma and theta components in the parametric return equation does not contribute to analytical risk.

In the following chapters, the estimation process for spread models is outlined by asset class. The below sections are detailed:

- The estimation universe;
- Calibration methodology;
- Mapping of exposures;
- A list of all spread factors for the asset class included in the model;
- Guidance on interpretation of the factors.

4 Credit Spread Factors

4.1 Government-related and covered bonds

Government-related bonds are issued by government-like agencies or international organizations, such as GSEs, regional or local authorities, foreign agencies, and supranational agencies. Although these bonds are not guaranteed by national governments, they have historically low default rates, good credit ratings, and trade with low spreads.

Covered bonds are issued by financial institutions, and are collateralized by a specific pool of assets like public sector loans or mortgages. Unlike asset-backed securities, they remain obligations of the issuers[18] and are considered more like corporate bonds. These bonds are common in Europe and Australia but less so in the United States. Due to their specific collateral pool, covered bonds usually trade at tighter spreads than corporate bonds.

The main risk for this asset class is the credit risk of the issuing organizations or their ability to manage the covered pools. Most segments trade at very tight spreads, making DxS less effective. Therefore, market-factor approach is used, with one cohort-based factor measuring the average spread movement of the corresponding market segment, in absolute or relative space, depending on the spread levels. In larger asset classes, sector/maturity[19] factors are used to capture the term structure effect of credit spreads.

Table 20 lists an overview of the different currency and asset classes which are covered in the Government-related and Covered bonds sub-models in this chapter. In the following section, model structure details of each government-related and covered bond risk model across multiple currencies are listed.

Table 20: Currency Market Segments and Indices

Currency	Market Segment	Index
USD	US Agencies	BlackRock Index Services (BIS) USD Agency Index
	Government Related	BlackRock Index Services (BIS) USD Corporate IG Index
EUR	Government Related	BlackRock Index Services (BIS) EUR Corporate IG Index
	Covered	
GBP	Government Related	BlackRock Index Services (BIS) GBP Corporate IGY Index
	Covered	
CAD	Provincials	FTSE TMX Canada All Corporate Bond Index
AUD	Government Related	BlackRock Index Services (BIS) AUD Investment Grade Index
	Covered and ABS	
JPY	Government Related	Nomura BPI index
CNY	Policy Bank	iBoxx ALBI China Onshore index
INR	INR Local	BlackRock Index Services (BIS) INR India Corporate Index

4.1.1 US Agencies

U.S. Agency securities are debt instruments issued by government-sponsored enterprises (GSEs) and federal agencies, such as Fannie Mae, Freddie Mac, Ginnie Mae, Federal Home Loan Banks (FHLB), and Federal Farm Credit Banks

[18] In contrast to ABSs or MBSs, the covered pool serves mainly as credit enhancement and not as a means to obtain exposure to the underlying assets.

[19] In general, when it comes to spread risk modeling, callable bonds are not treated differently. Their risk analytics (OAS and spread duration) are based on the option-based pricing methodology and are supposed to respect the embedded optionality, except that maturity_to_worst is used instead to determine its maturity-related factor mapping.

(FFCB). Some have explicit government guarantees (like Ginnie Mae), while others have implicit backing, making their credit quality just below U.S. Treasuries. These securities provide funding for housing and agriculture and offer investors high-quality, liquid fixed income investments that typically yield more than Treasury securities of similar maturities.

Table 21: USD Agencies - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) USD Agency Index
Calibration Methodology	$$\frac{R_i^{\text{spread}}}{\text{exposure}_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$ where j indicates the factor cohort. exposure_i is spread duration or $-\text{spd_dur}_i$.
Exposure Mapping	Spread duration-based dummy exposures based on Weighted Average Life (WAL) of a bond.
List of factors	The model has 4-maturity based factors – 2yr, 5yr, 10yr and 30 yr. The cohort and factor mapping criteria is shown below: • Agency 2Y: WAL < 3 • Agency 5Y: $3 \leq$ WAL < 7 • Agency 10Y: $7 \leq$ WAL < 15 • Agency 30Y: $15 \leq$ WAL < 100
Factor Interpretation	Factor return captures weighted average absolute spread movement of the corresponding market segment.

4.1.2 USD Government-Related

USD-denominated foreign government agencies, regional and local authorities, foreign sovereigns, and supranational bonds represent debt instruments issued by non-U.S. public sector and multilateral entities, but denominated and settled in U.S. dollars. These instruments provide investors with exposure to a broad range of sovereign and quasi-sovereign credit while avoiding direct foreign-exchange risk associated with non-USD issuance.

Foreign sovereign bonds are issued by national governments outside the United States, whereas foreign government agency and regional or local authority bonds are issued by government-related entities such as state-owned agencies, provinces, municipalities, or public infrastructure authorities. Supranational bonds are issued by multilateral institutions, such as development banks or international financial organizations, established by multiple governments to support cross-border economic and development objectives.

Table 22: USD Government-Related - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) USD Corporate IG Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$ for AAA and DxS for sub-AAA bonds.
Exposure Mapping	Spread duration for Agency AAA bonds and DxS for Sub-AAA bonds
List of factors	Rating-based factors: US Agency AAA, US Agency, US Non-corporate AAA, US Non-corporate
Factor Interpretation	• For spread duration exposure: factor return captures weighted average absolute spread movement of the market segment. • For DxS exposure: factor return captures weighted average percentage spread movement of the market segment.

4.1.3 EUR Government bonds

EUR-denominated foreign sovereign, government agency, regional and local authority, and supranational bonds are debt securities issued by non-domestic public sector or multilateral entities and denominated in euros. These instruments provide exposure to a broad range of sovereign and quasi-sovereign credit within the euro currency framework, without the foreign-exchange risk associated with non-EUR issuance.

Foreign sovereign bonds are issued by national governments outside the investor's home jurisdiction, while foreign government agency and regional or local authority bonds are issued by government-related entities such as agencies, states, provinces, or municipalities. Supranational bonds are issued by multilateral institutions formed by multiple governments to support regional or international policy and development objectives. Credit risk for these securities is driven by issuer fundamentals, institutional structures, and, where applicable, explicit or implicit government or multilateral support.

Table 23: EUR Government - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) EUR Corporate IG Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$ for Pfandbrief bonds and DxS for non-Pfandbrief bonds.
Exposure Mapping	Spread duration for AAA bonds and DxS for sub-AAA bonds
List of factors	EU Agency AAA,EU Agency, EU Non-corporate AAA and EU Non-corporate
Factor Interpretation	• For spread duration exposure: factor return captures weighted average absolute spread movement of the corresponding market segment. • For DxS exposure: factor return captures weighted average percentage spread movement of the corresponding market segment.

4.1.4 EUR Covered

Covered bonds (Pfandbriefe) in the EUR market are secured by high-quality assets like residential mortgages or public sector loans and backed by the issuing bank. They follow specific legislative frameworks in different European countries, with Germany's Pfandbrief market being the largest and most established. The cover pool assets must meet strict quality standards and overcollateralization levels, providing enhanced security for investors. Key features include the on-balance sheet nature of the assets, stringent regulatory oversight, and preferential treatment under European banking regulations. Covered bonds offer a yield premium over government bonds while maintaining high credit ratings due to their dual-recourse nature and strict legal frameworks.

Table 24: EUR Covered - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) EUR Corporate IG Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$ for Pfandbrief bonds and DxS for non-Pfandbrief bonds.
Exposure Mapping	Spread duration
List of factors	EU Collateralized Pfand, EUR Covered Spain, EUR Covered Ireland, EUR Covered Italy, EUR Covered Portugal, EUR Covered Core, EUR Covered Non-Europe
Factor Interpretation	• For spread duration exposure: factor return captures weighted average absolute spread movement of the corresponding market segment. • For DxS exposure: factor return captures weighted average percentage spread movement of the corresponding market segment.

4.1.5 GBP Government-Related

GBP-denominated foreign sovereign, government agency, regional and local authority, and supranational bonds are debt securities issued by non-U.K. public sector or multilateral entities and denominated in British pounds sterling. These instruments provide exposure to sovereign and quasi-sovereign credit while avoiding foreign-exchange risk relative to GBP-based investors. Issuers include national governments, government-related agencies, sub-sovereign entities such as states or municipalities, and supranational institutions established by multiple governments. Credit risk is driven by issuer fundamentals, institutional and legal frameworks, and, where applicable, explicit or implicit government or multilateral support.

Table 25: GBP Government-Related - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) GBP Corporate IG Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$ for AAA and DxS for sub-AAA bonds.
Exposure Mapping	Spread duration for AAA bonds and DxS for sub-AAA bonds
List of factors	The model has 4 factors - UK Agencies AAA, UK Agencies, UK Non-corporate AAA and UK Non-corporate
Factor Interpretation	• For spread duration exposure: factor return captures weighted average absolute spread movement of the market segment. • For DxS exposure: factor return captures weighted average percentage spread movement of the market segment.

4.1.6 GBP Covered

GBP covered bonds are debt securities issued in British pound sterling by financial institutions, such as banks or mortgage lenders. These bonds are secured by a dedicated pool of high-quality assets, typically residential mortgages or public sector loans, which serve as collateral. Investors are drawn to GBP covered bonds due to their dual recourse structure, offering claims against both the issuer and the underlying asset pool. This provides a higher level of security compared to unsecured debt instruments.

Table 26: GBP Covered bonds - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) GBP Corporate IG Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$ for Pfandbrief bonds and DxS for non-Pfandbrief bonds.
Exposure Mapping	Spread duration for Pfandbrief bonds and DxS for non-Pfandbrief bonds
List of factors	UK Collateralized Pfand and UK collateralized
Factor Interpretation	• For spread duration exposure: factor return captures weighted average absolute spread movement of the market segment. • For DxS exposure: factor return captures weighted average percentage spread movement of the market segment.

4.1.7 CAD Provincials

Canadian provincial bonds are debt securities issued by the provincial governments of Canada to finance various public expenditures, such as infrastructure projects, healthcare, and education. These bonds are considered high-quality investments, offering fixed interest payments and a return of principal at maturity. Investors are attracted to Canadian provincial bonds due to their combination of security, predictable income, and the opportunity to support regional development initiatives. However, it is important to assess the credit worthiness of individual provinces, as economic conditions can vary across regions.

Table 27: CAD Provincials - Model Overview

Section	Details
Estimation Universe	FTSE Canada Short/Mid/Long Term Provincial Bond Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$.
Exposure Mapping	Spread duration-based dummy exposures based on Weighted Average Life (Maturity Bucket) of a bond.
List of factors	3 maturity-based factors: short, intermediate, and long. The factor mapping is described below: • **Short**: WAL $<$ 5 • **Intermediate**: 5 $\leq$ WAL $<$ 10 • **Long**: 10 $\leq$ WAL
Factor Interpretation	Factor return captures weighted average absolute spread movement of the corresponding market segment.

4.1.8 AUD Government Related

AUD government-related bonds include National Housing Finance and Investment Corporation (NHFIC) Bonds for community housing projects, state government enterprise bonds from state-owned corporations, Australian post bonds for postal service operations, and bonds issued by public universities. These bonds are rated slightly lower than direct government bonds but often offer better yields to compensate for the slightly higher risk.
In the Australian market, semi-government bonds issued by local authorities may be inflation-linked. In this case, the following factor structure is used: AUD Interest Rates, AUD Inflation, AUD Semi Government spread factor. Additionally, AU Credit Govt Guaranteed Idio factor is used.

Table 28: AUD Government-Related - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) AUD Investment Grade Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$.
Exposure Mapping	Spread duration-based dummy exposures based on sub-class of bonds: Semi-Government and Supranational
List of factors	AUD Semi-Government and AUD Supranational
Factor Interpretation	Factor return captures weighted average absolute spread movement of the corresponding market segment

4.1.9 AUD Covered and ABS

AUD covered bonds are issued by Australian banks and financial institutions. They are attractive due to their dual recourse structure, high-quality collateral, strict regulatory oversight, and higher yields than direct government bonds. Common asset-backed securities include Residential Mortgage-Backed Securities (RMBS), auto loans, equipment leases, and credit card receivables. ABS offer higher yields and liquidity due to access to secondary markets.

Table 29: AUD ABS Covered - Model Overview

Section	Details
Estimation Universe	BlackRock Index Services (BIS) AUD Investment Grade Index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$, where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$.
Exposure Mapping	Spread duration-based dummy exposures
List of factors	AUD ABS and Covered
Factor Interpretation	Factor return captures weighted average absolute spread movement of the corresponding market segment

4.1.10 JPY Government-Related

JPY government-related bonds include those issued by Japan Housing Finance Agency (JHF), Japan Finance Organization for Municipalities (JFM), Japan International Cooperation Bank (JBIC), and Development Bank of Japan. These bonds fund housing, local government infrastructure, and regional development projects. Investors are attracted by their higher yields compared to government bonds and greater liquidity than corporate bonds.

Table 30: JPY Government-Related - Model Overview

Section	Details
Estimation Universe	Nomura BPI index
Calibration Methodology	$$\frac{R_i^{spread}}{exposure_i} = \sum_j \beta_j I^{\text{Market Segment}_j} + \epsilon_{i,t}$$ where j indicates the factor cohort. $exposure_i$ is Spread duration or $-spd_dur_i$.
Exposure Mapping	Spread_duration based dummy exposures for Municipalities (Muni bonds) and banks (debentures)
List of factors	The model has one factor: JP GovGuar Muni
Factor Interpretation	Factor return captures weighted average absolute spread movement of the corresponding market segment

4.1.11 CNY Policy Banks

The China onshore bond market has opened up significantly in recent years, allowing foreign investors to efficiently invest. As the second largest fixed income market globally (around $15.5 trillion[20] USD), it is crucial for the Aladdin risk model. Eligible Chinese policy bank bonds are issued by state-owned banks like the China Development Bank, Agricultural Development Bank of China, and Export-Import Bank of China. These bonds aim to provide social benefits, stimulate the economy, and support local industries. They are issued in the Inter-Bank and exchange bond markets, with maturity being the major risk driver.

Table 31: CNY Policy — Model Overview

Section	Details
Estimation Universe	iBoxx ALBI China Onshore index
Calibration Methodology	One factor for each maturity bucket. $$\frac{R_i^{\text{spread}}}{-D_i} = \sum_j \beta_j \, I_i^{\text{maturity}_j} + \varepsilon_i$$ This model does not use the DxS methodology due to the weak empirical relationship between spread volatility and spread at low levels.
Exposure Mapping	Estimation is done in the absolute spread space due to tight spreads in this asset class.
List of factors	• CNY Policy 2Y: WAL to worst < 3 • CNY Policy 5Y: $3 \leq$ WAL to worst < 7 • CNY Policy 10Y: $7 \leq$ WAL to worst < 15 • CNY Policy 30Y: $15 \leq$ WAL to worst < 100
Factor Interpretation	Factor returns represent weighted average spread movements of the corresponding maturity cohort.

[20] Allianz Global Investors

4.1.12 INR Local

The INR local bond universe comprises state, municipal, and territory-issued debt securities, classified as local bonds in accordance with Reuters taxonomy. This segment is dominated by State Development Loans (SDLs), which represent the primary form of sub-sovereign borrowing by state governments, alongside a smaller and evolving municipal bond market supporting urban infrastructure financing.

Annual SDL issuance exceeds $100 billion based on RBI-reported state borrowing programs, underscoring their central role in public financing.[21]

SDLs are issued via auctions conducted by the Reserve Bank of India (RBI), which is the country's central bank responsible for monetary policy and government debt management, and exhibit near-sovereign credit characteristics with a modest yield premium over central government securities.

In contrast, municipal bonds remain small in scale ($\sim$ \$0.5–\$2 billion outstanding) and relatively illiquid. Overall, the market is characterized by institutional investor dominance, fragmented issuance, and relatively shallow secondary market liquidity, with maturity acting as a key driver of risk and return dynamic.[22]

Table 32: INR Local - Model Overview

Section	Details
Estimation Universe	Local bonds covered in BlackRock Index Services (BIS) INR India Corporate Index
Calibration Methodology	One factor for each maturity bucket: $$\frac{R_i^{spread}}{-D \times S_i} = \sum_l \beta_l I_i^{maturity_l} + \epsilon_i$$
Exposure Mapping	Spread duration-based dummy exposures based on Weighted Average Life (Maturity Bucket) of a bond.
List of factors	• INR LOCAL 2yr: WAL to worst < 3 • INR LOCAL 5yr: $3 \leq$ WAL to worst < 7 • INR LOCAL 10yr: $7 \leq$ WAL to worst < 15 • INR LOCAL 30yr: $15 \leq$ WAL to worst < 100
Factor Interpretation	Factor returns represent weighted average spread movements of the corresponding maturity cohort.

4.1.13 Money Market

Money market securities require a different risk decomposition paradigm because they are very short-term, and non-coupon-bearing fixed income instruments. As such, the typical 11 key rate points does not apply as the short-end of the curve is not granular enough while the long-end is mostly irrelevant. Therefore, dedicated sets of interest rate and spread risk factors are in place for these securities to capture the short-end yield curve movement in a more accurate fashion.

A brief overview of the money market model is provided in Table 33.

[21] RBI Report on "State Finances: A study of budgets of 2025-26", Pg 16-18
[22] Municipal Bonds issuance as of March, 2026, is $\sim$ \$0.5 billion

Table 33: Money Market - Model overview

Factor Type	OIS Factor	Money Market Spread to OIS Factor
Exposure Universe	Commercial paper (CP) Deposit securities (certificate of deposit, time deposit, tri-party repos, TFN) Muni, Muni CP, MVRDN (US only)	
Parametric Return Equation	$-OAD \times \Delta OIS - spd_dur \cdot \Delta spread$	
Exposure Type	Option-adjusted duration (OAD)	Spread duration
List of factors	2W, 1M, 2M, 3M, 6M, 12M, 2Y, 3Y, 5Y	2W, 1M, 2M, 3M, 6M, 12M
Curve or Product Coverage	OIS curve for USD, EUR, GBP, CAD, AUD	CP A1/P1, CP A2/P2, deposit (USD, EUR, GBP) 1M muni swap (USD only) ARR (CAD, AUD)
Factor Estimation Methodology	Third-party market data	Spread between the rate of individual money market product listed above (data from third-party) and OIS rate
Factor Volatility Calculation	Volatility of absolute factor changes	
Note	1. The model uses OAD instead of KRD exposures as the money market instruments covered by the model do not have coupons and there is little to gain from using key-rate factors. 2. Each money market security covered by the risk model is mapped to one risk-free rate factor and one corresponding spread factor with tenors that are closest to the security. DxS framework is not used to model the systematic component of spread volatilities as the advantage of this approach is not clear in money market. 3. As the EU OIS has now been replaced by ESTR in EURO market, the EUOIS money market factors are now leveraging ESTR data.	

Note EM-currency denominated CDs map to rates factors and an appropriate ARR factor.

4.2 Developed Corporates

Corporate bonds are issued by companies to fund their operations. In developed countries, these bonds are usually issued in local currency. Large multinational companies often issue bonds in both domestic and foreign currencies to support their international business and reduce financing costs. Corporate bonds are the second largest asset class in global fixed income markets, after government bonds. The spread risk models cover seven major currencies: USD,

EUR, GBP, JPY, AUD, CAD, and CNY.

Credit risk, which is the risk of default, is the main factor affecting corporate bond spread movements. Corporate bond credit ratings, assigned by agencies[23] like Moody's, Fitch, and S&P, indicate the level of credit risk in the security. These ratings are crucial for measuring credit risk. With recent advancements in modeling, as mentioned in section 2.2.2, DxS has become the standard for measuring spread risk. DxS incorporates rating granularity into factor exposures, specifically the OAS component. Ratings are no longer an explicit factor attribute. Back-testing has shown that the DxS model effectively represents spread risk dynamics across various market regimes. The typical risk drivers considered are:

- Industry sectors: Industry is a key driver of credit risk. Companies in the same industry face similar demand and supply shocks, affecting their financial health. The industry's overall state and business cycle also impact credit quality. Therefore, industry factors are often considered first, with a focus on selecting the right level of granularity for industry factors.
- Maturity: Maturity influences credit risk like interest rate risk. The credit curve shape depends on short-term vs long-term projections. The 5-year tenor is the most liquid for most issuers.
- Capital structure: Capital structure determines the payout order if the issuer defaults. Bonds are generally classified as senior/secured, junior/subordinated, and capital securities. Not all issuers offer every type, but DxS captures this well. In larger markets like USD, EUR, and GBP, separate factors exist for capital securities, which are more equity-like and structured[24] differently. Capital securities are hybrid, with features of both bonds and stocks, often having a fixed-to-float structure and being callable.
- Country of issuer domiciles: Countries are generally less crucial in cash bond markets, which are dominated by domestic issuers. However, in Europe, the sovereign-spillover effect was significant during the 2011 Sovereign crisis and other uncertain times. Therefore, the EUR corporate model includes detailed country factors. In USD and GBP corporate risk models, a non-domestic factor is included to differentiate foreign and domestic issuers.
- Style factors: These factors capture sources of return not covered by industry sectors or maturity cohorts. They include well-known investment themes and technical factors based on historical data, enhancing overall explanatory power without compromising non-style factors. Some of the style factors include:
  - Liquidity: Liquidity is a key risk driver in all financial securities, especially in fixed income due to their illiquid trading nature. Our model uses bond trading volume to estimate liquidity levels.
  - Value: Value measures cheapness, defined as the residual bond spread after accounting for probability of default, duration, rating, and capital structure.
  - Quality: Quality measures credit risk of bond issuers and is constructed using the probability of default.
  - Reversal: Reversal is a technical factor reflecting the short-term historical performance of a bond. A short-term reversal effect is found in the US corporate bond space and particularly strong in IG.
  - Momentum (HY): Momentum is a technical factor reflecting the long-term historical performance of a bond. A long-term momentum effect is found in the US corporate HY universe but not in the IG universe.
  - Volatility: Volatility is another technical factor reflecting the historical downside risk of a bond. Bonds with higher risk tend to have higher return.

Additionally, a separate return-based distressed factor is included to cover the risks of all distressed bonds, as DxS becomes ineffective[25] once the bond is traded based on default projections.
The model structure details of each corporate spread risk model will be listed in the following sections.

[23] There are three main rating agencies, Moody's, Fitch, and S&P. Typically, rating agencies rate both specific issues and issuers. They are mostly the same yet could differ especially when the bond sits relatively higher or lower in the capital structure. Our model uses bond ratings if available and then fall back to issuer ratings. In addition, ratings from the three agencies are combined to obtain the final rating.

[24] Capital securities are considered hybrid-securities that have features of both bonds and stocks. They typically have fixed-to-float type of structure with fixed rated coupon at issuance and then floating after a few years. In addition, they are often "callable" which allows the issuer to call the bond at the beginning of the floating-rated period.

[25] High yield bonds are rare in other currency markets and hence there is no IG vs HY distinction outside of USD and EUR.

Table 34: USD corporates - estimation coefficients for bonds not covered by TRACE

Category	Intercept	Log(age)	Log(size)
IG	-12	-0.25	0.93
HY	-11	-0.11	0.84

4.2.1 USD Corporates

Model Data Inputs BlackRock Index Services (BIS) USD Corporate IG and HY Indices excluding emerging market bonds.

Model Estimation Process Non-distressed: A two-stage regression approach is used for non-distressed bonds. Industry, Maturity, Seniority, Yankee, and Liquidity factors are estimated separately for IG and HY universes respectively.

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{Industry_j} \cdot I_i^{Industry_j} + \sum_k \beta_k^{mat_k} I_i^{mat_k} + \sum_l \beta_l^{Seniority_l} I_i^{Seniority_l} + \beta_m^{Yankee_l} I_i^{Yankee_l} + \varepsilon_i \quad (4.25)$$

$$\varepsilon_i = \sum_m \beta_{style(m)} \cdot z_i^{style(m)} + \eta_i \quad (4.26)$$

Distressed: One (Market) factor

$$R_i^{total} = \beta_{distressed} \cdot I_i^{distressed} + \varepsilon_i \quad (4.27)$$

Exposure Mapping

- For non-distressed bonds, DxS based dummy variables exposures are used for non-style factors, indicating factor cohort membership.
- $z_i^{liquidity}$ is calculated as the normalized log monthly trailing trading volumes based on TRACE. Bonds not covered by TRACE and newly issued bonds (bonds issued within the past 20 business days) use fitted z-scores from the following formula

$$z_i^{liquidity} = intercept + \beta_1 \log(age_i) + \beta_2 \log(size_i) \quad (4.28)$$

where age_i is the age of the bond (defined as the number of months since issue date) and $size_i$ is the issue size (in thousands) of the bond. The formula is based on one-off calibrations separately fitted for IG and HY respectively, with coefficients listed in Table 34.

- The other style factor exposures are calculated as normalized raw style scores defined as follows. Reversal is defined as trailing 3-month %-spread change. Momentum is a style specific to the HY universe and is defined as negative trailing 12-month %-spread change excluding the most recent 3 months. There is no overlap between the lookback windows of these two factors. Volatility is defined as the worst %-spread change in the past 6 months. Quality is defined as negative logarithm of probability of default (PD).
- Value exposure is slightly more complicated than the other style factors and is itself generated using a regression model as follows:

$$\log(OAS) \sim \log(PD) + \log(OAD) + rating + seniority \quad (4.29)$$

where Value is defined as the residual term in the above regression model. The regression is run on cross-sections and coefficients are updated accordingly.

- Both Quality and Value exposures need PD to be calculated, which is generated by our internally developed PD model. As of November 2020, the PD model does not cover companies without public equities, and debts issued by private companies are not covered. To ensure coverage of all bonds, the following regression model is used to fit missing PDs.

$$\log(PD) \sim \log(OAS) + I_{financial} + \log(OAS) \cdot I_{financial} + rating + rating \cdot I_{financial} \tag{4.30}$$

Again, the regression is run on cross-sectional data and coefficients are updated accordingly. If the above value cannot be calculated due to missing input(s), the Z-score of the missing log(PD) will be filled with 0.

- For all the missing style exposures except Liquidity and Quality, the following procedure is used to fill missing values:

  1. A missing exposure will first be filled with the average exposure of the bonds in the same seniority group issued by the same issuer.
  2. If such number doesn't exist, the missing exposure will be filled with the average exposure of the bonds issued by the same issuer.
  3. If both numbers don't exist, the missing exposure will be filled by the following cross-sectional regression model:
  $$Exposure \sim Industry + \log(OAS) + rating + \log(amount_issued) + seniority \tag{4.31}$$
  4. If none of the above is available, the missing exposure will be filled with the average style exposure.

Distressed bonds are identified using the following criteria:

(OAS > max (3500 bps, 4 * market factor spread[26])) or if the bond is non-performing[27].
For distressed bonds, price-ratio based exposures are used to account for the distinct behavior of bonds with different prices. A specific price-ratio function is employed to better differentiate the risk for distressed bonds of varying prices.

$$\text{exposure}_i^{\text{distressed}} = f\left(\frac{p_i}{p_f^{\text{distressed}}}\right) = 0.060842 \cdot \left(\frac{p_i}{p_f^{\text{distressed}}}\right)^2 - 0.63139 \cdot \frac{p_i}{p_f^{\text{distressed}}} + 1.570545 \tag{4.32}$$

where $p_f^{distressed}$ is %market-cap weighted average price for all distressed bonds and p_i is the price of the security.

List of factors

- Non-distressed: Table 35 provides a list of non-distressed factors.
- Distressed: One (Market) factor, i.e. US Distressed Bonds.
- US IG Market and US HY Market serve as fallback systematic factors within the Investment Grade (IG) and High Yield (HY) segments, respectively.

  Securities are mapped to these fallback factors when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

[26] Market factor spread is defined as the weighted average OAS of all bonds in the estimation universe with OAS<3500bps. This is added to avoid the model the over-classify bonds into distressed bucket when the market is in super distress with spreads largely inflated, such as in financial crisis.

[27] In addition, a "stickiness" concept is incorporated within this criterion to avoid hysteresis in instances where a bond whose OAS falls on the boundary frequently jumps between categories. Specifically, once a bond is classified as distressed, it will remain classified as distressed as long as the below criteria is met: OAS > 3000 bps

Table 35: USD corporates – non-distressed factors

Factor Type	Sub-Cohort	Investment Grade	High Yield
Industry	Industrial	US IG Auto	US HY Auto
		US IG Media Cable	US HY Cyclical
		US IG Media Noncable	US HY Gaming
		US IG Aero/Envi/Pack	US HY Home Construction
		US IG Manufacturing/ Machinery	US HY Retail
		US IG Building Materials	US HY Healthcare
		US IG Chemicals	US HY Noncyclical
		US IG Computer Hardware	
		US IG Cyclical	US HY Pharmaceuticals
		US IG Independent	US HY Aerospace
		US IG Integrated	US HY Media Cable
		US IG IT Services Consulting	
		US IG Midstream/ Refining	US HY Capital Goods
		US IG Healthcare	US HY Chemicals/Paper
		US IG Metals	US HY Metals
		US IG Paper	US HY Media Non Cable
		US IG Pharmaceutical	US HY Other Industry
		US IG Retail	US HY Tech Other
		US IG Semiconductor	US HY Hardware
		US IG Software	
		US IG Transport	US HY Technology
		US IG Airlines	US HY Transport
		US IG Wireless	US HY Wireless
		US IG Wirelines	US HY Wirelines
		US IG Noncyclical	US HY Independent Energy
		US IG Other Ind	US HY Oil Field Services
			US HY Midstream/ Refining
	Financial	US IG Banking	US HY Banking
		US IG Life Insurance	US HY Finance Companies
		US IG Health Insurance	US HY Insurance
		US IG Other Fin	US HY Other Fin/REITS
		US IG PnC Insurance	
		US IG REIT	

Continued on next page

Factor Type	Sub-Cohort	Investment Grade	High Yield
	Utility	US IG Electric US IG Gas	US HY Utility
Maturity	Maturity to Worst < 3.5	US IG Mat Industrial 2 US IG Mat Financial 2 US IG Mat Utility 2	US HY Maturity 2
	$7.5 <$ Maturity to Worst ≤ 15	US IG Mat Industrial 10 US IG Mat Financial 10 US IG Mat Utility 10	US HY Maturity 10
	Maturity to Worst > 15	US IG Mat Industrial 20 US IG Mat Financial 20 US IG Mat Utility 20	US HY Maturity 20
Seniority	Subordinated	US IG Subordinate	US HY Subordinate
	Capital Security	US IG Capital Security	US HY Capital Security
Yankee	Foreign	US IG Yankee	US HY Yankee
Style	All	US IG Liquidity US IG Value US IG Quality US IG Reversal US IG Volatility	US HY Liquidity US HY Value US HY Quality US HY Reversal US HY Volatility US HY Momentum

Factor Interpretation

- Factor returns represent %spread movements for all non-distressed universe. In detail,
  - Spread is over a treasury of equivalent maturity .
  - Industry factors are first-order factors, capturing weighted average spread movements of the corresponding 5-year industry cohort.
  - Maturity, Seniority, and Yankee factors capture the net effect of factor cohort, controlling for other factors. For example, 2-year maturity factor represent the weighted average difference of spread movements of 2-year maturity bucket and 5-year maturity bucket, holding other effects the same.
  - Liquidity captures the differential spread movements of more liquidity bonds vs less liquidity bonds. The factor return can be interpreted as the spread movements of the 1-std liquidity bond against the average liquid bond.
  - Value measures relative bond cheapness given probability of default, duration, rating and seniority.
  - Quality captures how unlikely the bond issuer will default in one year. Good quality firms are more unlikely to default.
  - Reversal captures the short-term mean-reversion effect in the US corporate bond market.
  - Momentum captures the long-term momentum effect in the HY cohort.
  - Volatility is a measurement of downside risk. Bonds with higher downside risk overall have higher expected return.
- The distressed factor measures the weighted average total return of distressed bonds.

4.2.2 USD Preferred Equity

Model Data Inputs The data source for preferred Equity model is ICE BofAML Core Plus Fixed Rate Preferred Index (MLCPFIXPR). The associated option adjusted spread, spread duration, and spread return are generated by Aladdin analytics.

Model Estimation Process The preferred universe (ICE BofAML Core Plus Fixed Rate Preferred Index, MLCPFIXPR) is run through the USD corporate bond model (Eq 4.25) to obtain residuals. These residuals are then fed into a second-pass regression calibrated solely on the preferred universe with two rating-based factors of IG and HY.

$$\epsilon_i = \beta_{IG} \cdot I_i^{IG} + \beta_{HY} \cdot I_i^{HY} + \lambda_i \tag{4.33}$$

Exposure Mapping DxS exposures are used for all factors.

List of factors Two rating-based factors of IG and HY.

4.2.3 EUR Corporates

Model Data Inputs BlackRock Index Services (BIS) EUR Corporate IG and HY Indices

Model Estimation Process A two-stage regression approach is used for non-distressed bonds.
Step 1: Industry, Maturity, Seniority, and Liquidity factors are estimated using bonds from non-peripheral countries, separately for IG and HY universes. GBP high yield and distressed bonds are included in the Pan Euro HY and distressed universe, while GBP IG corporate bonds have their own set of factors.

$$\frac{R_i^{\text{spread}}}{-D_i \times S_i} = \sum_j \beta_j^{industry_j} \cdot I_i^{\text{industry}_j} + \sum_k \beta_k^{mat_k} \cdot I_i^{\text{mat}_k} + \sum_l \beta_l^{seniority_l} \cdot I_i^{\text{seniority}_l} + \beta_{\text{liquidity}} \cdot z_i^{\text{liquidity}} + \varepsilon_i \tag{4.34}$$

where $i \in$ non-peripheral countries.
Step 2: Coefficients estimated from step 1 are applied to estimate the whole estimation universe to obtain fitted residuals $\hat{\epsilon}_i$. And then one set of country factors (for both IG and HY universes) are estimated.

$$\hat{\varepsilon}_i = \frac{R_i^{\text{spread}}}{-D_i \times S_i} - \left(\sum_j \beta_j^{Industry_j} \cdot I_i^{\text{industry}_j} + \sum_k \beta_k^{mat_k} \cdot I_i^{\text{mat}_k} + \sum_l \beta_l^{seniority_l} \cdot I_i^{\text{seniority}_l} + \beta_{\text{liquidity}} \cdot z_i^{\text{liquidity}} \right) \tag{4.35}$$

$$\hat{\varepsilon}_i = \sum_m \beta_m \cdot I_i^{\text{country}_m} + \xi_i \tag{4.36}$$

where $i \notin \{\text{distressed universe}\}$
Distressed: One (Market) factor

$$R_i^{\text{total}} = \beta_{\text{distressed}} \cdot I_i^{\text{distressed}} + \varepsilon_i \tag{4.37}$$

Exposure Mapping For non-distressed bonds, DxS based dummy variables exposures are used for all factors, indicating factor cohort membership, except for the liquidity factor, where $z_i^{liquidity}$ is calculated as the normalized log monthly trailing trading volumes based on a third-party vendor. Bonds not covered or newly-issued bonds (bonds issued within past 30 days) use fitted z-scores from the following formula

$$z_i^{\text{liquidity}} = \text{intercept} + \beta_1 \log(\text{age}_i) + \beta_2 \log(\text{size}_i) + \beta_3 I_i^{\text{new issue}} \tag{4.38}$$

where $I_i^{newissue}$ is a dummy variable indicating whether the bond is newly issued within the past 30 days. The equation is fitted for European IG/HY Corporate respectively, using data from January 2013 to April 2014. The fitted coefficients are listed in Table 36.

Table 36: EUR corporates - estimation coefficients for bonds without volume data

Type	Intercept	Log(age)	Log(size)	New Issuance
IG	-6.57	-0.38	1.06	1.13
HY	-7.01	-0.36	1.04	1.19

Distressed bonds are identified largely based on whether its implied rating is CCC or below, except for bonds close to maturity (WAL< 0.35). These bonds will not be considered distressed if they have an investment grade (BBB+) agency rating. Distressed bonds are mapped to the distressed factor with adjusted %NAV as exposure.

List of factors

- Non-distressed: The list of non-distressed factors is shown in Table 37.
- Distressed: One (Market) factor, namely Pan Euro Distressed
- EU Credit IG and Pan Euro Credit HY serve as fallback systematic factors within the Investment Grade (IG) and High Yield (HY) segments, respectively.

  Securities are mapped to these fallback factors when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Non-Distressed

Table 37: EUR corporates – non-distressed factors

Factor Type	Sub-Cohort	Investment Grade	High Yield

Continued on next page

Table 37: EUR corporates – non-distressed factors (Continued)

Industry	Industrial	EU Auto IG	Pan Euro Auto HY
		EU Basic IG	Pan Euro Basic HY
		EU Cap Goods IG	Pan Euro Cap Goods HY
		EU Cyclical IG	Pan Euro Cyclical HY
		EU Tech Comm IG	Pan Euro Tech Comm HY
		EU Transport IG	Pan Euro Transport HY
		EU Noncyclical IG	Pan Euro Noncyclical HY
		EU Ind Other IG	Pan Euro Ind Other HY
Industry	Financial	EU Banking IG	Pan Euro Banking HY
		EU Fin Other IG	Pan Euro Fin Other HY
		EU Insurance IG	Pan Euro Insurance HY
	Utility	EU Energy IG	Pan Euro Energy HY
		EU Utility IG	Pan Euro Utility HY
Maturity	Maturity to Worst <3.5	EU Maturity IG 2	Pan Euro Maturity HY 2
	7.5 < Maturity to Worst ≤ 15	EU Maturity IG 10	Pan Euro Maturity HY 10
	Maturity to Worst > 15	EU Maturity IG 20	Pan Euro Maturity HY 20
Seniority	Subordinated	(NONE)	Pan Euro AT1 HY
	Capital Security	EU Capital Security IG	Pan Euro Capital Security HY
Liquidity	All	EU Corp Liquidity IG	Pan Euro Liquidity HY

Continued on next page

Table 37: EUR corporates – non-distressed factors (Continued)

Country	Non-Peripheral	EU Corp France	Pan Euro France HY*
		EU Corp Germany	Pan Euro Germany HY*
		EU Corp Netherlands	Pan Euro Netherlands HY
		EU Corp UK	Pan Euro UK HY
		EU Corp Luxembourg	Pan Euro Luxembourg HY
		EU Corp Sweden	Pan Euro Sweden HY
		EU Corp Switzerland	Pan Euro Switzerland HY
		EU Corp Finland	Pan Euro Finland HY
		EU Corp Denmark	Pan Euro Denmark HY
		EU Corp Norway	Pan Euro Norway HY
		EU Corp Austria	Pan Euro Austria HY
		EU Corp Belgium	Pan Euro Belgium HY
		EU Corp Non-domestic	Pan Euro Non-domestic HY
	Peripheral	EU Corp Spain	Pan Euro Spain HY
		EU Corp Italy	Pan Euro Italy HY
		EU Corp Ireland	Pan Euro Ireland H
		EU Corp Portugal	Pan Euro Portugal HY
		EU Corp Greece	Pan Euro Greece HY
		EU Corp Europe Other	Pan Euro Other HY

Factor Interpretation Factor returns represent %spread movements for all non-distressed universe. In detail:

- Industry factors are first-order factors, capturing weighted average spread movements of the corresponding 5-year industry cohort.
- Maturity, Seniority, and Country factors capture the net effect of factor cohort, controlling for other factors. For example, 2-year maturity factor represent the weighted average difference of spread movements of 2-year maturity bucket and 5-year maturity bucket, holding other effects the same.
- Liquidity factor capturing the differential spread movements of more liquidity bonds vs less liquidity bonds. The factor return can be interpreted as the spread movements of the 1-standard deviation liquidity bond against the average liquid bond.
- The distressed factor measures the weighted average total return of distressed bonds.
- (*) represents mapped factors. Refer Section 9.4.2 for more details.

4.2.4 GBP Corporates

Model Data Inputs iBoxx Sterling Index

Model Estimation Process

- Industry, Maturity, Seniority, Yankee, and Liquidity factors are estimated in a multi-variate regression for the IG universe only.

$$\frac{R^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{industry_j} \cdot I_i^{industry_j} + \sum_k \beta_k^{mat_k} I_i^{mat_k} + \beta^{seniority} I_i^{seniority} + \beta^{Yankee} I_i^{Yankee} + \beta_{liquidity} z_i^{liquidity} + \varepsilon_i \tag{4.39}$$

- Special note: The HY and distressed universe of GBP and EUR corporate bonds are bundled together to calibrate Pan-Euro high yield and distressed factors. Please see the EUR Corporates section for details.

Exposure Mapping

- DxS based dummy variables exposures are used for all factors, indicating factor cohort membership, except liquidity factor, where $z_i^{liquidity}$ is calculated as the normalized log monthly trailing trading volumes based on a third-party vendor. Bonds not covered or newly-issued bonds (bonds issued within past 30 days) use fitted z-scores from the following formula

$$z_i^{liquidity} = intercept + \beta_1 log(age_i) + \beta_2 log(size_i) + \beta_3 I_i^{new_issue} \tag{4.40}$$

  where age_i is the age of the bond (defined as the number of months since issue date) and $size_i$ is the issue size (in thousands) of the bond, and $I_i^{newissue}$ is a dummy variable indicating whether the bond is newly issued within the past 30 days. The equation is fitted for GBP IG Corporate, using data from January 2013 to April 2014. The fitted coefficients from equation 4.39 are listed in table 38.

Table 38: GBP corporates - estimation coefficients for bonds without volume data

Intercept	Log(age)	Log(size)	New Issuance
-6.86	**-0.26**	**1.17**	**1.70**

List of factors

- Table 38 shows the list of factors.
- UK Credit serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 39: GBP corporates – factor list

Factor Type	Sub-Cohort	Factor
Industry	Industrial	UK Auto
		UK Basic
		UK Cap Goods
		UK Cyclical
		UK Tech Comm
		UK Transport
		UK Noncyclical
		UK Ind Other
	Financial	UK Banking
		UK Insurance
		UK Fin Other
	Utility	UK Utility
		UK Energy
Maturity	Maturity to Worst < 3.5	UK Maturity 2
	$7.5 <$ Maturity to Worst ≤ 15	UK Maturity 10
	Maturity to Worst > 15	UK Maturity 20
Seniority	Capital Security	UK Capital Security
Yankee	Foreign	UK Non Domestic
Liquidity	All	UK Corp Liquidity IG

Factor Interpretation Factor returns represent %spread movements. In detail,

- Industry factors are first-order factors, capturing weighted average spread movements of the corresponding 5-year industry cohort.
- Maturity, Seniority, and Yankee factors capture the net effect of factor cohort, controlling for other factors. For example, 2-year maturity factor represent the weighted average difference of spread movements of 2-year maturity bucket and 5-year maturity bucket, holding other effects the same.
- Liquidity factor capturing the differential spread movements of more liquidity bonds vs less liquidity bonds. The factor return can be interpreted as the spread movements of the 1-standard deviation liquidity bond against the average liquidity bond.

4.2.5 CAD Corporates

Model Data Inputs Scotia Capital Domestic Corporate Index excluding Provincials

Model Estimation Process Industry and Maturity factors are estimated in a multi-variate regression.

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{industry_j} \cdot I_i^{industry_j} + \sum_k \beta_k^{mat_k} \cdot I_i^{mat_k} + \varepsilon_i \tag{4.41}$$

Exposure Methodology DxS exposures are used for all CAD Corporate factors.

List of factors

- Table 40 shows the list of factors.
- CA Cash Bonds serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 40: CAD corporates – factor list

Factor Type	Sub-Cohort	Factor
Industry	Financial	CA Banking
		CA Insurance
		CA Other Financial
		CA Securitized
		CA Real Estate
	Industrial	CA Consumer
		CA Cyclical
		CA Energy
		CA Infrastructure
		CA Other Ind
		CA Communication
	Utility	CA Utility
Maturity	Maturity to Worst < 3.5	CA Mat 2
	Maturity to Worst > 7.5	CA Mat 10

Factor Interpretation Factor returns represent %-spread movements for entire universe, in detail:

- Industry factors are first-order factors, capturing weighted average spread movements of the corresponding 5-year industry cohort.
- Maturity factors capture the net effect of the maturity bucket (against 5-year maturity bucket), controlling for Industry factors.

4.2.6 AUD Corporates

Model Data Inputs BlackRock Index Services (BIS) AUD Investment Grade Index

Model Estimation Process Industry and Maturity factors are estimated in a multi-variate regression.

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{industry_j} \cdot I_i^{industry_j} + \sum_k \beta_k^{mat_k} \cdot I_i^{mat_k} + \varepsilon_i \tag{4.42}$$

Exposure Mapping DxS exposures are used for all factors.

List of factors

- Table 41 shows the list of factors.
- AU Industrial serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 41: AUD corporates – factor list

Factor Type	Sub-Cohort	Factor
Industry	Financial	AU Financial
	Industrial	AU Industrial
Maturity	Maturity to Worst < 3.5	AU Maturity 2
	Maturity to Worst > 7.5	AU Maturity 10

Factor Interpretation Factor returns represent %-spread movements for entire universe, in detail:

- Industry factors are first-order factors, capturing weighted average spread movements of the corresponding 5-year industry cohort.
- Maturity factors capture the net effect of the maturity bucket (against 5-year maturity bucket), controlling for Industry factors.

Special Note Exposure Universe – The spread risk factors are calibrated using AUD cash bonds yet are used to measure the spread risk of both cash bonds and CDS of Australia corporate issuers.

4.2.7 CNY Corporates

Model Data Inputs BlackRock Index Services (BIS) China Bond Index

Model Estimation Process Industry and Maturity factors are estimated in a multi-variate regression, separately for IG and HY bonds.

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{industry_j} \cdot I_i^{industry_j} + \sum_k \beta_k^{mat_k} \cdot I_i^{mat_k} + \varepsilon_i \tag{4.43}$$

Exposure Mapping DxS exposures are used for all factors, except for the Distressed factor.

List of factors

- Table 42 details the List of factors.
- CNY IG Other Industrial serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 42: CNY corporates – factor list

Type	Factor Type	Sub-Cohort	Factor
IG	Industry	Metals	CNY Metals
		Consumer Cyclicals	CNY Home Construction
			CNY Retail
		Energy	CNY Energy
		Financials	CNY Bank
			CNY Finance
		Industrials	CNY Other Industry
			CNY Transport
		Real Estate	CNY Building
			CNY REIT
		Technology	CNY Tech
		Utility	CNY Utility
	Maturity	WAL to Worst < 2	CNY 1Y
		$2 <=$ WAL to Worst< 4	CNY 3Y
		$4 <=$ WAL to Worst< 7	CNY 5Y

Factor Interpretation Factor returns represent %spread movements for entire universe, in detail:

- Industry factors are first-order factors
- Maturity factors capture the net effect of the maturity bucket, controlling for Industry factors.

Special Note The HY model has only a single market factor, due to the low number of quotes observed in this space.

Median Absolute Deviation (MAD) is a data treatment applied to IG – Industry, HY and Distress universe to rule out the noisy security-level relative spread change, which is the dependent variable in the regression. The cut-offs are iterated over until most of the noise is treated. The chosen treatment is to limit the returns/ %spread change at ±3MAD when it is between 3 and 10 MAD and dropping returns/ %spread change over ± 10MAD.

4.2.8 JPY Corporates

Model Data Inputs Nomura BPI Corporate Index

Model Estimation Process Bonds have exposure to one of the following factor types- Government Guaranteed/ Municipal bonds or Corporate bonds.
Government Guaranteed/ Municipal bonds are modelled using 1 broad factor.

$$\frac{R_i^{spread}}{-D_i} = \beta_{Govgr} \cdot I_i^{Govgr} + \varepsilon_i \tag{4.44}$$

Corporate bonds are modeled using industry and maturity factors.

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{industry_j} \cdot I_i^{industry_j} + \sum_k \beta_k^{mat_k} \cdot I_i^{mat_k} + \varepsilon_i \tag{4.45}$$

Exposure Mapping There are 2 types of exposures for the IG factors

- Exposure is spread duration for Government Guaranteed/ Municipal bonds
- Exposure is duration times spread (DxS) for Corporate bonds

List of factors

- Table 43 shows the factors used within the JPY Corporate model.
- JP Corp Market serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 43: JPY corporates – factor list

Cohort	Factor type	Mapping Criteria	Factor
Government Guaranteed/ Muni		Currency = JPY and country = Japan and (Security is either Government Guaranteed or Municipal bond)	JP Corp GovGuar Muni
Corporate	Industry	Nomura sector = Communication Technology	JP Corp Comm Tech
		Nomura sector = Consumer Cyclicals	JP Corp Cyclical
		Nomura sector = Consumer Staples	JP Corp Noncyclical
		Nomura sector = Finance	JP Corp Finance
		Nomura sector = Government	JP Corp Government
		Nomura sector = Other Industry	JP Corp Other Industry
		Nomura sector = Materials	JP Corp Materials
		Nomura sector = Utility Energy	JP Corp Energy Utility
	Maturity	years to maturity less than 3.5 years	JP Corp Maturity 2
		years to maturity between 7.5 and 15 years	JP Corp Maturity 10
		years to maturity greater than or equal to 15 years.	JP Corp Maturity 20

4.3 Emerging-market Bonds

Emerging market debt refers to bonds from countries with sovereign ratings below BBB. In recent years, both sovereign and corporate debt in these markets have grown significantly. As a result, EM spread risk models need more frequent updates due to the increasing volume of issuance.

Generally, hard currency emerging market bonds, especially those denominated in USD, are relatively liquid. In contrast, local currency-denominated emerging market corporate bonds are fragmented and less traded by international investors. Therefore, EM spread risk factors are calibrated using USD- and EUR-denominated EM debt to model the spread risk of all hard-currency EM debt and local-currency EM corporate debt[28].

[28]The spread risk of all non-USD EM securities is measured by a parallel set of EM spread risk factors mapped to USD EM spread factors to allow more transparency in risk decomposition. Local currency EM government bond is modelled using local currency treasury curve as mentioned previously.

Issuing debt in a hard currency has traditionally enhanced liquidity in bonds. This allows investors to focus on credit risk without worrying about local currency inflation. Generally, the spread movement of EM debt reflects the issuer's credit risk, making the country the main risk driver for EM sovereign debt. While most cross-sectional spread behavior can be captured with fewer factors, country-specific events require individual country factors, especially for larger markets. Therefore, country-based spread risk factors are used over the hard currency base sovereign curve to capture the credit risk of individual EM countries. Region-based EM factors are used as fallback when there is insufficient liquidity for specific sovereign issuers[29].

For EM domiciled corporates, the country of domicile is the main driver of spread movement due to the local economy and geopolitical forces. Other factors like industry and maturity also play a significant role and are included in the model to capture the spread behavior of EM corporates more comprehensively. The EM corporate issuance universe is divided into four regions: Central & East Europe, Mideast & Africa, Latin America, and Southeast Asia. Spread risk factors are researched and calibrated for each region individually.

Emerging Market Sovereign Bonds

There are a number of EM Sovereign spread models, depending on the currency of issuance of the bonds. As highlighted previously, the USD EM sovereign risk model has a layered structure with the following curves (see Table 46):

- USD nominal base curve;
- Regional factors;
- 3 region-based maturity factors, detailed in Table 44
- Country-specific factors.

Table 44: EM Hard currency Maturity factor Criteria

Maturity	Criteria
1 year	Years to Maturity < 2
2 year	$2 <=$Years to Maturity < 4
5 year	$4 < =$Years to Maturity < 7

Two-stage Regression: Factor returns are estimated in two stages.

- Stage 1 Country Factors are estimated in the first stage of regression. Factor returns are expressed in percentage spread terms (DxS space – spread duration times OAS) and calibrated using security-level excess returns. Specifically, security-level daily excess returns (total returns minus interest rate and carry returns) are converted to the implied percentage spread change:

$$\Delta\%OAS_{i,t-1} = \frac{r_{i,t}}{OAS_{i,t-1} \times spread_{dur_{i,t-1}}} \tag{4.46}$$

- Stage 2 Regional-based maturity factors are calibrated in the second stage of regression using the residuals from the first stage. Factor returns for each maturity cohort are computed as the weighted average of the first stage residuals of all bonds in the cohort. This approach allows the model to capture the remaining risk after estimating country factors. The maturity factor structure is defined as follows:
  - Maturity factors will be calibrated using filtered universe of countries (shown in Table 46).

[29]The list of countries is constantly being reviewed and expanded as needed.

– Filtered universe of countries (which will be considered for maturity factors calibration) will be created by checking if a country lies in at least two maturity bucket and within each maturity bucket there must be at least two bonds.
– Regional 1yr, 3yr and 5yr maturity factors will be estimated at second stage, with exposure as DxS.

4.3.1 USD EM Sovereign & USD EM Sovereign Asia

Model Data Inputs

For USD EM SOV, JPMorgan Global EMBI Index excluding quasi-sovereign EM bonds from China, Korea, and India. For USD EM SOV Asia, ICE BofAAsian Dollar Index (MLADOLUS) is used. The calibration process is the same for all the USD EM models.

Model Estimation Process

Two-stage regression approach: The first stage estimates country factors; the second stage calibrates maturity factors using the residual from the first stage.

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{country_j} \cdot I_i^{country_j} + \epsilon_i \tag{4.47}$$

$$\epsilon_i = \sum_k \beta_k^{mat_k} I_i^{mat_k} + \eta_i \tag{4.48}$$

Similarly, 4 regional spread risk factors are calibrated using EM bonds in respective regions. And one global EM spread risk factors calibrated using all emerging market sovereign bonds in the estimation universe.

Exposure Mapping

DxS exposures are used for all factors

Maturity Factor Methodology

The following countries are used to calibrate the maturity factors. Additionally, only bonds that have mapping to these country factors will have exposure to the maturity factors. The list of countries used to calibrate the regional maturity factors is shown in Table 45.

Table 45: Countries used in calibration of region-specific maturity factors

Region	Country
AF	ANGOLA
	KENYA
	NIGERIA
	SAFRICA
EE	HUNGARY
	KAZAKHSTAN
	ROMANIA
	TURKEY
	UKRAINE
	UZBEKISTAN
ME	BAHRAIN
	EGYPT
	JORDAN
	OMAN
	QATAR
	SARABIA
	UAE
SA	BRAZIL
	CHILE
	COLOMBIA
	DOMINICA
	GUATEMALA
	MEXICO
	PANAMA
	PERU
	SALVADOR
SEA	CHINA
	INDONESIA
	MALAYSIA
	PHILIPPINES

List of factors

- Table 46 shows the topology and factors used to model risk in emerging market sovereign bonds.
- The following fallback factors are used when securities do not qualify for exposure to other systematic factors:
  - Emerg Mkt Africa – Africa

– Emerg Mkt E Europe – Central and Eastern Europe
– Emerg Mkt Mid East – Middle East
– Emerg Mkt S America – Latin America
– Emerg Mkt SE Asia – Southeast Asia

Additionally, EMBIG DxS serves as the global fallback systematic factor.

Securities are mapped to these fallback factors when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 46: Emerging markets sovereign bonds – factor list

Factor Type	Central Eastern Europe	South East Asia	Middle East	Africa	Latin America
Country	Belarus	China	Armenia	Algeria*	Argentina
	Bulgaria	Hong Kong	Azerbaijan	Angola	Belize
	Croatia	India	Bahrain	Cameroon	Bolivia
	Georgia	Indonesia	Egypt	Ethiopia	Brazil
	Hungary	Korea	Iraq	Gabon	Chile
	Kazakhstan	Macau	Jordan	Ghana	Colombia
	Latvia	Malaysia	Kuwait	Ivoire*	Costa Rica
	Lithuania	Maldives	Lebanon	Kenya	Dominica
	Poland	Mongolia	Oman	Morocco*	Ecuador
	Romania	Papua New Guinea	Qatar	Mozambique	Guatemala
	Russia	Pakistan	Saudi Arabia	Namibia	Honduras
	Serbia	Philippines	Tajikistan	Nigeria	Jamaica
	Slovakia	Singapore	UAE	Senegal	Mexico
	Turkey	Sri Lanka		South Africa	Panama
	Ukraine	Thailand*		Tunisia*	Paraguay
	Uzbekistan	Vietnam		Zambia	Peru
					Salvador
					Suriname
					Trintobago*
					Uruguay
					Venezuela
Region (Market)	Central and Eastern Europe	Asia	Middle East	Africa	Latin America
Maturity	1Y, 3Y, 5Y	1Y, 3Y, 5Y	1Y, 3Y, 5Y	1Y, 3Y, 5Y	1Y, 3Y, 5Y

*Factors are currently mapped to other country or regional factors.

Factor Interpretation

All factors measure percentage spread movements, capturing the weighted average spread movements of respective factor cohort, be it country, region, maturity, or the whole universe.

4.3.2 Distressed Bond Methodology

The USD EM Sovereign & USD EM Sov Asia Model classify the securities as distressed bonds when the security is non-performing (either flagged as non-performing or modeled as NPERF) and its DxS exposure is missing or zero:

$$\text{(flag_performing = N OR model = NPERF) AND (DxS is null or 0)} \tag{4.49}$$

The treatment of such securities is as follows:

1. Risk analytics are hard-wired with OAS of 2500bps and spread duration of 4;
2. Unit exposures.

Finally, factor levels are interpolated between start date and end date for factors specified for level interpolation.

4.3.3 EUR Emerging Market Sovereign

Model Data Inputs

JPMorgan EUR EMBI Index

Model Estimation Process The model is calibrated using the same methodology as the USD EM Sovereign model.

See equation 4.47 in the previous section for more details.

Exposure Mapping

DxS exposures are used for all factors

List of factors

- Table 47 shows the factors used for modeling risk for EUR-denominated EM Sovereign bonds. Country factors are chosen to ensure at least one bond with exposure, but sometimes there may be no bond from a country in the calibration index. When this happens, the factor is called "thin" and cannot be calibrated. To address this, a proxy universe of bonds is defined to allow for factor calibration. These proxy bonds borrow properties from all assets in the corresponding regional model. A similar approach is used for other asset-class factors whose estimation cohorts may occasionally be empty.
- The following fallback factors are used when securities do not qualify for exposure to other systematic factors:
  - Emerg Mkt Africa – Africa
  - Emerg Mkt E Europe – Central and Eastern Europe
  - Emerg Mkt Mid East – Middle East
  - Emerg Mkt S America – Latin America
  - Emerg Mkt SE Asia – Southeast Asia

  Additionally, EMBIG DxS serves as the global fallback systematic factor.

  Securities are mapped to these fallback factors when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 47: EUR Emerging markets sovereign bonds – factor list

Factor Type	Central and Eastern Europe	South East Asia	Middle East	Africa	Latin America
Country	Albania	China	Saudi Arabia	Benin	Argentina
	Bulgaria	Indonesia	UAE	Ivoire	Chile
	Croatia	Philippines		Morocco	Colombia
	Czech Republic			South Africa	Mexico
	Estonia			Senegal	Peru
	Hungary			Tunisia	
	Kazakhstan			Egypt	
	Macedonia				
	Montenegro				
	Poland				
	Romania				
	Russia				
	Serbia				
	Turkey				
	Ukraine				
Region (Market)	Central and Eastern Europe	Asia	Middle East and Africa		Latin America

Factor Interpretation

All factors measure percentage spread movements, capturing the weighted average spread movements of respective factor cohort, be it country, region, or the whole universe.

4.3.4 Distressed Bond Methodology

The EUR EM Sovereign Model classifies the securities as distressed bonds when the security is non-performing (either flagged as non-performing or modeled as NPERF) and its DxS exposure is missing or zero:

$$(\text{flag_performing} = \text{N OR model} = \text{NPERF}) \text{ AND } (\text{DxS is null or } 0) \tag{4.50}$$

The treatment of such securities is as follows:

1. Risk analytics are hard-wired with OAS of 2500bps and spread duration of 4;
2. Unit exposures.

4.3.5 Emerging Market Corporate Bonds

Model Data Inputs

1. Asia: JP Morgan Asia Credit Index
2. Latin America/Central & East Europe/Mideast & Africa: JP Morgan CEMBI Board Diversified Index

Model Estimation Process

The calibration methodology differs between Asian and the other regions.

1. Asia Corporates

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{country_{industry_j^{Asia}}} \cdot I_i^{country_{industry_j^{Asia}}} + \sum_k \beta_k^{mat_k^{Asia}} \cdot I_i^{mat_k^{Asia}} + \epsilon_i \tag{4.51}$$

   The regression equation estimates country and maturity factors.

2. Latin America, Central and East Europe and Mideast & Africa

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{country_{industry_j^m}} \cdot I_i^{country_{industry_j^m}} + \sum_k \beta_k^{industry_k^m} \cdot I_i^{industry_k^m} + \sum_l \beta_l^{mat_l^m} \cdot exp_i^{mat_l^m} + \epsilon_i \tag{4.52}$$

   where $m \in LatinAmerica, Central\&EastEurope, MidEast\&Africa$, indicating that three separate regressions are run using respective universe for the three regions to calibrate country, industry and maturity factors.

$$s.t. \sum_k \beta_k w_k = 0 \tag{4.53}$$

   Three regional-based market factors are estimated using:

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_m \beta_m \cdot I_i^{region_m} + \epsilon_i \tag{4.54}$$

Exposure Mapping

All factors are DxS based dummy variables, indicating the membership status of the security belonging to the factor cohort, except the maturity factor exposures in the three Non-Asia corporate models, which are continuous, measuring the proportional distance of the security's maturity to the maturity factor. In detail,

$$exp_i^{mat} = \begin{cases} f^{mat_2} & : mat_i \leq 2 \\ \frac{5-mat_i}{3} \cdot f^{mat_2} & : 2 < mat_i \leq 5 \\ \frac{mat_i-5}{5} \cdot f^{mat_{10}} & : 5 < mat_i \leq 10 \\ f^{mat_{10}} & : mat_i \geq 10 \end{cases} \tag{4.55}$$

where mat_i is the maturity of the security and $f^{mat_k}, k \in \{2, 10\}$ refers to the 3 maturity factors, with k indicating the maturity tenor. This continuous approach allows the maturity exposures to transition smoothly between different maturity factors as it matures, resulting in more stable risk forecasts

List of factors

- Table 48 shows the list of factors used to model corporate bonds in Asian Emerging Markets.

Table 48: Asian markets corporate bonds – factor list

Class	Country	Sub-Class//Industry	Factor
Maturity		10-year	Asia Mat Corp 10
		30-year	Asia Mat Corp 30
Industry	China	Properties	China Property IG
		Corporates	China Corp IG
		Properties	China Property HY
		Corporates	China Corp HY
		State Owned Enterprises	China State Owned
	Hong Kong	Financials	Hong Kong Fin IG
		Industrials	Hong Kong Ind IG
		Financials	Hong Kong Fin HY
		Industrials	Hong Kong Ind HY
	Korea	Financials	Korea Fin IG
		Industrials	Korea Ind IG
		Financials	Korea Fin HY
		Industrials	Korea Ind HY
		State Owned Enterprises	Korea State Owned
	India	Corporates	India Corp IG
		Corporates	India Corp HY
		State Owned Enterprises	India State Owned
	Indonesia	Corporates	Indonesia Corp IG
		Resources	Indonesia Resources IG
		Corporates	Indonesia Corp HY
		Resources	Indonesia Resources HY
	Philippines	Corporates	Philippines Corp IG
		Corporates	Philippines Corp HY
	Singapore	Corporates	Singapore Corp IG
		Corporates	Singapore Corp HY
	Other	Corporates	Asia Corp IG
		Corporates	Asia Corp HY

Table 49: Regional corporate bonds – factor list

Region	Latin America	Middle East and Africa	Central & Eastern Europe
Market	Latin America	Middle East and Africa	Europe
Country	Brazil	United Arab Emirates	Russia
	Chile	Israel	Turkey
	Colombia	Nigeria	E Other Country
	Mexico	South Africa	
	Peru	MIDAF Other Country	
	LA Other Country		
Industry	LA Consumer	MIDAF Financial	E Financial
	LA Financial	MIDAF Metals and Mining	E Metals and Mining
	LA Industrial	MIDAF Oil and Gas	E Oil and Gas
	LA Metals and Mining	MIDAF Other Sectors	E Other Sectors
	LA Pulp and Paper		
	LA Telecom		
	LA Utilities		
	LA Other Sectors		
Maturity	EM Corp LA Mat 2	EM Corp MIDAF Mat 2	EM Corp E Mat 2
	EM Corp LA Mat 10	EM Corp MIDAF Mat 10	EM Corp E Mat 10

4.4 CDS and CDS Indices

Since the early 2000s, global credit markets have evolved significantly, driven by trends toward globalization and the growing use of credit derivatives such as credit default swaps (CDS). A CDS is a financial contract in which the seller agrees to compensate the buyer if a loan default or other credit event occurs. In exchange for this protection, the buyer makes an initial ("upfront") payment and/or periodic ("spread") payments.

CDS indices aggregate a basket of the most liquid CDS contracts into a standardized product, offering greater liquidity and narrower bid-offer spreads. The two primary families of corporate CDS indices (CDX and iTraxx) are published by IHS Markit, now part of S&P Global (since 2022). These indices are rebalanced every six months, creating a new "series" that adds or removes issuers from the previous version. The most recent series is referred to as the "on-the-run" series. If a credit event occurs for an issuer, its CDS contract is removed from the index, resulting in a revised version of the same series.

Credit risk or the risk of default is the main driver of spread movements in both single-name CDS and CDS indices. The single-name CDS risk model incorporates multiple sets of spread risk factors, calibrated through multi-step, multivariate regressions using traded CDS data worldwide to capture credit risk.

The CDS index model is relatively new compared to the single-name CDS model, reflecting the post-2008–09 shift in market liquidity from single-name CDS to CDS indices. The risk model for CDS indices uses spread risk factors based on movements in the on-the-run series of each index, with a smoothing mechanism to prevent distortions caused by index rebalancing or version changes.

4.4.1 Single-name CDS

After the rapid initial growth of the CDS market, ISDA made significant changes to contract structures, including the creation of "Determinations Committees" to define credit events or successions and to standardize terms and coupons.

See ([3]) for further details. Before the GFC, many modeling frameworks priced corporate bonds and CDS from a single issuer using a common factor, usually the implied hazard rate. These practices have reduced as it became clear that single-name CDS often do not represent a perfect hedge for issuer credit risk.

Model Data Inputs Daily composite CDS quotes[3] for global CDS issuers from US, Europe, Japan, and Australia.

Model Estimation Process Each CDS contract is first categorized into one of the four rating categories based on Implied Rating model. See Table 50 for the categories.

Table 50: CDS – single-name CDS calibration methodology

Implied Rating	Rating Category
AA- and above	Super Investment Grade (SIG)
BBB- to A+	Investment Grade (IG)
CCC+ to BB+	High Yield (HY)
CCC and below	Distressed

Then, distressed and non-distressed CDS use different methodologies.
For non-distressed CDS:

- Step 1 estimating global factors by regressing global five-year CDS daily spread returns on rating category dummies:

$$\frac{\Delta Spread_i}{Spread_i} = \sum_l \beta_l \cdot I_i^l + \epsilon_i, \quad l \in \{\frac{SIG}{IG}, HY\} \tag{4.56}$$

- Step 2 the residual from step 1 is used to estimate the region/rating-based maturity factors:

$$\frac{\Delta Spread_i}{Spread_i} - \sum_l \hat{\beta}_l \cdot I_i^l = \sum_j \beta_j \cdot I_i^{region_{rating_{maturity_j}}} + \epsilon_i \tag{4.57}$$

- Step 3 the residual from step 2 is used to estimate the region/rating-based industry factors:

$$\frac{\Delta Spread_i}{Spread_i} - \sum_l \hat{\beta}_l \cdot I_i^l - \sum_j \hat{\beta}_j \cdot I_i^{region_{rating_{maturity_j}}} = \sum_k \beta_k \cdot I_i^{region_{rating_{industry_k}}} + \epsilon_i \tag{4.58}$$

For distressed CDS: Factors are estimated in a regression with one dummy variable, with market cap as weight, using total returns.

$$R_i^{total} = \sum_l \beta_l \cdot I_i^{region} + \epsilon_i, \quad l \in \{US, EU\} \tag{4.59}$$

Special features:

- EU and UK are clustered as one region in model calibration.
- Due to limited issuance in Australia and Japan markets, although issuers from these countries are included in the global factor estimation for non-distressed CDS, they are excluded from the 2nd and 3rd steps in the non-distressed calibration and excluded from the distressed CDS calibration. Thus, AU and JP CDS do not have their own risk factors, instead they share the same model as corporate bonds. See section (4.2.6) and section 4.2.8 sections above for details.

Exposure Mapping

- For non-distressed CDS, dummy variables exposures are used for all factors, indicating factor cohort membership.
- For distressed CDS, price-ratio adjusted % NAV are used as exposures.

List of factors The CDS model has a total of 91 risk factors:

- Two global factors, segmenting the universe into investment and HY universes
- 16 maturity factors (8 for US, and 8 for EU)
- 71 industry factors (50 for US, 21 for EU)
- Two distressed factors (US and EU)

In addition, the model contains a set of maturity and industry factors for GBP-denominated CDS, but they are mapped to the corresponding EU factors as EU and UK are clustered as one region in the model estimation.

Table 51: CDS – single-name CDS factor list

Global Factor by Rating			
		SIG / IG	HY
		Global 5	Global HY 5
Maturity Factor by Rating / Region			
		SIG / IG	HY
2 Year Maturity		{US, EU} Maturity 2	{US, EU} Maturity HY 2
5 Year Maturity		{US, EU} Maturity 5	{US, EU} Maturity HY 5
10 Year Maturity		{US, EU} Maturity 10	{US, EU} Maturity HY 10
20 Year Maturity		{US, EU} Maturity 20	{US, EU} Maturity HY 20
Industry Factor by Rating / Region			
	United States		
	SIG	IG	HY
Automotive	US Ind SIG	US Auto IG	US Auto HY
Basic Industries		US Basic IG	US Basic HY
Capital Goods		US Cap Goods IG	US Cap Goods HY
Consumer Cyclicals		US Cycl IG	US Cycl HY
Energy		US Energy IG	US Energy HY
Home Construction		US Home Con IG	US Home Con HY
Media Cable		US Cable IG	US Cable HY
Median Non-Cable		US Non Cbl IG	US Non Cbl HY
Non-cyclical Industries		US Non Cyc IG	US Non Cyc HY
Other Industrial		US Other IG	US Other HY
Paper		US Paper IG	US Paper HY
Pharmaceutical		US Pharma IG	US Pharma HY
Pipelines		US Pipe IG	US Pipe HY
Technology		US Tech IG	US Tech HY
Telecom		US Telecom IG	US Telecom HY
Tobacco		US Tobacco IG	US Tobacco HY
Transportation		US Transp IG	US Transp HY
Banking and Brokerage	US Finance SIG	US Bank IG	US Bank HY
Financial Companies	US Fin Oth SIG	US Fin Com IG	US Fin Com HY
Foreign and Sovereign		US Foreign IG	US Foreign HY
Insurance		US Insur IG	US Insur HY
REIT		US Reit IG	US Reit HY

Continued on next page

Table 51: CDS – single-name CDS factor list (Continued)

Utility	US Util SIG	US Util IG	US Util HY
	Euro / UK		
	SIG	IG	HY
Communications and Tech	EU Ind SIG	EU Com Tec IG	EU Com Tec HY
Consumer Cyclicals		EU Cycl IG	EU Cycl HY
Non-cyclical Industries		EU Non Cyc IG	EU Non Cyc HY
Other Industrial		EU Ind Misc IG	EU Ind Misc HY
Industrial		EU Ind IG	EU Ind HY
Basic		EU Basic IG	EU Basic HY
Finance	EU Fin SIG	EU Finance IG	EU Finance HY
Government		EU Govt IG	EU Govt HY
Utility and Energy	EU Util Ene SIG	EU Util Ene IG	EU Util Ene HY
Distressed Factor by Region			
US Distressed			
EU Distressed CDS			

Factor Interpretation Factor return represents % spread movement for all non-distressed factors. In detail,

- Rating-based global factors are first-order factors, capturing weighted average spread movements of the corresponding rating cohort.
- Maturity factors capture the net effect of factor cohort, controlling for global factors.
- Industry factors capture the net effect of factor cohort, controlling for global and maturity factors.
- The distressed factors measure the weighted average total return of distressed CDS.

Model Map In the CDS model, spread risk exposure is mainly loaded onto global and maturity factors, which may not be intuitive for users focusing on specific regions or industries. Additionally, the current factor structure is inconsistent with the corporate bond model, where industry is the primary factor. To address this, users can turn on the "model map" feature, which re-factors the original calibrated factors.

- The industry factor is calculated as the sum of the global factor, the 5-year maturity factor, and the original industry factor corresponding to the factor's rating/region combination. For example, under the model map, the US Cycl IG factor is the sum of Global 5, US Maturity 5, and the original US Cycl IG factors.
- The maturity factor (other than 5Y) is calculated as the difference between the original factor and the 5Y maturity factor corresponding to the factor's rating. For example, under the model map, EU Maturity HY 10 is the difference between the original EU Maturity HY 10 and EU Maturity HY 5.
- The global factors and the 5Y maturity factors are dropped out from the model to ensure model identifiability. The re-factored industry and maturity factors bear similar meaning as those in the corporate bond models.

The re-factored industry and maturity factors bear similar meaning as those in the corporate bond models[30].

[30]CDS risk model is one of the oldest risk models actively in use. The factor structure is not consistent with those developed/revamped

4.4.2 CDS Indices

Model Data Inputs On-the-run series of major global CDS indices. See "List of factors" section for the full list.

Model Estimation Process Factor return is calculated as the daily spread movement of each on-the-run CDS index series. 5Y and 10Y spreads are used, when available, as they are the most liquid tenors.
Special features:

- For indices that are low liquidity, discontinued, or subject to other data limitations, the respective risk factors are set as mapped factors to their proxy indices. See section (4.4.2) for the full list.
- Factor return smoothing is applied in the event of index rebalancing or version change to make sure on each day factor return will be calculated from the same series and version of a particular CDS index.

Exposure Mapping DxS for all CDS index risk factors

List of factors

Table 52: CDS – CDS indices factor list

Family	Index	Tenor	
		5Y	10Y
CDX NA IG	IG Main Index	CDX.NA.IG.5	CDX.NA.IG.10
	High Volatility	CDX.NA.IG.HVOL.5	CDX.NA.IG.HVOL.10
	Industrials	CDX.NA.IG.INDU.5	CDX.NA.IG.INDU.10
	Consumers	CDX.NA.IG.CONS.5	CDX.NA.IG.CONS.10
	Energy	CDX.NA.IG.ENRG.5	CDX.NA.IG.ENRG.10
	TMT	CDX.NA.IG.TMT.5	CDX.NA.IG.TMT.10
	Financials	CDX.NA.IG.FIN.5	CDX.NA.IG.FIN.10
	BBB[1]	CDX.NA.IG.BBB.5	CDX.NA.IG.BBB.10
CDX NA HY/XO	HY Main Index	CDX.NA.HY.5	CDX.NA.HY.10
	HY B	CDX.NA.HY.B.5	CDX.NA.HY.B.10
	HY BB	CDX.NA.HY.BB.5	CDX.NA.HY.BB.10
	HY High Beta	CDX.NA.HY.HB.5	CDX.NA.HY.HB.10
	Cross Over (XO)		CDX.NA.XO.10
	HY Ex-BB[2]	CDX.NA.HY.EXBB.5	
CDX and iTraxx EM	CDX EM	CDX.EM.5	CDX.EM.10
	CDX EM Diversified	CDX.EM.DIV.5	CDX.EM.DIV.10
	CDX EM ex. EU	CDX.EM.EX.EU.5	CDX.EM.EX.EU.10
CDX and iTraxx EM	CDX LatAm Corp	CDX.LATAM.CORP.5	CDX.LATAM.CORP.10
	iTraxx CEEMEA	ITRAXX.CEEMEA.5	ITRAXX.CEEMEA.10

Continued on next page

later such as the corporate bond model. Model map is a tactical solution that allows consistent interpretation of risk factors across all risk models, without changing underlying factor calibration. When CDS model is revamped in the future, its factor calibration methodology will be consistent with other models, and by that time model map will no longer be needed.

Table 52: CDS – CDS indices factor list (Continued)

Muni CDX (MCDX)	Muni CDX	MCDX.NA.5	MCDX.NA.10

Table 53: CDS – list of iTraxx and other CDS index factors

Index Family	Index	Tenor	
		5Y	10Y
iTraxx Europe	Main Index	ITRAXX.EUR.5	ITRAXX.EUR.10
	Crossover	ITRAXX.XO.5	ITRAXX.XO.10
	Senior Financials	ITRAXX.FINSR.5	ITRAXX.FINSR.10
	Subordinated Financials	ITRAXX.FINSUB.5	ITRAXX.FINSUB.10
	Non-Financials	ITRAXX.NONFIN.5	ITRAXX.NONFIN.10
	iTraxx MSCI ESG Screened Europe[1]	ITRAXX.ESGEUR.5	
iTraxx Asia	Australia	ITRAXX.AUST.5	ITRAXX.AUST.10
	Japan	ITRAXX.JPN.5	ITRAXX.JPN.10

Factor Interpretation Factor return represents %-spread movement of each on-the-run CDS index series, subject to smoothing around index rebalancing and version change.

4.5 Leveraged Loans

Leveraged buyout (LBO) loans started in the mid-1980s and have since become the main way for corporate borrowers to get loans from banks and other institutional capital providers. Leveraged loans are used for several purposes:

- Supporting mergers and acquisitions (M&A), like leveraged buyouts
- Recapitalizing a company's balance sheet
- Refinancing debt or funding general corporate needs

Unlike traditional loans, leveraged loans are issued for specific purposes and are cheaper and more efficient to manage. Compared to high yield bonds, leveraged loans are always secured by collateral, have more restrictive covenants, and typically have shorter maturities. They are mostly floating-rated with optionality and share a similar investor base with high yield bonds.

The leveraged loan market has grown significantly worldwide, with the US being the dominant player. The European market is about one-fifth the size of the US market in terms of issuance. Currently, European loans are mapped to US loan factors as there is no dedicated model for them.

Industry segments play a crucial role in risk management because sectors go in and out of favor. Loans in desirable sectors, like telecom in the late 1990s or healthcare in the early 2000s, can help syndication. Loans to issuers in defensive sectors (like consumer products) are more appealing during economic uncertainty, while cyclical borrowers (like chemicals or autos) are more attractive during economic upswings. This supports the need for industry granularity in the risk model.

Second-lien loans have claims on collateral that are junior to first-lien loans and typically have less restrictive covenant packages. They are priced at a premium to first-lien loans, making them potentially riskier.

4.5.1 USD Leveraged Loans

Model Data Inputs S & P Leveraged Loan Index.

Model Estimation Process Non-distressed: Industry, and second lien factors are estimated in cross-sectional regressions, with weights being market cap multiplied by DxS.

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_k \beta_k^{industry_k} \cdot I_i^{industry_k} + \beta_{lien2} I_{lien2} + \epsilon_i \tag{4.60}$$

Distressed: The distressed factor is estimated as follows in a regression with one dummy variable, with market cap as weight.

$$R_i^{total} = \beta_{distressed} \cdot I_i^{distressed} \tag{4.61}$$

Exposure Mapping Exposure type is DxS for non-distressed loans. Each non-distressed loan will be mapped to one industry factor, and it will be mapped to the second lien factor only if it belongs to the second (or lower) lien in the debt structure.
Distressed loans are identified with the following criteria:

(price < 65) or (OAS >2500) or if a loan is non-performing [31]

The exposure to the distressed factor is price-ratio-adjusted % -NAV where a price-ratio-based function (i.e. price of security/average price of all non-distressed loans) is employed to better differentiate risks of different distressed loans:

$$exposure_i^{distressed} = f(\frac{p_i}{p_f}) = -1.364829 \cdot \frac{p_i}{p_f} + 2.364829 \tag{4.62}$$

List of factors

- Table 54 details the list of factors for USD leveraged loans model.
- US Loans Market serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

[31] Note a "stickiness" concept is applied only to a representative universe of loan securities. In detail, once a bond is classified as distressed, lately it only needs to satisfy the following less strict criteria to remain distressed: (price < 75) or (OAS > 2000).

Table 54: USD leveraged loans – factor list

Type	Class	Factors
Non-distressed	Industry	Auto
		Consumer Cyclical Service
		Entertainment
		Gaming
		Other Cyclical
		Retail
		Metal
		Other Basic Industry
		Capital Goods
		Cable & Media
		Other Communications
		Other Industry
		Software
		Tech Other
		Technology
		Energy
		Utility
		Other Financials
		Banking
		Insurance
		Pharmaceuticals
		Healthcare
		Other Noncyclical
	Structure	Second Lien
Distressed	Distressed Factor	Distressed Factor

Factor Interpretation Factor returns represent %-spread movements for all non-distressed universe. In detail,

- Industry factors are first-order factors, capturing weighted average spread movements of the corresponding industry cohort;
- Second lien factor captures the net effect of factor cohort, controlling for industry factors.

The distressed factor measures the weighted average total return of distressed bonds.

4.5.2 Other Leveraged Loans

Outside of US, leveraged loans are mostly only issued in Europe. Currently, risk models do not yet have dedicated factors for non-USD leveraged loans. Instead, 2 factors are created, mapped to USD leveraged loan factors, to proxy the spread risk of non-USD leveraged loans. A non-USD loan is mapped to the distressed factor if its agency rating is CCC or below. If the agency rating is not populated, then a loan is distressed when its price is lower than 90 or its spread is higher than 1500 bps.

List of factors

Table 55: Other leveraged loans – factor list

Type	Factor
Non-Distressed	Non-USD Non-Distressed Factor*
Distressed	Non-USD Distressed Factor*

(*) represents mapped factors. Refer Section 9.4.2 for more details.

4.6 Municipal Bonds

Municipal bonds, or "munis," are debt securities issued by US state and local governments to fund daily operations and public projects like schools, highways, hospitals, and sewer systems. There are both taxable and tax-exempt municipal bonds, with tax-exempt bonds being the majority. These tax-exempt bonds are attractive to retail investors because their coupon payments are exempt from federal and state income taxes for in-state residents. Retail investors hold about 70%[32] of the muni bond market. The market size of municipal bond market is substantial at around $4.2 trillion of the outstanding as of 2024YE[33]. The muni market is highly decentralized, with over two million bonds issued by tens of thousands of issuers, featuring a wide variety of structures and features. Compared to other bond markets, the muni market is relatively illiquid due to its fragmented nature and lack of transparency. Municipal bonds are generally divided into two categories: General obligation (GO) bonds and revenue (REV) bonds. GO bonds are used to finance municipal operations and are backed by the credit and taxing power of the state or municipality. REV bonds are issued to finance specific revenue-generating infrastructure projects, such as toll roads, bridges, airports, and housing projects. The interest and principal payments for REV bonds are tied to the specific sources of revenue from the projects, rather than the issuer. This makes REV bonds riskier due to the uncertainty around the project's success.

Many municipal bonds are pre-refunded, meaning they are backed by US Treasury bonds or other government securities, making them the lowest risk segment in the municipal universe. The muni market is traditionally considered one of the safest asset classes, with credit risk being the primary source of uncertainty. The BlackRock fixed income model uses the DxS approach to better capture variations and respond quickly to market changes. It controls for spread, regional factors for GO bonds, and industry factors for REV bonds. Pre-refunded bonds are modeled separately due to their tight spreads and low risk. The taxable municipal bond market is smaller and funds activities not qualified for tax-exempt status. Risk models use maturity-based factors to model spread risk. Only non-callable bonds are used for factor estimation to simplify the process and improve model performance. See [4] for further details. For duration-based exposure models, the BlackRock Municipal Bond pricing model is used, details of which can be found in [5].

4.6.1 Muni Tax-exempt IG

The risk decomposition for tax-exempt munis, presents risk as exposure to the following curves:

- Nominal USD base curve;
- A spread curve capturing the basis between the nominal base curve and a AAA muni-specific curve, the details of which are provided in section Muni AAA Estimation below;
- Spreads above the AAA muni curve shown in Table 56

[32] MSRB
[33] SIFMA

Muni AAA Estimation A basis curve is computed for tax-exempt muni bonds using non-callable AAA MMD quotes from LSEG/Refinitiv. Then, a basis is estimated as the spread between this curve and the USD nominal base curve.

Model Data Inputs BlackRock Municipal IG Index.

Model Estimation Process Bonds will have exposure to one of the following factor types- Pre Refunded/ Escrowed, AAA or Industry. Bonds having exposure to Industry factors will also have exposure to maturity factors.

$$\frac{R_i^{spread}}{-D_i} = \begin{cases} \beta I_i^{pre-refunded} \\ \sum_k \beta_k I_i^{AAA_k} \end{cases} \tag{4.63}$$

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_k \beta_k \cdot I_i^{industry_k} + \sum_l \beta_l \cdot I_i^{mat_l} + \varepsilon_i \tag{4.64}$$

Exposure Mapping There are 2 types of exposures for the IG factors:

- Exposure is spread duration for Pre-Refunded and AAA factors
- Exposure is duration times spread (DxS) for Industry and Maturity factors

List of factors

- Table 56 details the List of factors for the Muni Tax-exempt IG model.
- IG Corp Other, IG State Other and AAA Rev Other serve as fallback systematic factors within the Revenue (REV), General obligation (GO) and AAA cohorts respectively.

  Securities are mapped to these fallback factors when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 56: Muni Tax-exempt IG Factors

Cohort	Sub cohort	Mapping Criteria	Factor
Escrowed	Escrowed	sector = [Escrowed to Maturity , Pre refunded]	US Muni IG Escrowed

Continued on next page

Table 56: Muni Tax-exempt IG Factors (Continued)

AAA	AAA	Rating = AAA & sector = Tax Backed State	US Muni AAA State Tax
		Rating = AAA & sector = Tax Backed Counties	US Muni AAA Local Tax
		Rating = AAA & sector = Tax Backed Authority	US Muni AAA Go Other
		Rating = AAA & sector = [Public Power , Gas , Water & Sewer, Cogeneration]	US Muni AAA Utility
		Rating = AAA & sector not in above AAA segments	US Muni AAA Rev Other
Industry	GO	state = California & sector = Tax Backed Authority/County/State	US Muni IG California
		state = New York & sector = Tax Backed Authority/County/State	US Muni IG New York
		state = New Jersey & sector = Tax Backed Authority/County/State	US Muni IG New Jersey
Industry	GO	state = Connecticut & sector = Tax Backed Authority/County/State	US Muni IG Connecticut
		state = Illinois & sector = Tax Backed Authority/County/State	US Muni IG Illinois
		state = Texas & sector = Tax Backed Authority/County/State	US Muni IG Texas

Continued on next page

Table 56: Muni Tax-exempt IG Factors (Continued)

	GO	state = Massachusetts & sector = Tax Backed Authority/County/State	US Muni IG Massachusetts
		state = Florida & sector = Tax Backed Authority/County/State	US Muni IG Florida
		state = Pennsylvania & sector = Tax Backed Authority/County/State	US Muni IG Penn
		sector = Tax Backed Authority/County/State where state not in the above-mentioned state	US Muni IG State Other
		sector = School Districts	US Muni IG School Dist.
		sector = Development Districts	US Muni IG Dev. Dist.
	Revenue	state = [Puerto Rico , American Samoa, Guam, U.S. Virgin Islands , Northern Mariana Islands]	US Muni IG Territories
		sector = [Public/Private University , Charter Schools, Institution, Student Housing, Student Loan, Other Education]	US Muni IG Education
		sector = Airport	US Muni IG Airport
		sector = [Toll, Parking, Ports, Transport Other, Mass Transit]	US Muni IG Transport Other
		sector = [Hospital, Continuing Care Retirement Community , Skilled Nursing Facility , Other Healthcare]	US Muni IG Health
		sector = [Tobacco State , Tobacco Local]	US Muni IG Tobacco
		sector = Single Family Housing	US Muni IG Single Family Housing
		sector = Multi Family Housing	US Muni IG Multi Family Housing
		sector = [Water & Sewer, Sanitation]	US Muni IG Essential Services
		sector = [Public Power, Gas, Cogeneration]	Us Muni IG Public Utility
		sector = Hotel	US Muni IG Hotel
		sector = Prepaid Gas	US Muni IG Prepaid Gas
		sector not in above	US Muni IG Corp Other
Maturity	Maturity	Maturity to Worst < 3y	US Muni IG Maturity 2
		7y <= Maturity to Worst < 15y	US Muni IG Maturity 10
		Maturity to Worst ≥ 15y	US Muni IG Maturity 30

Factor Interpretations Factor returns represent %-spread movements of the universe

4.6.2 Muni Tax-exempt HY

Model Data Inputs BlackRock Municipal HY Index

Model Estimation Process Non- distressed - Industry and Maturity factors

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_k \beta_k \cdot I_i^{industry_k} + \sum_l \beta_l \cdot I_i^{mat_l} + \varepsilon_i \tag{4.65}$$

Distressed - One (Market) factor:

$$R_i^{spread} = \beta_{distressed} I_i^{distressed} + \epsilon_i \tag{4.66}$$

Exposure Mapping

- For non-distressed bonds, DxS based dummy variables exposures are used for all factors, indicating factor cohort membership.
- Distressed bonds are identified with the following criteria: **(Rating is CCC or below but not NR (not rated)) or (OAS > 650 bps) or if bond is non-performing)**

  For distressed bonds, price-ratio based exposures are employed to account for the differential behavior of differently priced distressed bonds. More specifically, the following price-ratio based function is employed for distressed universe to better differentiate the risk for distressed bonds of different prices.

$$\text{exposure}_i^{\text{distressed}} = f\left(\frac{p_i}{p_f^{\text{distressed}}}\right) = 1.04\left(\frac{p_i}{p_f^{\text{distressed}}}\right)^2 - 4.41\left(\frac{p_i}{p_f^{\text{distressed}}}\right) + 4.37 \tag{4.67}$$

  where $p_f^{distressed}$ is %market-cap weighted average price for all distressed bonds and p_i is the price of the security.

List of factors

- Non-Distressed: Table 57 details the List of factors for the Muni tax-exempt HY model.
- Distressed: One (Market) factor, i.e. US Distressed Bonds
- HY Commercial Land Dev and HY Corp Other serve as fallback systematic factors within the Revenue (REV) and General obligation (GO) cohorts respectively.

  Securities are mapped to these fallback factors when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 57: Non-distressed HY Muni bond factors

Cohort	Sub cohort	Mapping Criteria	Factor

Continued on next page

Table 57: Non-distressed HY Muni bond factors (Continued)

Industry	GO	sector = Tax Backed Authority/County/State	US Muni HY Tax
		sector = School Districts	US Muni HY School Dist.
		sector = Development Districts & state = California	US Muni HY California Dev. Dist.
		sector = Development Districts & state = Florida	US Muni HY Florida Dev. Dist.
		sector = Development Districts & state = Colorado	US Muni HY Colorado Dev. Dist.
		sector = Development Districts & state = [Arizona , Texas , South Carolina , Nevada, Hawaii]	US Muni HY Other Land Dev Dist.
		sector = Development Districts where state not in above	US Muni HY Commercial Land Dev. Dist.
	Revenue	state = Puerto Rico	US Muni HY Puerto Rico
		state = [American Samoa, Guam, U.S. Virgin Islands , Northern Mariana Islands]	US Muni HY Territories Other
		sector= Charter Schools	US Muni HY Charter School
		sector = Student Housing	US Muni HY Student Housing
		sector = [Public/Private University , Institution, Student Loan, Private School, Other Education]	US Muni HY Education Other
		sector = Airport	US Muni HY Airport
		sector = [Toll, Parking, Ports, Transport Other, Mass Transit]	US Muni HY Transport Other
		sector = Hospital	US Muni HY Hospital
		sector = [Continuing Care Retirement Community , Skilled Nursing Facility , Other Healthcare]	US Muni HY Health Other
		sector = [Tobacco State , Tobacco Local]	US Muni HY Tobacco
		sector = [Single Family Housing , Multi Family Housing]	US Muni HY Housing
		sector = Water Sewer	US Muni HY Essential Services
		sector = [Public Power, Gas, Cogeneration, Sanitation]	US Muni HY Public Utility
		sector = Hotels	US Muni HY Hotels
		sector = [Resource Recovery , Other]	US Muni HY Fin. Project
		sector = Metal	US Muni HY Metal
		sector = Paper	US Muni HY Paper
		sector not in above	US Muni HY Corp Other

Continued on next page

Table 57: Non-distressed HY Muni bond factors (Continued)

Maturity	GO Maturity	Maturity to Worst $<$ 3y and sub-cohort = GO	US Muni HY GO Maturity 2
		7y $\leq$ Maturity to Worst $<$ 15y and sub-cohort = GO	US Muni HY GO Maturity 10
		Maturity to Worst $\geq$ 15y and sub-cohort = GO	US Muni HY GO Maturity 30
	Revenue Maturity	Maturity to Worst $<$ 3y and sub-cohort = Revenue	US Muni HY REV Maturity 2
		7y $\leq$ Maturity to Worst $<$ 15y and sub-cohort = Revenue	US Muni HY REV Maturity 10
		Maturity to Worst $\geq$ 15y and sub-cohort = Revenue	US Muni HY REV Maturity 30

Factor Interpretation

- Factor returns represent %spread movements for all non-distressed universe.
- The distressed factor measures the weighted average spread return of distressed bonds

4.6.3 Muni-taxables

Model Data Inputs BlackRock USD Municipal Taxable Index

Model Estimation Process One factor for each maturity bucket

$$\frac{R_i^{spread}}{spd_dur_i} = \sum_j \beta_j \cdot I_i^{mat_j} + \varepsilon_i \tag{4.68}$$

Exposure Mapping Spread duration based dummy exposures based on Weighted Average Life (Maturity Bucket) of a bond.

Table 58: Taxable municipal bonds – factor list

Maturity Bucket (WAL)	Factor
0<WAL<7	Tax Muni 5Y*
7<WAL<15Y	Tax Muni 10Y*
WAL>15Y	Tax Muni 30Y

Factor Interpretation

- Factor returns are the weighted average of the absolute spread movements of the factor cohort.
- (*) represents mapped factors. Refer Section 9.4.2 for more details.

5 Securitized Spread Factors

Fixed-income securitized products are debt securities backed by loans, lease payments, or future earnings. Securitization involves repackaging illiquid assets into fixed income securities for efficient distribution in capital markets. These products differ from unsecured fixed income securities as they are backed by the performance of the underlying collateral, not the originator's credit. Securitized markets have improved capital market efficiency, allowing issuers to diversify risks and investors to access otherwise inaccessible assets. However, their complexity and opacity have led to misunderstandings and risk misassessment, contributing to the Global Financial Crisis (GFC). Post-GFC, significant regulations and structural changes have occurred, especially for securitized products. Complex structures and low-quality collateral pools, like Collateralized Debt Obligations (CDOs) and subprime mortgages, have largely disappeared. The market remains large, with residential mortgages, Commercial Mortgage Backed Securities (CMBS), and consumer Asset Backed Securities (ABS) as key segments. Risk models for securitized products are closely linked to analytics models, which can significantly impact risk assessments. Understanding the analytics model is crucial for constructing risk model factors and exposures, and models are updated as needed.

5.1 US Agency Mortgages

US residential mortgages are the largest sector among securitized products. They include agency Mortgage Backed Securities (MBS) issued by government-sponsored enterprises (GSEs) like Fannie Mae, Freddie Mac, or Ginnie Mae, and non-agency MBS issued by private institutions. Agency MBS are backed by the US government, so the credit risk is assumed by the agency, not the investor.

Residential mortgage loans often come with free prepayment options, allowing borrowers to pay back principal at any time. Risk models for agency mortgages focus on the cash flow uncertainty due to prepayments. Agency mortgages can be further categorized into different subsectors.

- Agency fixed-rate MBS pass-throughs are the bulk of agency MBS securities. Agencies pool mortgage loans with similar characteristics, like coupon and tenor, and sell them to investors. Interest and principal payments are passed directly to bondholders. The main challenge is modeling prepayment behavior, influenced by various macro/micro-economic forces and consumer behaviors. Key risk drivers include "moneyness" (whether the prepayment option is "in-the-money" or "out-of-the-money"), home prices, borrower credit, loan size, and employment market. Advanced modeling techniques and economic data are used to capture prepayments in detail. Risk models focus on top-level drivers like moneyness, agency, and tenor.
- The MBS market includes highly liquid To-be-Announced (TBA) contracts for deferred delivery of MBS. These contracts don't specify the actual pool, allowing the seller to choose the "cheapest to deliver" from eligible pools. This promotes liquidity and efficiency with standardized terms and transparency. TBAs are issued by the three major agency issuers and cover various maturity types (30-year, 20-year, 15-year, and 10-year). They are defined by six active contract coupon rates ("moneyness") and three settlement dates.
- Agency ARMs are adjustable-rate mortgages in the US, with rates floating and indexed to market interest rates like Libor. They often start as "hybrid ARMs," with a fixed rate for the first two to five years before switching to an adjustable rate. Prepayment behavior for ARMs is more influenced by short-term interest rates compared to fixed-rate mortgages. The overall size of the ARM market has significantly decreased in recent years and continues to decline. Risk models primarily use loan type factors to model Agency ARMs.
- Agency CMOs are complex mortgage securities that repackage and direct principal and interest payments from a collateral pool to different types and maturities of securities, meeting investor needs. Unlike pass-throughs, Collateralized Mortgage Obligations (CMOs) use various tranching rules to reallocate prepayment and interest rate payments. However, post-GFC, the market has lost appetite for complex securities like CMOs, and their market size has diminished. Risk models use cohort-based factors to capture the spread movements of major CMO tranche types like Planned Amortization Class (PACs) and Sequentials.

- IO/POs are stripped mortgage securities that separate principal (PO) and interest (IO) cash flows from collateral pools. There are trust IO/POs from pass-throughs and structured IO/POs from CMO tranching. IOs and POs let investors take a leveraged view on prepayments, as they are more sensitive to prepayment than the underlying collateral. This market has shrunk significantly in recent years, with limited trading and liquidity, posing challenges for risk modeling. Currently, spread risk is modeled using moneyness factors from IOs/POs indices.
- Specified pools are mortgage-backed securities with special loan characteristics that generally correlate with favorable prepayment behavior. Compared to the same coupon TBA, specified pools offer varying degrees of protection from refinance and extension risk, with superior convexity. Pool returns are modeled as TBA return plus payup return, which is explained using a combination of payup moneyness factors and payup story factors.
- Mortgage Servicing Rights (MSRs) are not securities but contractual agreements to service existing mortgages. They are often sold by the original lender to specialized servicing parties. The servicing fee is typically a fixed portion of the outstanding principal balance, making MSRs similar to IOs and sensitive to prepayments and interest rates. However, unlike IOs, servicers must perform various business activities to service the loans. The MSR market has grown rapidly, with banks, mortgage companies, and financial investors as main participants. Currently, BlackRock's fixed income model proxies MSR spread risk to IO at-the-money factors.

It is worth noting that, in the return decomposition of agency mortgages, there is one more component that captures the price movement due to mortgage basis (discussed in section 3.2.5). Hence, the spread return for mortgage securities is defined as:

$$R^{spread} = R^{Total} - (R^{Time} + R^{Curve} + R^{Convexity} + R^{Vol} + R^{mortbasis}) \tag{5.69}$$

5.1.1 Prepayment Model Versioning

In order to compute the pricing related analytics, a modeling framework as described in section 6.1 is used. An overview of the framework for mortgage modeling is provided in [6]. Details of the prepayment model for Agency mortgages can be found in [7]. Credit characteristics are modeled as described in [8].
Prepayment models are crucial for accurate risk model exposures and factor returns, such as KRDs, OAS, spread duration, and mortgage basis. When a new prepayment model is released, risk models can become "out-of-sync" or inconsistent. For example, changes in prepayment speeds can significantly alter risk analytics, making historical data unreliable.
To address this, the risk model factors are "version-based." This means that a new set of risk factors is created with each new prepayment model, including at least two years of historical data backfilled under the new model. This ensures consistency and accuracy in risk models. Currently, mortgage basis, TBA spread factors, CMO spread factors, and IOPO spread factors are all version-based.[34].
In the following sections, model details are listed of each mortgage asset class.

5.1.2 Agency Pass-throughs

Agency pools are made with similar mortgage loans to make them easier to analyze and sell. Their risk is mainly affected by how close the average interest rate (WAC) is to the current mortgage rate, which follows the yield curve. The risk model uses this relationship to create stable risk factors over time, despite changing interest rates.

Model Data Inputs

- Uniform Mortgage-Backed Securities (UM) 30 year and 15-year TBAs. Ginnie Mae (GN/G2) 30-year TBAs

[34]For the two moneyness factors on the border (± 350bps), there could exist more than two TBAs that contribute to the factor calibration. It is also common to have no TBA belonging to the more extreme moneyness factor cohorts, e.g. out-of-the-money 350bps = -50bps wac when 3% is the par coupon while wac of TBAs typically ranges from 2% to 8%, the moneyness factor is proxied to the closest moneyness factor that can be calibrated.

Agency mortgage pass-throughs are the largest and most liquid part of the agency MBS market, thanks to TBA (to-be-announced) forward contracts that simplify trading. Over 90% of agency MBS trading happens in the TBA market. Therefore, TBAs are used to estimate Agency pass-throughs.
Specifically:

- UM 30-year TBAs estimate moneyness factors.
- GN 30-year TBAs estimate the GN Agency factor.
- From risk model v6.04, G2 TBAs estimate the GN factor due to their higher liquidity.
- UM 15-year TBAs estimate the 15-year tenor factor.

Since the market size of each TBA cohort is hard to find, market-cap weights from the BBG Barclays US MBS index are used as a proxy. From model version v6.1, ICE BofA Mortgage Master Index weights are used to maintain consistency between BLK and BEN factors.

Model Estimation Process Moneyness factors are derived from 30-year UM TBAs. The return of each factor is calculated using the two TBAs with the closest moneyness or WAC (weighted average coupon) to the factor. In detail,

$$\beta_{moneyness_k^{fac}} = w_k \cdot -\frac{R_{i^k}^{spread}}{spd_{dur_{i^k}}} + (1 - w_k) \cdot -\frac{R_{i^k+1}^{spread}}{spd_{dur_{i^k+1}}} \tag{5.70}$$

where $k \in \{-350bps, -300bps, ..., 300bps, 350bps\}$ refers to the 15 moneyness factors, i^k and $i^k + 1$ are the two neighboring TBAs with moneyness closest to that of the factor.

$$moneyness_{i^k} \leq moneyness_k^{fac} \leq moneyness_{i^k+1} \tag{5.71}$$

and w_k is the weight, calculated as

$$w_k = \frac{moneyness_{i^k+1} - moneyness_k^{fac}}{moneyness_{i^k+1} - moneyness_{i^k}} \tag{5.72}$$

For each security, moneyness is defined as the difference between its WAC (weighted average coupon) and the sum of the par coupon (specific to agency/tenor pair) and the primary secondary 30-year spread (PSS).[35] ,

$$moneyness_i = wac_i - (Par\ Cpn + PSS) \tag{5.73}$$

For the latest risk model (v6.1), the spot PMMS rate from the mortgage rate model is used to better capture moneyness. For the GN agency factor, the difference between the actual return and the predicted return of each G2 30-year TBA i is calculated using the moneyness factors.

$$\varepsilon_i = \frac{R_i^{spread}}{spd_{dur_i}} - \left(\overbrace{w_i \cdot \beta_{moneyness_{k^i}^{fac}} + (1 - w_i) \cdot \beta_{moneyness_{k^i+1}^{fac}}}^{\text{exposure to moneyness factors}} \right) \cdot (*) \tag{5.74}$$

Again,w_i is based on the distance of moneyness of the security to the two closest moneyness factors ($moneyness_{k^i} \leq moneyness_i \leq moneyness_{k^i+1}$), and

$$w_i = \frac{moneyness_{k^i+1} - moneyness_i}{moneyness_{k^i+1} - moneyness_{k^i}} = \frac{moneyness_{k^i+1} - moneyness_i}{50} \tag{5.75}$$

Next, GN factor return is calculated as the (%mkt-cap) weighted average residuals from all G2 30-year TBAs.

$$\beta_{GN} = \sum_i \%mktcap_i \cdot \varepsilon_i \tag{5.76}$$

Similarly, 15-year tenor factor is calibrated following the same procedure using 15-year UM TBAs.

[35] Also known as servicing spread. Refer to [6] for details.

Exposure Mapping Exposure measures are based on spread duration. Each security is mapped to (at most) 2 moneyness factors, with spread duration allocated to each moneyness factor as shown in (*). G2 securities are linked to GN factors, and securities with a tenor of 15 years or less are linked to the corresponding 15-year factor.

List of factors

Table 59: US agency mortgages – agency pass-through factor list

Type	Money-ness Category	15-year	30-year
Money-ness	At the money	USD TBA 15yr ATM*	USD TBA 30yr ATM
	In the money	USD TBA 15yr ITM 50bps*	USD TBA 30yr ITM 50bps
		USD TBA 15yr ITM 100bps*	USD TBA 30yr ITM 100bps
		USD TBA 15yr ITM 150bps*	USD TBA 30yr ITM 150bps
		USD TBA 15yr ITM 200bps*	USD TBA 30yr ITM 200bps
		USD TBA 15yr ITM 250bps*	USD TBA 30yr ITM 250bps
		USD TBA 15yr ITM 300bps*	USD TBA 30yr ITM 300bps
		USD TBA 15yr ITM 350bps*	USD TBA 30yr ITM 350bps
	Out of the money	USD TBA 15yr OTM 50bps*	USD TBA 30yr OTM 50bps
		USD TBA 15yr OTM 100bps*	USD TBA 30yr OTM 100bps
		USD TBA 15yr OTM 150bps*	USD TBA 30yr OTM 150bps
		USD TBA 15yr OTM 200bps*	USD TBA 30yr OTM 200bps
		USD TBA 15yr OTM 250bps*	USD TBA 30yr OTM 250bps
		USD TBA 15yr OTM 300bps*	USD TBA 30yr OTM 300bps
		USD TBA 15yr OTM 350bps*	USD TBA 30yr OTM 350bps
Other Attributes	Tenor	USD TBA 15yr v/s 30yr	
	Agency	USD TBA Ginnie	

Factor Interpretation

- Moneyness factors capture average spread movements of their groups. These groups change as mortgage rates fluctuate, causing securities with the same WAC to switch categories. The factors are calibrated using UM 30-year TBAs, which are the most liquid. This helps capture spread movements of 30-year UM mortgage pass-throughs. 15-year moneyness factors are mapped to the 30-year ones for better exposure mapping but are not added as real factors due to lack of statistical significance.
- The GN Agency factor captures the average difference in spread movements between G2 and UM Mortgage Pass-throughs, estimated using G2 30-year TBAs.
- 15-year tenor factors capture the average difference in spread movements between 15-year and 30-year Mortgage Pass-throughs, estimated using UM 15-year TBAs.

5.1.3 Mortgage Pools

Mortgage Pools are collections of mortgages used as collateral for mortgage-backed securities. They differ from TBAs in their characteristics. Understanding the risk profile of these securities involves modeling the collateral pool, which is held in a special purpose vehicle to keep the originator separate from bankruptcy risks. Purchasers receive interest payments and principal paydowns from the underlying mortgages.

Specified pools have unique characteristics that differ from TBA-compliant pools, often leading to favorable prepayment behavior. Compared to the same coupon TBA, specified pools offer varying degrees of protection from refinance and extension risk, with superior convexity.

Payup compensates issuers for including pools with different characteristics from TBA-compliant pools. Analysis of specified pools usually breaks down return into two parts.

1. Return on a similar TBA with same coupon and notional;
2. Return needed to make a specified pool equal to the equivalent TBA.

In the fixed income risk model, there are two types of mortgage pools. The first type is linked to the concept of a "story," which is a classification system used to explain potential borrower behavior. Examples of such stories include:

- Loan Balance
- MHA (TBA eligible)
- Investor
- FICO
- LTV
- GEO (geographical state) of borrowers
- MHA (TBA ineligible)
- Whether load has Jumbo characteristics
- Loan maturity

In the Agency MBS sector, 'Payup' is the premium a specified story pool commands over its corresponding TBA. Spec stories with flatter 's-curve' profiles lead to more predictable prepayment behavior and higher payups. In the spec pool risk model, pool return is modeled as TBA return plus payup returns, explained by Story Payup factors and Moneyness Payup factors.

Payup factors are aligned with these stories to intuitively explain differential returns above the baseline for similar TBAs. The second type of specified pool is linked to coupon vintage. The BFIRM model treats both types separately but uses a similar approach.

5.1.4 Model Details: Pools Associated with Stories

Model Data Inputs 13k pool securities with LTV, MHA, FICO, GEO, Jumbo, Investor, 10, 20 and loan balance story types.

Model Estimation Process Pool Return is modeled as TBA return plus Payup Return. The TBA component of return for a pool is adjusted so that the dollar exposure of the pool matches that of a TBA with the same coupon and notional.

$$R_{pool}^{total} = R_{tba} + R_{pool}^{payup} \tag{5.77}$$

$$R_{pool}^{total} = R_{pool}^{time} + R_{tba}^{interest_rate} + R_{tba}^{volatility} + R_{tba}^{mortgagebasis} + R_{tba}^{spread} + I_{pool} R_{pool}^{payup} \tag{5.78}$$

Spread return for TBA component of return is measured by mapping to TBA moneyness spread factors using TBA WAC and spread duration

$$Moneyness_{\text{tba spread factor}} = wac_{tba} - \text{ATMWAC} \tag{5.79}$$

$$R^{tba}_{spread} = w_{tba_{upper}} F_{tba_{mnyupper}} spd_{dur} + w_{tba_{mnylower}} F_{tba_{lower}} spd_{dur} + I_{GN} F_{GN} spd_{dur} + I_{15} F_{15} spd_{dur} \tag{5.80}$$

Payup return is explained by using combination of payup moneyness factors and payup story factors. Payup factor returns are modeled in return space and are calibrated on a weekly basis.
Mapping to payup moneyness factors is calculated using pool WAC

$$Moneyness_{\text{pool moneyness payup factor}} = wac_{tba} - \text{ATMWAC} \tag{5.81}$$

$$R^{payup}_{pool} = \beta_{mny} I_{mny} w_{mny_{upper}} F_{mny_{upper}} + \beta_{mny} I_{mny} w_{mny_{lower}} F_{mny_{lower}} + \beta_{story} F_{story} \tag{5.82}$$

w_x is weight of a security in a moneyness bucket, F_x represents factor return, β_x represents a payup factor return scaling factor and I_x is a dummy value that is activated (takes value 1) when the factor mapping eligibility criteria is met. For pools that are TBA eligible but are not mapped to an eligible story $I_{pool} = 0$.

Even though there are multiple loan balance stories with a wide range of payup between LLB85K and LB225K, their returns are captured by the same payup factors for simplicity. To account for differences in payup return at the cusip level, payup beta has been introduced. Research shows a linear relationship between current payup levels and future payup volatilities, which flattens out for high payup levels (above 80 ticks). Therefore, a specified pool is mapped to its payup factors using a payup beta.

$$\beta^{payup}_{i,k} = \frac{\max(0, \min(payup_i, 80))}{payup_k} \tag{5.83}$$

where $payup_i$ is the payup of the specified pool and $payup_k$ is the level of the factor the pool is mapped, computed as the weighted average of the payups of all specified pools mapped to this cohort.

For MHA (TBA ineligible) stories, payups can go to negative and thus they are floored at -80 and the beta is calculated as

$$\beta^{payup}_{MHAN,i,k} = \frac{\max(0, \min(payup_i + 80, 160))}{payup_k + 80} \tag{5.84}$$

Jumbo pools are an exception. They prepay slower in low-rate environments and faster in rising-rate regimes, resulting in negative payups. They respond to interest rate changes in the opposite direction from other specified pools. The linear relationship doesn't hold for Jumbo stories, so they load on moneyness payup factors with beta=-1 and the Jumbo story factor with beta=1.

Exposure Mapping Eligible specified pools first map to risk factors using the same exposures as TBA. Then, they load onto payup risk factors measured in total return space and calibrated from payup changes. Each specified pool security loads onto one story type payup factor and one or two moneyness factors, except for 10/20 story pools that only load on their own story payup factors.

List of factors

Table 60: Model Structure of Payup Factors

Type	Story	Payup Factors	
		Story Type	Moneyness

Continued on next page

Table 60: Model Structure of Payup Factors (Continued)

Low Loan Balance	LLB85K	Loan Balance	15 moneyness payup factors ranging from -350 to +350 bps
	MLB110K		
	MLB125K		
	HLB150K		
	SHLB175K		
	LB200K		
	LB225K		
MHA (TBA eligible)	M80-90LTV	MHA (TBA eligible)	
	M90-95LTV		
	M95-100LTV		
	M100-105LTV		
Investor	Investor	Investor	
FICO	FICO<660	FICO	
	FICO<700		
LTV	LTV>90	LTV	
GEO	FL	GEO	
	NY		
	PR		
	TX		
MHA (TBA ineligible)	M105-125LTV	MHA (TBA ineligible)	
	M125+LTV		
Jumbo	Jumbo	Jumbo	
10 Year	10	10	Not Applicable
20 Year	20	20	

Factor Interpretation

- Payup moneyness factors capture the difference in S-curve between a specified pool and TBA. Story factors represent the net effect of the story type, beyond what is captured by moneyness factors.

5.1.5 Model Details: Pools Associated with Coupon Vintage

The model captures the risk of the payup component in coupon vintage generic pools. Previously, the basis risk between these pools and TBAs wasn't explicitly captured. These pools are treated as TBA plus payup, with exposure to existing mortgage-related risk factors and coupon vintage payup risk factors.

Many coupon vintage generic pools trade at a pay-up above their same coupon TBA due to their collateral characteristics, which offer prepayment protection during market rallies and extension protection in a sell-off. Moneyness drives day-to-day payup changes, hence, moneyness factors are used in the payup model to capture payup activity variations. Each pool loads onto one or two moneyness payup factors. Despite lower payup return volatility than specified pools, capturing the risk of these pools with dedicated risk factors is crucial as they make up over 60% of the outstanding balance.

Model Data Inputs The calibration universe for vintage pool risk factors is the Merrill Lynch MBS index. The input data is pricing on production indices, following standard processes. Only cohorts with payups greater than four ticks are included in the calibration. Returns for factors without mapped cohorts are zero.

Model Estimation Process The total return of a pool is modeled as the same coupon TBA return plus the return due to pool payup.

$$R_{pool}^{total} = R_{tba} + R_{pool}^{vintage_payup} \tag{5.85}$$

$$R_{pool}^{total} = R_{pool}^{time} + R_{tba}^{interest\,rate} + R_{tba}^{volatility} + R_{tba}^{mortgage\,basis} + R_{tba}^{spread} + I_{pool} R_{pool}^{vintage\,payup} \tag{5.86}$$

TBA return includes several components: time return (due to passage of time), interest rate return (effect of yield curve movements), volatility return (changes in implied volatility), and mortgage basis return (changes in par coupon basis). TBA spread return is the excess return based on asset class characteristics, modeled using 15 moneyness factors and one each for agency and term. Payup return captures Pool-TBA basis return for qualifying pass-through pools.

Qualifying pools load on one or two of 15 payup moneyness factors, calibrated from a range of [-350, 350] in steps of 50. Moneyness is defined as the difference between pool WAC and the spot PMMS rate, using a 1-day lag to calibrate factor returns based on the previous observed level of moneyness.

$$Moneyness_{\text{pool moneyness payup factor}} = wac_{pool} - ATMWAC \tag{5.87}$$

$$R_{pool}^{vintage_payup} = \beta_{mny} I_{mny} w_{mny_{upper}} F_{vintage_{mny_{upper}}} + \beta_{mny_{lower}} I_{mny} w_{mny_{lower}} F_{vintage_{mny_{lower}}} \tag{5.88}$$

Change in payup is the day-over-day change in the forward payup of each security. Each security is mapped to one or two moneyness factors based on its moneyness, and their corresponding weights are calculated. The factor return is the weighted average of the change in payup.

$$MoneynessFactor_j = \frac{\sum_i w_i^j \cdot dpayup_i}{\sum_i w_i^j} \tag{5.89}$$

Where w_i^j is the current balance of security i times % of exposure of security i mapped to moneyness factor j.

List of factors

Table 61: Coupon Vintage Generics Moneyness factor structure

Type	Category	Coupon Vintage Payup

Continued on next page

Table 61: Coupon Vintage Generics Moneyness factor structure (Continued)

Moneyness	At the Money	USD VINT Payup ATM
	In the money	USD VINT Payup ITM 50bps
		USD VINT Payup ITM 100bps
		USD VINT Payup ITM 150bps
		USD VINT Payup ITM 200bps
		USD VINT Payup ITM 250bps
		USD VINT Payup ITM 300bps
		USD VINT Payup ITM 350bps
	Out of the money	USD VINT Payup OTM 50bps
		USD VINT Payup OTM 100bps
		USD VINT Payup OTM 150bps
		USD VINT Payup OTM 200bps
		USD VINT Payup OTM 250bps
		USD VINT Payup OTM 300bps
		USD VINT Payup OTM 350bps

Factor Interpretation Payup moneyness factors explain the basis return capturing the difference in S-curve of a Coupon Vintage specified pool and TBA in a generic manner.

5.1.6 IOs and POs

IO and PO tranches receive only select payments from the underlying collateral. This leads to enhanced convexity with respect to changes in interest rates. Since moneyness governs the interest rate sensitivity, the factor structure aligns with moneyness.

Model Data Inputs ML CMO Index

Model Estimation Process The same as that of TBA moneyness factors.

List of factors

Table 62: US Agency mortgages – IO & PO factor list

Moneyness Categories	Type IO	PO
At the money	IO ATM	PO ATM
In the money	IO ITM 50bps	PO ITM 50bps
	IO ITM 100bps	PO ITM 100bps
	IO ITM 150bps	PO ITM 150bps
	IO ITM 200bps	PO ITM 200bps
	IO ITM 250bps	PO ITM 250bps
	IO ITM 300bps	PO ITM 300bps
	IO ITM 350bps	PO ITM 350bps
Out of the money	IO OTM 50bps	PO OTM 50bps
	IO OTM 100bps	PO OTM 100bps
	IO OTM 150bps	PO OTM 150bps
	IO OTM 200bps	PO OTM 200bps
	IO OTM 250bps	PO OTM 250bps
	IO OTM 300bps	PO OTM 300bps
	IO OTM 350bps	PO OTM 350bps

Factor Interpretation Moneyness factors capture the average (absolute) spread movements of the corresponding moneyness cohort.

5.1.7 Agency Hybrid ARMs

Hybrid ARMs account for a small proportion of agency MBS index as compared to fixed rate pools. In recent years, ARM issuance has declined significantly from its peak in the early 2000s.

Model Data Inputs Since Bloomberg retired the hybrid ARM index and no similar indices exist, a new method was devised to select bonds for the estimation universe. Single-family hybrid ARM pools are filtered from the full universe of pools published by eMBS according to the following criteria:

- Conventional
  - Product: 3/1, 5/1, 7/1, 10/1
  - Reset index: 12M LIBOR
  - Cap structure (first reset-periodic-life): 2-2-5 (most common for 5/1), 2-2-6 (less frequently used for 5/1), 5-2-5 (7/1 and 10/1)
- Ginnie Mai II
  - Product: 3/1, 5/1
  - Reset index: 1Y CMT
  - Cap structure: 1-1-5
- All pools
  - First reset date is six plus months in the future
  - Not a mega/giant/platinum

These criteria filter out pools without standard terms and mirror the attributes used to define cohorts in the Bloomberg index. Due to computing constraints, not all pools can be included in the factor estimation universe.

To reduce the number of pools, the initial list is stratified by agency (conventional vs Ginnie Mae), product type, coupon range (in 50 bps increments), and vintage. Within each group, the largest balance pools are selected until their sum is at least 25% of the group's total. For small groups, the selected pools may account for much more than 25% of the balance. This method ensures a consistent distribution across agencies, coupons, and vintages between the full set of pools and the reduced set.

Model Estimation Process One factor for each Fixed Interest Rate Period.

$$\frac{R_i^{spread}}{spd_{dur,i}} = \sum_j \beta_j I_i^{ARM\ type_j} + \varepsilon_i \tag{5.90}$$

where $I_i^{ARMtype_j}$ is the dummy variable indicating security i belongs to type j, and $ARMtype_j \in \{3/1 \text{ Agency ARM}, 5/1 \text{ Agency ARM}, 7/1 \text{ Agency ARM}\}$. The % market value * spd_dur is the regression weight. Thin industry methodology ensues factors of thin cohorts are calibrated in a robust function.

Exposure Mapping Spread duration based on ARM type/agency.

Table 63: US agency mortgages – agency ARM factor list

ARM type	Agency	Jumbo
1/1	1/1 Agy Hyb ARM*	1/1 Jumbo Hyb ARM*
3/1	3/1 Agy Hyb ARM	3/1 Jumbo Hyb ARM*
5/1	5/1 Agy Hyb ARM	5/1 Jumbo Hyb ARM*
7/1	7/1 Agy Hyb ARM	7/1 Jumbo Hyb ARM*
10/1	10/1 Agy Hyb ARM*	10/1 Jumbo Hyb ARM*

Factor Interpretation

- Factor returns captures the weighted average cohort-level absolute spread movements.
- (*) represents mapped factors. Refer Section 9.4.2 for more details.

5.1.8 Agency CMO

Model Data Inputs ML CMO Index

Model Estimation Process

$$\frac{R_i^{spread}}{spd_dur_i} = \sum_j \beta_j \cdot I_i^{structure_{tenor_j}} + \varepsilon_i \tag{5.91}$$

where $I_i^{structure_tenor_j}$ is the dummy variable indicating security i belongs to structure tenor cohort j, and $structure_tenor_j \in$ {short PAC, long PAC, short Sequential, long Sequential}

Exposure Mapping Spread duration-based exposures on the tranche type/WAL, with TAC/NAS mapped to PAC factors.

List of factors

Table 64: US Agency mortgages – agency CMO factor list

Tranche	Maturity Bucket	Factor
PAC	$0 < \text{WAL} \leq 5$	Short PAC
	$5 < \text{WAL} \leq 10$	Long PAC
Sequential	$0 < \text{WAL} \leq 3$	Short Sequential
	$3 < \text{WAL} \leq 10$	Long Sequential

Factor Interpretation The risk factors capture the spread risk of the various tranches based on maturity.

5.1.9 CMO Floater

For CMO floater securities, cash flows depend on both mortgage rates and SOFR. The methodology is similar to PACs and Sequentials, but a term for "floating" duration is added to account for additional sensitivity to SOFR.

$$R^t_{swap} = -float^{t-1}_{dur} \cdot \Delta sofr = -(oad^{t-1} - spd^{t-1}_{dur} - \alpha \cdot mrb^{t-1}_{dur}) \cdot \Delta sofr \tag{5.92}$$

Where the coefficient α is set to 0.85[36], and the LIBOR shock corresponds to the point on the swap curve closest to the bond's WAL. Having handled this additional sensitivity, the realized spread return is computed for each floater as

$$R^t_{spd} = (\frac{P_t}{P_{t-1}} - 1) - (R^t_{rates} + R^t_{vol} + R^t_{mrb} + R^t_{swap}) \tag{5.93}$$

List of factors This factor is no longer supported since 7/31/2020 due to data lack of issuance. The model will map CMO Floater securities to CMO VXXX factor, where XXX depends on the pre-payment version being used.

5.1.10 MSR

As mentioned previously, currently risk models do not yet have dedicated factors for MSRs. Instead, two MSR factors are created for ARM-based MSRs and fixed rated MBS-based MSRs, which are mapped to the IO ATM factors.

List of factors

Table 65: US Agency mortgages – MSR factor list

Factors
USD_MSR_ARM600
USD_MSR_MBS600

5.1.11 CRT

Credit Risk Transfer (CRT), also known as GSE Risk-Sharing deals, refers to mortgage-backed securities issued by Fannie Mae (CAS) and Freddie Mac (STACR) starting in 2013. These securities transfer a portion of credit risk from the government-sponsored enterprises (GSEs) to private investors. Unlike Agency MBS, CRT securities do not guarantee full repayment of the original principal balance. By reducing their repayment obligations, GSEs significantly offload credit risk, lowering taxpayer exposure and mitigating systemic risk during financial downturns. Although issued by agencies, CRTs exhibit risk characteristics closer to non-agency MBS.

[36]This is under the assumption that Libor/TSY is nearly constant at 0.85

Model Data Inputs JPM CRT sub-indices[37]

Model Estimation Process The price return of a sub-index is calculated as percentage price change as follows:

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1 \tag{5.94}$$

The primary purpose of the model is to explain the price return. In the CRT model, the price return can be further decomposed into a systematic and an idiosyncratic component.

$$R_{i,t} = \underbrace{\sum_{j}^{m} \text{Exposure}_{\ i,t-1} * \beta_{j,t}}_{\text{systematic}} + \underbrace{\varepsilon_{i,t}}_{\text{Idiosyncratic}} \tag{5.95}$$

The systematic component refers to the part of the security return that is due to common movements in groups that the security belong to. The idiosyncratic component of the price return is the portion not explained by the systematic factors. It is designed to capture within-cohort variations or deal-specific risk.

Systematic factor returns are estimated using daily cross-sectional regressions. The regression is performed in price return space, with each security's weight determined by its current balance. The resulting factor returns, also known as factor deltas ($\beta_{j,t}$), quantify the daily movement of each rating-based factor. In the CRT model, the systematic component is primarily driven by rating-based factors, which represent the primary source of co-movement in CRT prices.

Exposure Mapping Each CRT security is assigned to a specific systematic factor based on its characteristics such as: rating[38], tranche, and program information (such as STACR or CAS). Each security is mapped to one of four rating-based factors: A, BBB, BB/B, or NR & First Loss.

$$R_{i,t} \sim \sum_{j} \beta_{j,t} I_i(i \in j) \tag{5.96}$$

where $j \in \{A, BBB, BB\ \&\ B, NR\ \&\ \text{First Loss}\ \}$, $I_i(i \in j)$ is the dummy variable indicating whether security i belongs to rating j.

If a security cannot be mapped directly using agency or tranche information, a fallback logic is applied. This fallback uses the original credit enhancement (CE) of the security to assign it to the appropriate factor. The mapping ensures that each security's risk exposure reflects its position within the broader factor cohort.

Additionally, the model refines exposures using a credit beta adjustment. This adjustment accounts for differences between individual deals and their factor cohort averages, ensuring that the risk forecast for each security is tailored to its unique risk profile. If every security in a rating group has the same exposure, they would all get the same risk forecast. However, as individual deals pay down over time, their volatility decreases. At the same time, newer deals in the same group may show higher volatility compared to the average for that group. To address this, the model utilizes a factor-ratio[39] based function to calculate credit beta[40], aiming to adjust the risk of deals according to the

[37] For most fixed income asset classes, well-established public indices provide a stable representative universe and robust analytics to capture spread movements. However, CRTs remain a relatively nascent market, and such indices are not yet available for this asset class. Moreover, the primary risk in CRTs—credit or default risk—is inherently reflected in price dynamics. To address this, the model calibrates systematic risk factors using indicative price levels from JPMorgan sub-indices, recognized as the most reliable source covering approximately 99% of the CRT universe. Each sub-index groups CRTs with similar characteristics such as vintage, rating, and LTV type. For instance, the sub-index "14/A/HLTV" represents high-LTV CRTs issued in 2014 with an A rating.

[38] Rating here is original agency rating, and it is defined based on tranches (i.e. either M1, M2, M3 or B) and programs (i.e. either STACR or CAS).

[39] Sub-cohort level factor divided by factor-level factor, where factor is ratio of current balance and original balance. Sub-cohort is defined based on factor ratios, i.e., securities with factor ratios falling into the same range belongs to the same sub-cohort.

[40] The floor and cap on credit beta is set to be 0.25 and 1.5, since cohorts beyond them are not well-populated.

difference from their corresponding factor cohort's average. This adjustment ensures that the exposure reflects both the security's group assignment and its unique characteristics, such as how much of its principal has been paid down.

The credit beta formula was developed by analyzing historical volatility data from IDC between 2015 and 2017. The specific formula and its coefficients are shown in Table 66.

$$\begin{aligned} \text{credit beta } &= a + b \cdot fr + c \cdot (fr)^2 + d \cdot (fr)^3 \\ \text{, where } fr &= \frac{\text{Security's factor}}{\text{Factor cohort's factor}}, \quad \text{factor } = \frac{\text{cur bal}}{\text{orig bal}} \end{aligned} \tag{5.97}$$

Table 66: Coefficients of credit beta formula

Rating	a	b	c	d
A	0.51	1.19	-1.63	0.93
Others	0.15	1.26	-2.12	1.71

In conclusion, the securities have unit risk exposures adjusted by credit beta.

List of factors There is only one type of systematic factors in the model: rating, defined by tranche and program. The model has 4 rating-based factors that tracks movements in different rating buckets, as rating turns out to be the most prominent impact on prices of CRTs. Each CRT security will load onto one of the rating factors.

Table 67: Systematic Model Structure

Factor Type	Factors	CRT Sub-indices	Underdying Tranches		Exposure Type
			STA CR	CA S	
Price-based Rating Factors	A	6 A ated sub-indices	M1 (Before 2017)		Credit beta*%NAV
	BBB	8 BBB rated sub-indices	M2 M1 After 2017	M1	
	BB/B	10 BB rated sub-indices	M3 M2 A fter 2017	M2	
	NR First loss	6 NR or First loss sub-indices	B	B	

If a CRT cannot be mapped to the corresponding factor based on its agency/tranche info, then it will be mapped based on the fallback logic. The fallback mapping logic uses the original credit enhancement as shown in below Table 68, i.e. if a CRT's original enhancement (CE) is within certain range, it will be mapped to the respective factor.

Table 68: Fallback Mapping

Factor	Criteria
A	CE > 3.3
BBB	CE

	$\in (1.5.3.3]$
BB/B	CE $\in (0.3, 1.5]$
NR First loss	CE $\in [0.0.3]$

Factor Interpretation Factor returns represent weighted average spread returns for each rating cohort, using the current balance as the weight.

5.2 US Non-agency Mortgages

Non-agency mortgages are mortgage-backed securities (MBS) issued by private institutions. They consist of pools from borrowers who don't meet Agency standards due to factors like loan size, documentation, borrower credit, and loan-to-value ratios. Many of these were "Alt-A" and "subprime" loans, which were key catalysts for the Global Financial Crisis (GFC) as they allowed less creditworthy borrowers to gain financing.

Post-GFC, new issuance in this market is rare, except for the prime sector, which includes high-quality mortgage loans that don't qualify for Agency standards due to larger loan sizes. However, the market remains significant, with legacy securities issued before the financial crisis still traded in the secondary market. Due to data limitations, non-agency risk factors track price movements of various non-agency sectors and focus on capturing the default risk of non-agency securities.

BlackRock has a model that targets the response of non-agency collateral to changes in interest rates and other related variables.

Model Data Inputs JPM Sub-indices of Non-agency MBS

Model Estimation Process The factor returns are aggregated from price returns of the corresponding non-agency sub-indices, with weights based on the monthly updated outstanding balance of the subindices to reflect changing market composition resulting from pay down. In detail,

$$R_i^{price} = \sum_i \beta_{type_k} I_i^{type_k} \tag{5.98}$$

where $I_i^{type_k}$ is the dummy variable indicating subindex i belongs to type k, and $type_k \in$ Prime 2.0, Prime Fixed, Prime ARM, Alt-

Exposure Mapping The exposure loading of a non-agency bond on its risk factor cohort is adjusted using a price-based function to better differentiate risks of securities within each factor cohort. In detail,

$$exposure_i^k = \frac{\sigma_i^k}{\sigma_f^k} = \frac{f^k(p_i)}{f^k(p_f^k)} \tag{5.99}$$

where p_i is the price of the security, p_f^k is the price of the factor cohort k (calculated as the weighted average prices of securities in the factor cohort), $f^k(p)$ is the priced based function defined as

$$f^k(p) = \sum_k \alpha_k I(k \in K) + 0.31 \cdot p - 9.43E^{-3} \cdot p^2 + 9.75E^{-5} \cdot p^3 - 3.60E^{-7} \cdot p^4 \tag{5.100}$$

where$I(k \in K)$is a dummy variable indicating the sector to which the cohort belongs to Table 69 below lists the resulting α_k for each cohort:

Table 69: US non-agency mortgages – coefficients in price-based function

Non-Agency Sectors	Prime 2.0 (New Issues)	Prime Fixed	Prime ARM	Alt-A Fixed	Alt-A ARM	OptionAR	Subprime
Intercept	2.45	2.66	3.22	2.33	3.17	3.67	2.37

List of factors

Table 70: US non-agency mortgages – factor list

Type	Factors	Non-Agency Sub-indices
Prime	Prime 2.0 (New Issues)	Prime 1012
	Prime Fixed	2 Prime Fixed sub-indices, 05/06 and 06/07
	Prime ARM	2 Prime ARM sub-indices, 05/06 and 06/07
Alt-A	Alt-A Fixed	2 Alt-A Fixed sub-indices, 05/06 and 06/07
	Alt-A ARM	2 Alt-A ARM sub-indices, 05/06 and 06/07
Option ARM	OptionARM	2 OptionARM sub-indices, 05/06 and 06/07
Subprime	Subprime	8 ABX sub-indices

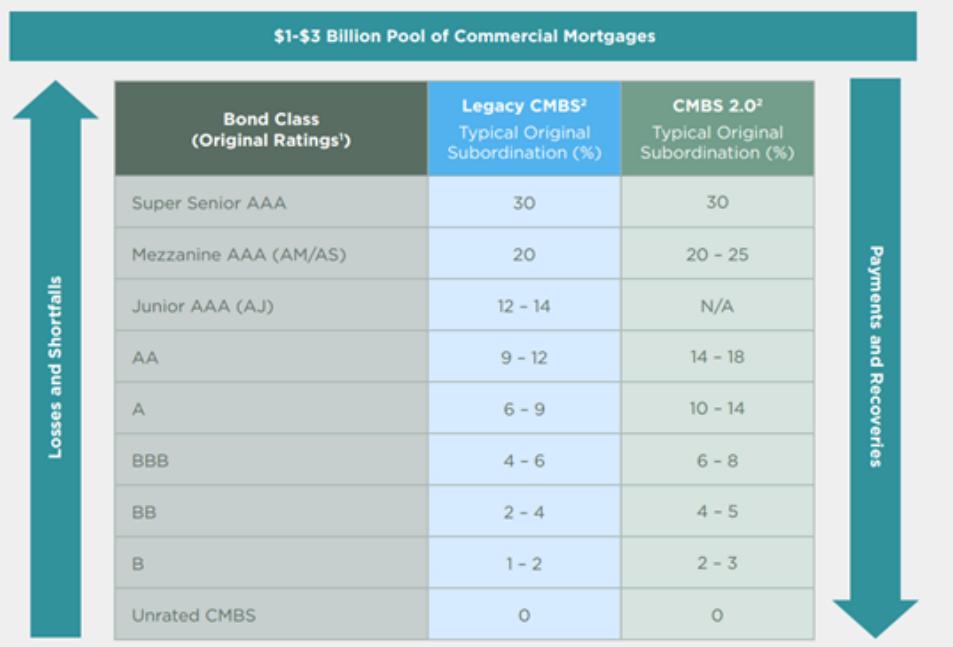
Factor Interpretation Factor returns represent the average %price movements of the corresponding non-agency factor cohort.

5.3 US CMBS

Commercial mortgage-backed securities (CMBS) are backed by commercial mortgage loans for properties like hotels, office buildings, shopping malls, and apartment buildings. A typical deal contains 75-150+ loans, with the largest 10-15 making up a significant portion. CMBS are divided into tranches, ranging from senior to subordinated, with lower tranches taking on more risk. Unlike residential mortgage-backed securities (RMBS), CMBS bonds are "call protected" with lock-out periods and prepayment penalties, so risk management focuses on credit risk.
The Great Financial Crisis (GFC) impacted the CMBS market, reducing liquidity and lowering prices of even the safest securities. The financial distress was most pronounced for deals issued in 2006/2007, which had lower underwriting standards and hence worse quality loans. Post-GFC, new issuance has increased under stricter regulations and more conservative collateral, with more credit enhancement for senior tranches. Refer to Figure 2 for a comparison of the deal structure pre/post crisis.
Exposures are calculated using partial durations from a pricing model like that outlined in section 6.1. Details of the collateral performance model can be found in [9].
To account for the structural changes the CMBS market has undergone, risk models have incorporated the following risk drivers in the model structure:

- Tranches are closely linked to the credit enhancement of the bond, indicating the level of credit risk. Super Senior tranches are divided into pre-last-cash-flows (PLCF, A1-A3 tranches) and last-cash-flow (LCF, A4 tranche). LCF has higher risk than previous tranches and deserves its own factor. Subordinated tranches are grouped together due to their low market representation and infrequent trading activities.
- Vintage factors are usually not included in the model to avoid data challenges as older bonds phase out and newer ones lack historical data. However, for CMBS, the difference between older and newer issuance is too significant to ignore. Therefore, vintage is directly included in the factor structure, with robust calibration and proxy mechanisms to address these data challenges.

A price-based distressed factor is included to capture the risk of highly depressed CMBS securities. Other factors are measured in relative spread space. The spread risk model of CMBS includes granular risk factors based on tranches



Bond Class (Original Ratings[1])	Legacy CMBS[2] Typical Original Subordination (%)	CMBS 2.0[2] Typical Original Subordination (%)
Super Senior AAA	30	30
Mezzanine AAA (AM/AS)	20	20 - 25
Junior AAA (AJ)	12 - 14	N/A
AA	9 - 12	14 - 18
A	6 - 9	10 - 14
BBB	4 - 6	6 - 8
BB	2 - 4	4 - 5
B	1 - 2	2 - 3
Unrated CMBS	0	0

Figure 2: Pre and Post Crisis CMBS Deal Structure

structure and deal vintage. Spread risk is captured using swap-based OAS instead of treasury-based OAS, considering the quoting convention of this asset class. As a result, excess return is defined as

$$R^{spread} = R^{total} - (R^{time} + R^{curve} + R^{convexity} + R^{vol} + R^{swap}) \tag{5.101}$$

Model Data Inputs BlackRock USD CMBS Index

Model Estimation Process Non-distressed:

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j \cdot I_i^{tranche_{vintage_j}} + \varepsilon_i \tag{5.102}$$

where $I_i^{tranche_vintage_j}$ is a dummy variable indicating the cohort that the security belongs to. The spread here is based off swap curves.
Distressed:

$$R_i^{total} = \beta_{distressed}\, I_i^{distressed} \tag{5.103}$$

Exposure Mapping

- For non-distressed bonds, dummy variables exposures are used for all factors, indicating factor cohort membership
- Distressed bonds are identified in the following criteria:
  Legacy Mezzanine or Legacy AM/AJ CMBS bonds with OAS > 700

In addition, to account for the differential risk characteristics, bond exposure (%NAV) is further adjusted as follows:

$$exposure_i^{distressed} = \frac{\sigma_i}{\sigma_f} = \frac{f(p_i)}{f(p_f)} \tag{5.104}$$

, where p_i is the price of the distressed security, p_f is the price of the distressed factor (calculated as the weighted average prices of securities in the factor cohort), and f(p) is the priced based function defined as

$$f(p) = \alpha_i + \beta_1 p + \beta_2 p^2 + \beta_3 p^3 + \beta_4 p^4 \tag{5.105}$$

List of factors

- Table 71 details the List of factors for the US ABS model.
- CMBS PreLCF post09 serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 71: US CMBS – factor list

Class	Type	Factor
Distressed	Legacy Mezzanine Legacy AM/AJ with oas>700	CMBS Distressed
Legacy CMBS issued in 2005 and before	PRELCF/PAC	CMBS PreLCF pre05
	LCF	CMBS LCF pre05
	AM and NOT DISTRESSED	CMBS AM pre05
	(AJ or MEZZ) and NOT DISTRESSED	CMBS AJ pre05
Legacy CMBS issued in 2006, 2007 and 2008 (before 9/1/2008)	PRELCF/PAC	CMBS PreLCF 06/07
	LCF	CMBS LCF 06/07
	AM and NOT DISTRESSED	CMBS AM 06/07
	(AJ or MEZZ) and NOT DISTRESSED	CMBS AJ 06/07
CMBS New Issues (2.0/3.0)	PRELCF/PAC	CMBS PreLCF post09
	LCF WAL ≤ 3	CMBS LCF post09 3yr
	LCF $3 <$ WAL ≤ 5	CMBS LCF post09 3-5yr
	LCF $5 <$ WAL ≤ 7	CMBS LCF post09 5-7yr
	LCF WAL > 7	CMBS LCF post09 7yr
	AM, AS, AJ	CMBS AS post09
	MEZZ	CMBS MEZZ post09
Agency	Agency CMBS	Agency CMBS

Continued on next page

Table 71: US CMBS – factor list (Continued)

IO	Legacy IO	CMBS IO pre2005*
		CMBS IO 2006-2008*
	New Issues	CMBS IO post09*

Factor Interpretation

- Factor returns represent %-spread movements for all non-distressed universe. In detail, Tranche × vintage factors capture weighted average spread movements of the CMBS issued from the same tranche and within similar range of vintage.
- The distressed factor measures the weighted average total return of distressed and legacy mezzanine CMBS.
- (*) represents mapped factors. Refer Section 9.4.2 for more details.

5.4 US ABS

Asset-backed securities (ABS), specifically consumer ABS, are backed by a pool of assets like auto loans, leases, credit card debt, and other consumer/commercial loans. They are similar to mortgage-backed securities (MBS) but have different collateral and risk/return characteristics.

Introduced in 1985, ABS has grown steadily with diverse collateral types, issuers, and investors. During the Global Financial Crisis (GFC), consumer ABS performed relatively well in terms of credit performance and remains one of the most stable sectors in structured finance.

Auto, credit card, and student loans make up most of the ABS market, with smaller sectors like consumer loans, equipment, timeshare, and manufactured homes also present. Accurate analytics models are crucial for capturing prepayment and default behaviors of different collateral types. Aladdin analytics models cover major ABS sectors like auto, student loans, and consumer loans. Collateral types and maturity are key factors in the risk model, with credit risk being the primary concern. Spread return is measured in swap OAS space, similar to CMBS.

$$R^{spread} = R^{total} - (R^{time} + R^{curve} + R^{convexity} + R^{vol} + R^{swap}) \tag{5.106}$$

Model Data Inputs Merrill Lynch Fixed and Floating ABS Index.

Model Estimation Process The spread is OAS over the swap rate. This is more consistent with the quoting standards of ABS and allows the model to explicitly capture the Treasury-Libor basis. Regressions are performed in relative spread change space, with the weight being notional weight * DxS. From the regression, the factor returns will be given by the betas:

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j^{collateral} \cdot I_i^{collateral} + \sum_k \beta_k^{mat_k} \cdot exp_i^{mat_k} + \varepsilon_i \tag{5.107}$$

Exposure Mapping Exposures to collateral type factors are DxS-based dummy exposures, with "S" being swap-based spread instead of treasury-based OAS. For maturity factors, the exposures are continuous, measuring the proportional distance of the security's maturity to the maturity factor. In detail,

$$exp_i^{mat} = \begin{cases} f^{mat_1} & mat_i \leq 1 \\ (2 - mat_i) f^{mat_1} & 1 < mat_i \leq 2 \\ (mat_i - 2) f^{mat_3} & 2 < mat_i \leq 3 \\ \frac{7-mat_i}{4} f^{mat_3} + \frac{mat_i - 3}{4} f^{mat_7} & 3 < mat_i < 7 \\ f^{mat_7} & mat_i \geq 7 \end{cases} \tag{5.108}$$

where mat_i is the maturity of the security and $f^{mat_k}, k \in 1, 3, 7$ refers to the 3 maturity factors, with k indicating the maturity tenor[41]. This continuous approach allows the maturity exposures to transition smoothly between different maturity factors as matures, resulting in more stable risk forecasts.
Further details on the asset pricing models used to compute partial durations can be found at[10], [11], [12] , [13]

List of factors

- Table 72 details the List of factors for the US ABS model.
- ABS Market serves as the fallback systematic factor.

  Securities are mapped to this fallback factor when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

Table 72: US ABS – Factor List

Class	SubClass	Factor
Collateral	Auto	ABS Auto Lease
		ABS Auto Prime
		ABS Auto Subprime
	Credit Card	ABS Credit Bank Card
		ABS Credit Retail Card
	Student Loans	ABS Student Loans Govt Gtd
		ABS Student Loans Private
	Other	ABS Equipment
		ABS Consumer
		ABS Floorplan
		ABS Other
Maturity[45]	1-year	ABS Maturity 1
	3-year	ABS Maturity 3
	7-year	ABS Maturity 7

[41] For example, a security with a maturity of 5 years will have 0.5 exposure to the 3-year factor and 0.5 exposure to the 7-year factor.
[45] Proportional exposures are adopted as detailedin the Exposure Mapping section

Factor Interpretation Factor returns represent percentage spread movements for the universe. In detail:

- Collateral type factors capture the weighted average spread movements of their respective factor cohort.
- Maturity factors capture the net effect of the corresponding maturity factor cohort by controlling for collateral type effect. For example, 3 year maturity factor represents the weighted average difference of spread movements of 3 year maturity vs 7 year maturity bucket, holding other effects the same.

5.5 Danish MBS

Danish Mortgage-Backed Securities (DMBS) are covered bonds used by banks in Denmark to fund mortgages. Unlike traditional covered bonds, DMBS are subject to prepayment risk, affecting cash flows and repayment schedules. Banks follow the match-funding principle, selling bonds with matching characteristics to fund loans. New bonds are sold whenever a loan is disbursed. Borrowers can prepay their loans by buying back the underlying bond from the market.

5.6 Pan-Euro ABS

The Pan European securitized bond market started in the UK and Netherlands in the late 1990s, adapting US structures. It includes various collateral types from multiple European countries. Securities are usually denominated in the same currency as the collateral. This asset class was once very safe, with high-rated tranches traded globally. Issuance peaked before the 2008 credit crisis and then dropped due to credit losses during the GFC. Interest revived before the 2020 pandemic due to low rates. Currently, floating rate bonds dominate, with RMBS being the majority. Fixed-rate issuance is less common and mainly includes UK Whole loan ABS and UK CMBS. Collateral type is the main source of spread risk.

- Collateral types – Generally speaking, bonds backed by the same type of underlying collaterals are linked to the similar fundamentals and often respond to similar economic events (e.g. RMBS vs CMBS). In addition, the following two additional attributes are help further differentiate the risk characteristics of different ABS security at a more granular level: Countries – Collaterals originated from the same country may move together as they are subject to similar country level risk/economic conditions (e.g. Germany vs Greece).
- Subtypes – Additional subtypes such as public vs non-public (wholes loans), prime vs subprime RMBS also play a significant role

In summary, the spread risk model of Pan European securitized products contains a granular set of spread risk factors that are closely tied to the distinct collateral type/subtypes in the Pan European securitized market, with DxS employed as spread risk exposures.

Model Data Inputs BlackRock EUR ABS Index + BlackRock GBP ABS Index

Model Estimation Process Factors based on collaterals and countries are estimated in cross-sectional regressions, with weights being market cap multiplied by DxS:

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_j \beta_j \cdot I_i^{\overline{region_{collateral_j}^{country}}} + \varepsilon_i \tag{5.109}$$

There are two generic market factors (one for floating and one for fixed) that capture the spread movements of the entire representative universe:

$$\frac{R_i^{spread}}{-D_i \times S_i} = \sum_k \beta_k \cdot I_i^{\text{coupon type}_k} + \varepsilon_i \tag{5.110}$$

Table 73: Pan-Euro ABS – factor list

Collateral Type	Factor
Whole Loan	Whole loan non-Pub
	Whole loan Pub
CMBS	Fixed CMBS
	CMBS
Auto	AUTO
Consumer	CONSUMER
RMBS	RMBS Ireland
	RMBS Spain
	RMBS Italy
	RMBS Portugal
	RMBS GB PRIME
	RMBS GB BTL
	RMBS GB SUB PRIME
	RMBS Netherlands
	RMBS Core Europe excl GB and NL
CLO	CLO SME[49]
Market	Fixed Rate ABS (generic factor)
	Float Rate ABS (generic factor)

Exposure Mapping Exposure type is DxS for all European Securitized Bonds. Each of them is mapped to one of the factors based on its collateral type, countries and certain other attributes as listed below. The bonds which are missing attributes fall back to the two generic market factors based on their coupon type (fixed or float).

List of factors

- Table 73 details the List of factors for the Pan-Euro ABS model.
- Pan Euro ABS Fix and Pan Euro ABS Flt serve as fallback systematic factors for fixed rate ABS and floating rate ABS, respectively.

  Securities are mapped to these fallback factors when they do not meet the criteria for exposure to other systematic factors. This ensures that every security maintains an associated risk representation through systematic risk factor loadings.

A description of the collateral models used for the asset classes listed above can be found in BlackRock German Auto ABS, BlackRock UK RMBS Model, BlackRock European Regional RMBS Model.

Factor Interpretation Factor returns represent %-spread movements for the corresponding universes. There are two types of prepayment options available depending on the type of the bond:

[49]Note that CLO SME is different from the traditional CLOs in that the collateral pool is fixed whereas traditional CLOs are structured in a way such that loans are actively managed in the trust by a portfolio manager. We will discuss traditional CLOs in section 5.7.

Table 74: Danish MBS – factor list

Model	Factor Type	Cohort	Criterion	Factor
Non-Callable	Market	Market		DKK MBS Noncallable
Callable	Moneyness	OTM	Moneyness <-0.5%	DKK MBS OTM
		ATM	-0.5%<=Moneyness <=0.5%	DKK MBS ATM
		ITM	Moneyness> 0.5%	DKK MBS ITM

- Callable bonds have an embedded call option that allows the borrower to prepay their mortgage at par prior to maturity. This option makes them subject to prepayment risk.
- Noncallable bonds only have the delivery option meaning borrowers can only prepay their mortgages by purchasing the bond in the open market and delivering it to the bank. This implies there is no inherent prepayment risk for these bonds, and they will be exposed to only credit risk.

Given the callable and noncallable bonds are exposed to different types of risk, they warrant separate modelling treatment.

Model Data Inputs Nykredit Danish Mortgage Bond Index (NYKILXGOV) and Nykredit Total Index (NYTRIDX). The associated option adjusted spread, spread duration, and spread return are generated by Aladdin analytics.

Model Estimation Process Factors are estimated separately for callable and non-callable bonds due to the differing borrower response to IR changes highlighted previously. In the case of non-callable bonds, a single market factor is estimated

$$R_i^{\text{noncallable spread}} = \beta_{noncallable} \cdot I_i^{noncallable} + \varepsilon_i \tag{5.111}$$

where $i \in \{\text{noncallable universe}\}$
Callable bond spread returns are estimated based on a three-factor model defined in moneyness space. The definition of moneyness in this context is the difference between the stated coupon rate and the prevailing mortgage rate, prevailing rate being proxied to DKK treasury rate 30yr + 1.3%. 1.3 % is the historically observed average spread between the prevailing rate and the 30yr DKK Treasury rate, based on 2 years of data. It is assumed to be static.

$$Moneyness = Coupon - (DKKTreasuryRate30year + 1.3\%) \tag{5.112}$$

Then,

$$\frac{R_i^{spread}}{-SprD_i} = \sum_k \beta_{moneyness} \cdot I_i^{moneyness_k} + \varepsilon_i \tag{5.113}$$

where $i \in \{\text{callable universe}\}$

Exposure Universe Estimation is done in absolute spread risk space i.e. spread duration (D) exposure for all factors. Each of them will be mapped to one of the factors based on its callability, and moneyness as listed below in Table 74.
A description of the collateral model used in spread duration calculation can be found in [14].

List of factors The List of factors to which Danish MBS are exposed, is shown below.
Additionally, the Danish MBS have exposure to the EUR Implied volatility factor, with the exposure defined as IR-volatility sensitivity.

Factor Interpretation Factor return captures weighted average absolute spread movement for the corresponding universes.

5.7 CLO

A collateralized loan obligation (CLO) is a type of structured credit that buys a diverse pool of bank loans made to businesses. These loans are typically floating-rate, senior secured, first-lien loans, rated BB or B. CLO assets are financed by a mix of debt and equity.
In a CLO deal, payments from the underlying loans are pooled and distributed to the deal's liability tranches. The debt issued by CLOs is divided into tranches, each with a different risk/return profile based on its priority of claim on the cash flows from the loan pool. Since the late 1980s, the CLO market has grown significantly, with about 85% issued in the US and 15% in Europe. US CLOs are mapped to a generic risk factor linked to leveraged loan spread risk, while European CLOs are mapped to two JPM European CLO subindices based on their ratings (investment grade/high yield).

List of factors

Table 75: CLO – factor list

Currency	Factor	Source
USD	CLO Generic[53]*	
EUR	IG CLO	JPM AAA CLO subindex
	HY CLO	JPM BB CLO subindex

(*) represents mapped factors. Refer Section 9.4.2 for more details.

5.8 CAD securitized

There is one risk factor (CAD Securitized IG) which represents the % spread movement of CAD Securitized bonds. It is currently nested within the CAD Corporate Bond risk model. See CAD Corporates section above for details.

5.9 Commercial Mortgage Loans

Model Data Inputs ICE BofA CMBS Index (CMBS)

Model Estimation Process The model has one systematic factor: property type. Each CML loads on a factor based on its property type, with DxS exposures. There is one factor for Hotel CMLs, one for non-hotel core Commercial CMLs, and a Market CML factor for unspecified property types. CML factor returns depend on property type, known as factor deltas $f^p_{CML,t}$.

$$R_{i,t} > \sum_j f^p_{CML,t} I_i \quad (i \in p) \tag{5.114}$$

Where $I_i(i \in p)$is the dummy variable indicating whether loan i belongs to property type p. Currently, the property types are categorized according to NCREIF classifications, including Hotel, Industrial, Multifamily, Office, Retail, and Others. Aggregating SASB deals to construct CML property type factors follows a stepwise process. First, we create a set of SASB rating-by-property-type risk factors as building blocks:

[53]Two factors are currently included in the model: USD CLO Generic & USD CLO Generic V1. The factors differ in that V1 uses an enhanced redemption model.

$$\{f^P_{AAA,t'}, f^P_{AA,t'}, f^P_{A,t'}, f^P_{BBB,t'}\} \tag{5.115}$$

SASB deals in the ML_{CMBS} universe are grouped by ratings and property types. Within each group, the deal spread delta is weighted averaged to form the building block factor. The weight of each deal reflects its market capitalization in the index. For example, $f^p_{AAA,t}$ is constructed by averaging all AAA deals from property type p.

Secondly, for each property type, the four rating building blocks are combined into one CML property type factor.

$$f^P_{CML,t} = w^P_{AAA,t} f^P_{AAA,t} + w^P_{AA,t} f^P_{AA,t} + w^P_{A,t}(f^P_{A,t} + w^P_{BBB,t} f^P_{BBB,t}) \tag{5.116}$$

In the equation above, w^p_t is the weights of each rating block to form property type p factor at time t.

Finally, for all the loans whose property types are not specified, one CML market factor is generated to map them on. The market factor is the weighted average of all property type CML factors according to each property type's weight in the Trepp universe:

$$f^{mkt}_{CML,t} = w^{Hot}_t f^{Hotel}_{CML,t} + w^{Ind}_t f^{Industrial}_{CML,t} @ + w^{Mul}_t f^{Multifamily}_{CML,t} + w^{Off}_t f^{Office}_{CML,t} + w^{Oth}_t f^{Other}_{CML,t} + w^{Ret}_t f^{Retail}_{CML,t} \tag{5.117}$$

In the equation above, w^p_t is the point-in-time capitalization weight of property type p in the entire Trepp SASB universe.

Assigning each property type one CML factor can overstretch the data capacity. For example, Multifamily is a thin sector with few deals in the Trepp universe. Combining property type factors moving together provides more robust outputs. The final factor structure supports a separate CML factor for hotels and combines all other property types into one "Core Commercial" CML factor, weighted according to each property type's point-in-time balance weight.

$$f^{Core}_{CML,t} = w^{Ind}_t/(1 - w^{Hot}_t)\ f^{Industrial}_{CML,t} @ + (w^{Mul}_t)/(1 - w^{Hot}_t) f^{Multifamily}_{CML,t} + (w^{Off}_t)/(1 - w^{Hot}_t) f^{Office}_{CML,t} + (w^{Oth}_t)/(1 - w^{Hot}_t) f^{Other}_{CML,t} + (w^{Ret}_t)/(1 - w^{Hot}_t) f^{Retail}_{CML,t} \tag{5.118}$$

Exposure Mapping Spread duration-based exposures based on property type mapped to property type factors.

Table 76: Commercial Mortgage Loan factor list

Property Type Factors	Exposure Type	Property Type included	Broad Sector
Hotel	DxS	Hotel	Market
Core Commercial	DxS	Office, Retail, Industrial, Multifamily, Others	Market

6 Idiosyncratic Factors

Idiosyncratic risk refers to the portion of security returns that cannot be attributed to systematic factors. Essentially, it represents the residual return ($\epsilon_{i,t}$), and its volatility ($\sigma(\epsilon_{i,t})$) is the idiosyncratic risk which is modeled. Similar to the systematic model, the idiosyncratic risk model has the form.

$$\sigma(\epsilon_{i,t}) = e_{i,t}\sigma(u_{i,t}) \tag{6.119}$$

where the exposure to idiosyncratic risk ($e_{i,t}$) varies by asset class and $\sigma(u_{i,t})$ is the forecasted residual volatility. Generally, the measurement choice for idiosyncratic risk is consistent with that of the systematic model. Tables 77 to 80 provide the full list of estimation universes and exposure units for the idiosyncratic model. For example:

- For government bonds, $e_{i,t}$ is duration, and idiosyncratic volatility is estimated in yield space.
- For sub-AAA USD corporates, $e_{i,t}$ is DxS (duration times spread), and idiosyncratic volatility is estimated in relative spread space.
- For distressed securities, $e_{i,t}$ is 1, and idiosyncratic volatility is estimated directly in total return space.

6.1 Model Structure

Securities with similar characteristics, such as rating and issue size, generally exhibit similar levels of idiosyncratic risk. For instance, financial bonds typically have higher idiosyncratic volatility (in relative spread space) compared to utility bonds. Consequently, the idiosyncratic risk model uses observable security characteristics to estimate the idiosyncratic risk of individual securities.

Mathematically, idiosyncratic risk is modeled as the product of three components:

$$\sigma_{i,t} = \hat{\theta_{j,t}}(1 + \hat{\delta}_{i,t})M_{j,t} \tag{6.120}$$

where $\hat{\Sigma}$ represents the forecasted mean absolute weekly residual for the cohort j that the security i belongs to, indicating ex-ante idiosyncratic factor volatility. The term $(1+\hat{\delta}_{i,t})M_{j,t}$ measures security i's "beta" to the idiosyncratic factor volatility of j. Specifically, $(1 + \hat{\delta}_{i,t})$ is the forecasted relative absolute weekly residual of the security, and $M_{j,t}$ is a cohort-level time-varying scalar that converts weekly absolute residual return forecasts into monthly volatility forecasts.

In detail, estimation universe is first divided into major cohorts, denoted as $j = 1, 2, 3, \cdots, J$. Table 77 to Table 80 list the cohort logic for idiosyncratic risk models by asset classes. These major cohorts become the idiosyncratic risk factor cohorts, with factor return tracking realized mean absolute residual computed as the cap-weighted average of security absolute weekly residual.

$$\theta_{j,t} = \sum_{i \in j} w_{i,t} \|u_{i,t}\| \tag{6.121}$$

The forecasted mean absolute residual is estimated using the history of past realized mean absolute residuals,

$$\hat{\theta_{j,t}} = \sqrt{\frac{1-\lambda}{1-\lambda^n}} \sum_{s=t-n}^{t-1} \lambda^{t-s} \theta_{j,s}^2 \tag{6.122}$$

where $\sigma = 0.917$ (8-week half-life) and $n = 104$ by default[54].
Next, a structural model is used to forecast the absolute residual return of each security relative to the cohort mean absolute residual forecast. The relative absolute residual return of each security is defined as:

$$\delta_{i,t} = \frac{\|u_{i,t}\| - \theta_{j,t}}{\theta_{j,t}} \tag{6.123}$$

Relative absolute residual $\hat{\delta_{i,t}}$ is regressed on security sub-cohort dummy variables and trailing realized specific volatility in a pooled cross-sectional regression using 104 weeks of data with 26-week half-life [55]

$$\delta_{i,t} = \sum_{sc \in j} \beta_{sc} I_{sc \in sc} + \sum_k \phi_k^j X_k + \gamma \tilde{\sigma_{i,t}} = \xi_{i,t} \tag{6.124}$$

where sub-cohort dummies ($I_{sc \in j}$) represent more detailed groupings of securities with similar characteristics, often aligning with the first-order factor cohorts in the systematic risk model. The X_k terms are other potential continuous

[54] The weighting scheme is shared across systematic and idiosyncratic risk factors and the default setting varies by client.
[55] The 26-week half-life here is fixed and not subject to change in user settings. We choose 26-week half-life instead of 8-week half-life in order to get more stable estimation results.

or dummy variables that improve the explanatory power of the regression. The trailing realized residual volatility is denoted as $\tilde{\sigma_{i,t}}$. This regression is performed weekly for each cohort, and the resulting coefficients are used to forecast the relative absolute residual ($\hat{\delta_{i,t}}$).

To forecast the weekly absolute specific return ($|\hat{u_{i,t}}|$), the average level of idiosyncratic risk ($\hat{\theta_{j,t}}$) is multiplied by the relative level of idiosyncratic risk ($(1+\hat{\delta}i,t)$). The goal is to predict the standard deviation ($\sigma(ui,t)$). To convert $|\hat{u_{i,t}}|$ into standard deviation units, a scaling factor is used. Normally, the standard deviation of $\frac{\hat{u_{i,t}}}{|\hat{u_{i,t}}|}$ is $\sqrt{\frac{\pi}{2}}$. However, residual returns are usually fat-tailed, so the standard deviation is greater than $\sqrt{\pi/2}$. Additionally, since the forecast is weekly but the horizon is monthly, the square root of time scaling cannot be simply applied due to negative serial correlation in weekly residuals. Therefore, adjustments for both kurtosis and serial correlation are made by defining the standardized return with two-lag autocorrelation.

$$V_{i,t} = \frac{u_{i,t} + u_{i,t-1} + u_{i,t-2}}{(\sqrt{3}\theta_{j,t}(1+\hat{\delta_{i,t}})} \tag{6.125}$$

Then the scaling factor $M_{j,t}$ is calculated as the average realized standard deviation of $V_{i,t}$ for $i \in j$. Finally, the security's idiosyncratic risk forecast is estimated as

$$\sigma_{i,t} = \hat{\theta_{j,t}}(1+\hat{\delta}_{i,t})M_{j,t} \tag{6.126}$$

6.2 Model Variations for Government and Inflation-linked bonds

Most fixed income securities are illiquid and their residuals often contain event-related jumps that are unlikely to reoccur soon. Therefore, idiosyncratic risk is modeled using a structural framework, focusing on average cohort-level residuals to mitigate the impact of large outliers and learn from similar bonds. This results in a more robust and forward-looking volatility forecast. However, it is less transparent and has limited ability to adjust to sudden changes in idiosyncratic risk. For government and inflation bonds, which are very liquid, a hybrid approach is used combining time series and structural estimates for a more accurate forecast.

Specifically,

$$\sigma_{i,t} = \gamma\sigma_{i,t}^{TS} + (1-\gamma)\sigma_{i,t}^{STR} \tag{6.127}$$

where $\sigma_{i,t}^{TS}$ and $\sigma_{i,t}^{STR}$ denote time series and structural idiosyncratic risk forecasts, and γ is the weight of the time-series component (see Appendix to computation details). If the security does not have enough history or if its residuals are not well-behaved, more weight will be put on the structural estimate, and vice versa. Mathematically,

$$\sigma_{i,t}^{TS} = (\frac{1-\lambda}{1-\lambda^n}\sum_{s=t-n+1}^{t}\lambda^{t-s}u_{i,s}^2)^{\frac{1}{2}} \tag{6.128}$$

Where λ is the decay parameter and γ_i is the number of periods used under the default weighting scheme. A serial correlation adjustment with lags of 3 weeks [56] is applied further to scale the forecast into a monthly horizon. $\sigma_{i,t}^{STR}$ is the same as defined in equation 6.127. γ is computed as follows:

First, $Z_{i,t}$ is computed to measure how well-behaved the residual history is.

$$Z_{i,t} = \|\frac{sd(u_{i,t}) - robsd(u_{i,t}))}{robsd(u_{i,t})}\| \tag{6.129}$$

where $sd(u_{i,t})$ is an equally-weighted sample standard deviation, and $robsd(u_{i,t})$ is a robust sample standard deviation calculated based on the first and third sample quartile of the residual return history $robsd(u_{i,t}) = 1/135 \times (Q3–Q1)$. For residuals that exhibit excess kurtosis and outliers, $robsd(u_{i,t})$ tend to differ a lot from $sd(u_{i,t})$ and $Z_{i,t}$ will have large values.

[56] $\sigma_{t,h}^2 = h\sigma_t^2[1 + 2(1-\frac{1}{h})\phi_u^2 + 2(1-\frac{3}{h})\phi_u^3]$, where $\phi_u^k$is the kth order serial correlation in residuals

The weight of the time series forecast is jointly determined by $Z_{i,t}$ and the length of available residual return history n as follows:

$$\gamma = [min(1, max(0, \frac{n-26}{52}))] \times [min(1, max(0, exp(0.25 - Z_{i,t})))] \tag{6.130}$$

Thus, only time series forecast is used when the residuals have more than 78 weeks of history and $Z_{i,t} \leq 0.25$. As the length of history decreases or $Z_{i,t}$ increases, the weight of structural forecast increases, and only structural forecasts are used if the length of history ≤ 26.

Additionally, treasury futures instruments are exposed to the idiosyncratic risk as treasury bonds. The model selects the cheapest-to-deliver treasury bond and uses the idiosyncratic risk from that bond multiplied by the duration of the future.

6.3 USD Tax-exempt Municipal Bonds

Due to the large size of the municipal bond universe, security returns and analytics are aggregated at the obligor level before model calibration. Obligor is chosen over issuer because municipal issuers often act as conduits to the true underlying entity (the obligor), which funds interest and principal payments. If a security doesn't belong to any obligor, issuer is used instead. It is also ensured that bonds from different municipal bond groupings (escrowed, general obligation, and revenue) are not aggregated together. To achieve this,"synthetic obligors" are created by combining raw obligors with these groupings.

6.3.1 Estimation Universe and Cohort Construction

Table 77 through Table 80 summarize the estimation universe, as well as factor and sub cohorts for each asset class covered in this model:

- Table 77 is for government bonds.
- Table 78 is for global credits.
- Table 79 is for structured debt.
- Table 80 is for other asset classes.

Table 77: Panel A: Idiosyncratic model - government bonds

Class	Estimation Universe	Factor Cohort	Exposure Unit	Sub Cohorts	Other Variables
Government Bonds	Barclays Global Treasury Index	USD	OAD	Maturity buckets (Maturity<=2.5, 2.5<Maturity<=7, 7<Maturity<=15, Maturity>15)	None
		GBP			
		JPY			
		EUR Sovereigns, excluding PT, ES, IT, GR		Countries	Maturity buckets
		EUR Peripherals (PT, ES, IT, GR)			
		Other DM and EM Currencies			
Inflation-linked Bonds	BBG Global Inflation Linked BBG Global Inflation Linked US Index (Legacy Lehman) BBG Global Inflation Linked ex-US Index (Legacy Lehman)	USD	OAD	Countries	Maturity buckets
		GBP			
		JPY			
		EUR			
		Other DM and EM Currencies			

6.4 Idiosyncratic Model Structure

Table 78: Panel B: Idiosyncratic model – global credit

Class	Estimation Universe	Factor Cohort	Exposu Unit	Sub Cohorts	Other Variables
US Credit	Barclays US Investment Grade and High Yield Indices	Muni Taxables	spd_dur	None (Intercept only)	None
		Government Related AAA (foreign agencies, local authorities, sovereign, supranationals)		Type (Agency or Non-corporate)	None
		Government Related sub-AAA	DxS	Type (Agency or Non-corporate)	None
		Financial IG		Industry/Rating (IG, HY)	Subordination (CapSec, Subordinated, Senior), Yankee Status, Maturity buckets
		Utility IG			
		Industrials IG			
		Financial HY			
		Utility HY			
		Industrials HY			
		Distressed	% NAV	None (Intercept only)	None
EU/UK Credit	BlackRock CHF Corporate HY Index BlackRock EUR Corporate HY Index BlackRock EUR Corporate IG Index BlackRock GBP Corporate HY Index BlackRock NOK Corporate HY Index BlackRock SEK Corporate HY Index IBoxx Sterling Index	EU Government Related AAA	Spd_dur	Type (Agency or Non-corporate)/Currency (EUR or GBP)	None
		EU Covered Pfandbriefe		Currency (EUR or GBP)	None
		EU Government Related sub-AAA	DxS	Type (Agency or Non-corporate)/Currency (EUR or GBP)	None
		EU Covered non-Pfandbriefe		Currency (EUR or GBP)	None
		EU Financial IG		Industry/Rating (IG, HY)	Subordination (CapSec or not), Domestic or Foreign Issuer, Maturity buckets
		EU Industrials IG			
		EU Financial HY			
		EU Industrials HY			
		UK Financial IG			
		UK Industrials IG			
		EU Distressed	% NAV	None (Intercept only)	None

Continued on next page

Table 78: Panel B: Idiosyncratic model – global credit (Continued)

JP Credit	Nomura BPI Index	Government Guaranteed	spd_dur	Type (Muni or Government Guaranteed or Bank Debenture)	None
		Financial IG	DxS	Industry/Rating (Super IG, IG)	Maturity buckets
		Utility IG			
		Industrials IG			
AU Credit	BlackRock AUD Investment Grade Index	Government related	Spread duration	Maturity buckets	None
		Financial IG	DxS	Industry/Maturity buckets	
		Industrials IG			
CA Credit	DEX Canadian Bond Market Index	Provincials	spd_dur	Maturity buckets	None
		Government Agency			
		Financial IG	DxS	Industry/Maturity buckets	
		Utility IG			
		Industrials IG			
CDS	All CDS issuers	US Financial	DxS	Currency/ Industry/ Rating (IG, HY)	None
		US Utility			
		US Industrials			
		JP Consumer			
		JP Financial			
		JP Industrials			
		EU Consumer			
		EU Financial			
		EU Industrials			
		AU Financial			
		AU Industrials			

Table 79: Panel C: Idiosyncratic model – securitized products

Class	Estimation Universe	Factor Cohort	Exposure Unit	Sub Cohorts	Other Variables
US ABS	Merrill Lynch Fixed/Floating ABS index	Auto	DxS	Auto subsectors	Maturity buckets
		Credit Card		Bank Cand, Retail Card	
		Student Loans		Student Loans Govt Gtd/Private	
		Other		Consumer, Equipment, SBA Loans, Other	
US CMBS	BlackRock USD CMBS Index Grade Index	Legacy Super Senior	DxS	Vintage-based tranches	None
		Legacy AM/AJ			
		New Issue Super Senior			
		New Issue AS			
		New Issue Mezzanine			
		Distressed	%NAV	None	None

Table 80: Panel D: Idiosyncratic model – other asset classes

Class	Estimation Universe	Factor Cohort	Exposure Unit	Sub Cohorts	Other Variables
Emerging Markets Bonds	JP Morgan Global EMBI Index ICE BofA Asian Dollar Index BBG EM USD Aggregate Index JPM Euro EMBI Diversified	Asia	DxS	Regions/Rating (IG, HY)	Sovereign or Corporate, Maturity buckets
		Latin America			
		Central/ Eastern Europe			
		Middle East and Africa			
US Leveraged Loans	SP Leveraged Loan Index	Non-Distressed	DxS	Rating	None
		Distressed	%NAV	None (Intercept only)	
USD Tax-exempt Municipal Bonds	SP Municipal Bond Index	Escrowed	spd_dur	None	
		General Obligation	DxS	Region (consistent with systematic model structure)	None
		Revenue		Industry (consistent with systematic model structure)	

6.4.1 Fallback Logic

All securities in the estimation universe that meet data requirements are included in the structural model for each asset class. For those not covered, a cohort proxy using the average idiosyncratic risk of all securities within the same cohort is used. A fallback cohort structure is followed, starting with the most granular cohorts and moving to coarser levels if necessary.

6.4.2 Idiosyncratic Correlations

The idiosyncratic component is independent of systematic factors by model assumption. However, there is still idiosyncratic correlation among securities within one issuer or deal, which varies by asset class. For corporate-like instruments (bonds, CDS, etc.), the model captures intra-issuer correlation as a ratings-based function that increases with lower issuer ratings, as these issuers are more susceptible to event risk. For EM and government bonds, most issuer-level risk is captured by the systematic risk model, so the correlation is assumed to be zero. Securitized products' idiosyncratic correlation is modeled at the deal level. For CMBS, idiosyncratic correlation is stronger for legacy deals than new issues. For US ABS, the model uses a tranche-based structure, with higher intra-deal correlations for lower tranches that have less credit enhancement. The resulting correlations are summarized in Table 81.

Table 81: Idiosyncratic model - correlation structure

Class	Estimation Universe	Factor Cohort	Exposure Unit	Sub Cohorts	Other Variables
Emerging Markets Bonds	JPMorgan EM Sovereign Index, JACI Index	Asia	DxS	Regions/Rating (IG, HY)	Sovereign or Corporate, Maturity buckets
		Latin America			
		Central/ Eastern Europe			
		Middle East and Africa			
US Leveraged Loans	SP Leveraged Loan Index	Non-Distressed	DxS	Rating	None
		Distressed	%NAV	None (Intercept only)	
USD Tax-exempt Municipal Bonds	SP Municipal Bond Index	Escrowed	spd_dur	None	
		General Obligation	DxS	Region (consistent with systematic model structure)	None
		Revenue		Industry (consistent with systematic model structure)	

6.4.3 Other exceptions

Other models also employ idiosyncratic components, but the above idiosyncratic factor structure framework is not adopted for these exceptions due to practical considerations such as insufficient or shallow data. Exceptions include non-agency MBS and credit risk transfer deals, where a relatively static approach is used to estimate idiosyncratic risks based directly on realized idiosyncratic volatilities. Treasury future securities are mapped to the idiosyncratic factor for the underlying treasury issuance.

7 Model Testing & Surveillance

All models are tested extensively using historical data to evaluate the model performance over different market environment to the extent possible. The section below provides a brief overview of the various diagnostics and back-tests performed on the BFIRM models prior to their release. These models are also tested on an on-going basis and as such, model users are advised to attend the Quarterly Backtesting calls for a discussion of the performance of BFIRM in more recent data.

7.1 Model Estimation Diagnostics

The model estimation is a core component of the model construction process. Several diagnostics are appraised during this process to check the explanatory power of the models on different cohorts and also individual factor efficacy:

- Aggregate-level parametric fit between realized and parametric returns (of relevant indices, portfolios and certain sub-cohorts) is measured through R-squared which quantifies how closely portfolio/cohort level parametric returns tracks the realized returns over certain time period. This is a standard measure used to compare the

new model to existing model, where new model's performance is usually better than the old model, especially when gauged on the sub-cohort where the model enhancement focuses on.

- Security-level parametric fit between realized and model-forecast returns is measured via R-squared that quantifies the proportion of cross-sectional variation in the security returns explained by the set of common factors in the model.
  - Cross-sectional R-squared can be measured on any time period of interest of any sub-cohort of concern. For example, when old model performs unsatisfactorily during an eventful period on certain industry, cross-sectional R-squared can then be calculated on this cohort using new model to evaluate new model's capability to address such issues.
  - Factor's contributions to total cross-sectional R-squared over time can also be assessed as an additional diagnostic to highlight model performance contribution from different factors, in order to better connect the contribution of different factors to important market events over time.
- Individual factor efficacy is assessed using a history of t-statistics for each factor by calculating the proportion of significant t-statistics over different periods. This metric serves as a good proxy for the persistence of individual factor effects. The majority of factors are significant more than 10% of time over the research history, with most well in excess of this threshold.
- Exposure investigation is required in most of fixed income risk models' research process, in order to choose the most suitable modeling space. As mentioned in section 2.3, the relationship between spread volatility and spread is reviewed to determine whether DxS is a good choice of exposure type than duration itself and also if certain floors or caps on exposures are needed. In addition, sometimes improvement on sub-cohort level R-squared is calculated as a supporting evidence to validate the floors and caps on exposures.
- Other types of diagnostics are also employed in certain models. For example, in the European Corporate Bond Risk Model, multi-collinearity effect is investigated among factor exposures. Aiming to avoid this issue, a two-stage regression framework is adopted in this model eventually.

7.2 Assessment of Volatility Forecasts

A well-calibrated model should provide risk forecasts that are stable and accurate. One way of testing the accuracy of the risk forecasts is to compare the volatility of realized returns to forecast risks. This forms the basis of the bias statistic and other diagnostics used to evaluate the BFIRM models.

- Bias statistics of the risk factors is generated and reviewed for certain asset classes. The bias statistics helps measure accuracy of volatility prediction of pure factor portfolios for different systematic factors.
- Index level risk forecast is evaluated in some models during research process to check how closely it tracks the realized volatilities. An additional purpose of this measurement is to compare DxS based exposures with duration-based exposures, to demonstrate advantages of DxS that it is more responsive to the current market environment.
- Cohort-level absolute residuals and their volatilities are produced for some models. This diagnostic helps confirm whether new systematic model reduces overall residuals, which cross-validates the model's performance. Moreover, it helps identify trend in idiosyncratic model's research process to identify potential factor structure.

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7.3 Model Surveillance

In this section, the quality control (QC) processes for model input and output, which are monitored daily, are detailed.

Input QC

QC is conducted at security level for all factor blocks and at factor-level for systematic factor block. The day-over-day changes in input data is monitored to ensure stability of the universe coverage.

Note that this QC will generate a diagnostic summary. Any violation will trigger a warning but will not halt or fail the estimation process.

Factor Output QC

- Achieving T-1 Delivery

  After the deltas and levels are estimated, we check if the output include delta and level for T as a measurement of the success of achieving T-1 delivery. If delta or level does not exist for T, this QC will fail the estimation process.

- Compare with T-1 Generated Outputs

  The deltas and levels generated as of T are compared against the values generated as of T-1 for the same date, to gauge the magnitude of changes on the input data. The update is considered significant if on any given date, the absolute difference on levels exceeds pre-determined criteria.

  Note that this QC will generate a diagnostic summary. Any violation will trigger a warning but will not fail the estimation process.

Model Performance Monitoring As part of the daily model calibration process, back-testing diagnostics are generated and internally reviewed to ensure alignment between model predictions and realized returns across selected indices, sub-indices, and sample portfolios. This evaluation assesses whether parametric return and risk forecasts effectively track realized returns across various portfolio types, including aggregate market indices, sub-indices, and actual portfolios. Additionally, the model is included in the Quarterly Backtesting Call, during which volatility and VaR forecasts from Aladdin Risk Models are comprehensively back-tested and discussed with clients. The following performance measures on model predictions are examined as part of this review.

- Parametric returns versus realized returns on several relevant indices.
- Checking absolute and active risk biases.
- Cross sectional bias statistics.
- 1-week 99%-VaR backtests on several relevant indices.

A comprehensive description of the model backtesting process can be found in the following whitepaper: “Model Performance Monitoring of Fixed Income Models”.[57]

[57]Link to ”Model Performance Monitoring of Fixed Income Models” .

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9 Appendix: Additional Details

9.1 Credit Beta Illustration

In addition to risk contribution, the report includes a column for stand-alone risk. Stand-alone risk from a risk factor is calculated as the product of factor exposure and factor volatility. There are two equivalent ways of decomposing stand-alone risk for DxS based risk factors. Stand-alone risk can be expressed as the product of the risk factor percentage spread volatility times the portfolio DxS factor exposure:

$$\sigma_{stand_alone} = \sigma(\frac{\Delta OAS_f}{OAS_f}) \cdot (\sum_i OAS_i \cdot spr_dur_i) \tag{9.131}$$

An equivalent representation of stand-alone risk is as the product of three terms – the risk factor spread volatility times credit beta times the portfolio spread duration exposure to the risk factor.

$$\sigma_{stand_alone} = \sigma(\Delta OAS_f) \cdot \beta p, f \cdot (\sum_i OAS_i \cdot spr_dur_i) \tag{9.132}$$

Credit $\beta_{p,f}$ in equation 9.132 is defined as:

$$\beta_{p,f} = \frac{\frac{\sum_i spd_dur_i \cdot OAS_i}{\sum_i spd_dur_i}}{OAS_f} \tag{9.133}$$

Namely, credit beta is the spread duration weighted spread of the securities in the portfolio exposed to factor f divided by the spread of factor f. Credit beta is a unit-less multiplier that scales the factor volatility by the ratio of the portfolio to factor spread level. The advantage of the credit beta representation of stand-alone risk is that the units of factor volatility and factor exposure are more intuitive. Units of factor volatility and exposure are basis points and spread duration contribution respectively.
The risk factor decomposition reports are configurable to express factor volatilities and exposures in units consistent with either equation 9.132 or 9.133. Note that credit betas are only relevant for DxS based factors. Non-DxS based spread factors always have an implicit credit beta of one.

9.2 Historical and implied volatility in risk models

The second of class of upstream models, namely the asset pricing models, highlight a clarification that should be made in the interpretation of risk-indicating market observables between:

- Backward-looking "historical" volatilities;
- Forward-looking, market-expectation based implied volatility

The risk model uses backward-looking historical volatilities in the construction of its risk estimates. The factor-factor covariance matrix used to estimate systematic risk is computed by sampling factor return data with user-specified parameters for the sampling frequency and half-life. A user can adjust the rate at which the model responds to changes in the market environment by appropriate selection of these parameters in the PRT tools. They may decouple the parameters used for historical volatility estimation from those used for correlation estimation.
Forward-looking implied volatility factors are not indicators of realized volatility. Rather, they represent the sampling of interpolated forward-looking implied volatility quotes used in asset pricing models, conveying information on the market consensus on the future distribution of interest rates. Indeed, historical volatility estimates are employed for forward-looking implied volatility factors.
A similar dichotomy exists with correlation, whereby implied correlation which may be imputed from assets such as basket options, or basket credit derivatives (synthetic credit derivatives like tranches CDX indices), but this differs from the historical correlations observed from sampling factor time-series.

9.3 Calculation of covariance matrix and exposures

BlackRock's fixed income model when used in tools such as PRT are designed to allow a user to compute risk across a portfolio with a range of assets. To support such use, the models and supporting software perform the following operations when a user enters a portfolio:

1. For each asset, exposures are loaded. The details of the calculation of exposures are presented in later chapters of this document. For many fixed income assets, use is made of theoretical asset pricing models to compute numerical partial derivatives.
2. Factor return histories related to the exposures are then loaded into the software.
3. A factor-factor covariance matrix is computed from the factor return history, using a sampling and weighting scheme defined by the user.

There are several important consequences of this workflow. Firstly, no single covariance matrix with all Aladdin factors is ever computed. The cohort of factors from which covariance is estimated is assembled dynamically at the time of loading the portfolio. Secondly, the coverage of assets is not limited to assets which may be modeled using the factors described in this whitepaper. Rather, the coverage is defined by a combination of the available set of factors for which a history has been computed in Aladdin, the mapping software which associates securities with the theoretical asset pricing models and the specification of the asset pricing models, in particular which market-related factors they use to quantify a change in theoretical price.

9.3.1 Calculation of risk

In matrix notation, the risk estimate namely variance, for a portfolio with weights $w = \begin{bmatrix} w_1 \\ \vdots \\ w_N \end{bmatrix}$, is given by

$$Std_{port} = \sqrt{w\left(X\Sigma X^T + \Omega\right)w^T} \tag{9.134}$$

where $X = \begin{bmatrix} x_{1,1} & \cdots & x_{1,k} \\ \vdots & \ddots & \vdots \\ x_{N,1} & \cdots & x_{N,K} \end{bmatrix}$, is the matrix of exposures of securities $S^i \in \left\{S^1, \ldots, S^N\right\}$ to factor-factor covariances, Σ , and Ω is the diagonal matrix of uncorrelated idiosyncratic variances for the N securities,

$$\Omega = \begin{bmatrix} \varepsilon_1^2 & 0 & 0 \\ 0 & \varepsilon_i^2 & 0 \\ 0 & 0 & \varepsilon_N^2 \end{bmatrix}$$

The factor-factor covariance matrix covariance matrix is estimated from returns on factors $\left\{f^{(1)}, f^{(1)}, \ldots, f^{(k)}\right\}$ as follows:

$$\Sigma = R^T R = \begin{bmatrix} \Sigma_{1,1} & \cdots & \Sigma_{j,1} & \Sigma_{K,1} \\ \vdots & \vdots & \vdots & \vdots \\ \Sigma_{1,K} & \vdots & \Sigma_{j,k} & \Sigma_{K,k} \\ \Sigma_{1,K} & \cdots & \Sigma_{j,k} & \Sigma_{K,K} \end{bmatrix}$$

where $R_t = \sqrt{\frac{1-\lambda}{1-\lambda^{m+1}}} \begin{pmatrix} f_t^{(1)} - \bar{f}^{(1)} & f_t^{(2)} - \bar{f}^{(2)} & \cdots & f_t^{(K)} - \bar{f}^{(K)} \\ \sqrt{\lambda^1}\left(f_{t-1}^{(1)} - \bar{f}^{(1)}\right) & \sqrt{\lambda^1}\left(f_{t-1}^{(2)} - \bar{f}^{(2)}\right) & \cdots & \sqrt{\lambda^K}\left(f_{t-1}^{(K)} - \bar{f}^{(1)}\right) \\ \vdots & \vdots & \vdots & \vdots \\ \sqrt{\lambda^m}\left(f_{t-m}^{(1)} - \bar{f}^{(1)}\right) & \sqrt{\lambda^m}\left(f_{t-m}^{(2)} - \bar{f}^{(2)}\right) & \cdots & \sqrt{\lambda^m}\left(f_{t-m}^{(K)} - \bar{f}^{(K)}\right) \end{pmatrix}$ and $\lambda = \frac{\ln(0.5)}{H}, H$ being the half-life in terms of number of observations of the factor return, and $\bar{f}^{(j)}$ is the mean return of factor $f^{(j)}$. The user may elect to assume that $\left\{\bar{f}^{(j)} = 0 : j = 1 : K\right\}$
The software allows a user to specify the parameters used for the estimation of covariances from those used for correlations.

9.3.2 Exposure Calculation for Fixed income Securities

Exposures used in equation 9.134 are frequently computed as partial derivatives with respect to changes in a factor, using numerical derivatives. If it is assumed that the theoretical price of a security, P_{m}, at time t, is given as a function of p market variables $x_0, x_p, \ldots, x_p$

$$P_{\mathrm{m}} = P_{\mathrm{m}}\left(x_0, x_p, \ldots, x_p, t\right)$$

The sensitivity of the price, from $x_p \to \widehat{x_p}$, to a change in the level of one of the market factors, is approximated using a Taylor series by

$$\frac{P_m\left(x_0, \widehat{x_p}, \ldots, x_p, t\right) - P_{\mathrm{m}}\left(x_0, x_p, \ldots, x_p, t\right)}{P_m\left(x_0, x_p, \ldots, x_p, t\right)} \approx$$
$$\frac{1}{P_m\left(x_0, x_p, \ldots, x_p, t\right)} \cdot \frac{\delta P_m}{\delta x_p} \cdot \left(\widehat{x_p} - x_p\right) + \frac{1}{2} \cdot \frac{1}{P_{\mathrm{m}}\left(x_0, x_p, \ldots, x_p, t\right)} \cdot \frac{\delta^2 P_{\mathrm{m}}}{\delta x_p^2} \cdot \left(\widehat{x_p} - x_p\right)^2 + \cdots$$

Now, generalizing to allow all the market variables to move simultaneously gives us

$$\frac{dP_{\mathrm{m}}}{P_{\mathrm{m}}} \approx \sum_{p=1}^{P} \frac{1}{P_{\mathrm{m}}} \cdot \frac{\delta P_{\mathrm{m}}}{\delta x_p} \cdot dx_p + \sum_{p}^{P} \sum_{q}^{P} \frac{1}{2} \cdot \frac{1}{P_{\mathrm{m}}} \cdot \frac{\delta^2 P_{\mathrm{m}}}{\delta x_p \delta x_{\mathrm{q}}} \cdot dx_p \cdot dx_q + \cdots$$

Where

$$dx_p = \widehat{x_p} - x_p$$

Ignoring second order and higher terms, gives us the motivation for the use of partial derivatives w.r.t the change in a factor as exposures in a risk calculation for assets whose theoretical pricing function is dominated by the first order (linear) factor contribution. It explains the significant role that theoretical pricing functions play in the estimation of risk using BFIRM.
Of course, the factors that most influence fixed income security pricing (excluding credit) are related to the yield curve. Section 9.4.1 describes the approaches taken in the BFIRM model to estimate yield curves.
The theoretical pricing models used for individual asset classes when generating exposures can be found in [15].

9.3.3 Factor creation in Aladdin

A question that is frequently posed by users of BlackRock's fixed income model, is what is the universe of factors representing the entirety of the model, and which factors does a particular asset type use in its modeling?
Aladdin generates exposures on-the-fly in response to a request for an analytic result on a portfolio. As a first step, the system maps from a known "street" identifier such as an ISIN into a fully specified description of the asset which may be accepted by the analytical models, including both terms and conditions and references to the market factor time-series required for calculations. It is the combination of the terms and conditions, the analytical pricing model and the market data which defines the exposures for an asset.
The following issues make these questions complex to address:

- OTC derivative assets are not a pre-defined universe, but rather must be defined by a user;
- A model is required to be able to generate analytics for securities historically, in addition to now;
- Fundamental approaches to the modeling and benchmarking of the fixed income markets have changed through time, such as the transition away from Libor to Alternative Reference Rates, and the movement in the synthetic credit space towards the standardization of CDS quoting with upfront pricing;
- An asset may have path dependent features, such as a conversion term, which means that the asset loses dependency on certain underlying assets as it matures, meaning that the exposures may be path dependent.

The full set of factors to which a user portfolio is exposed are defined therefore by the set of exposures created by the pricing models required to compute analytics across the portfolio. The equations in section 9.3 are used to compute the covariance matrix dynamically using the parameters specified by the user for half-life and sampling.

9.4 Model History and Proxy Mechanism

9.4.1 Backward (Historical) and Forward Proxy

All systematic fixed income factors are updated daily. The model attempts to ensure factor histories are as long as permitted by the underlying market data. Motivations for this are as follows:

- Users wish to understand the behavior of the factor through multiple market regimes. Capturing such regimes involves having many years of history.
- By having a long factor history, users may elect to use the model at different horizons and to work with a variety of sampling frequencies in the estimation of factor-factor covariances.

However, the calibration of risk models presents practical hurdles and frequently the lengths of history for different factors are unequal. This is especially true for asset-specific risk factors or emerging market interest rates where the relevant capital markets have come into existence or gained liquidity more recently than others. In such situations, a proxy mechanism is adopted where factors with a shorter history are proxied to other factors with longer history based on certain statistical and economic criteria. This is known as a backward (historical) proxy. Occasionally, there are factors for which there is no reasonable proxy and the factor history beyond a certain data point will be flatlined[59].
A forward proxy mechanism wherein a factor whose source data is planned to be retired can be mapped to a different relevant factor on a going forward basis from a certain defined date. A good use case for this type of proxy are Libor-like factors which have been retired from many markets. In such cases, the Libor factor is forward proxied to its ARR counterpart. For example, SOFR factors are used as a forward proxy for USD Libor factors and SONIA factors are used for GBP Libor factors since the retirement of the respective Libor rates.

9.4.2 Mapped Factors

For asset classes that do not yet have dedicated model coverage, or for certain sub-cohorts within an asset class for which risk factors cannot be robustly calibrated due to statistical significance not being justified, lack of data, or insufficient data quality, a proxy mechanism called factor mapping is designed, which allows certain factors to exist without an actual calibration process of its own time series. Such factors are called mapped factors, whose return and volatility are effectively those of other factors (the mapped-to factors). Factor mapping allows the risk model to maintain both granularity in exposure mapping and robustness in factor calibration [60]. Below are some of the most common ways mapped factors are set up.
For yield curve-related factors (rates, swaps, money market, etc.):

[59] We will share the details related to factor history availability and proxy when we go over the risk models/factors of individual asset classes in later sections.
[60] Essentially, factor mapping allows the model to dis-entangle exposure mapping (i.e. which factors a security maps to) from factor calibration (i.e. how factors are defined).

- Mapping to risk factors of another currency (e.g. UYU interest rate factors are mapped to BRL interest rate factors.)
- Mapping to risk factor of another tenor (e.g. LKR interest rate factors above the 5Y tenor are all mapped to the LKR 5Y factor.)
- Mapping to risk factors of a similar or related asset class (e.g. GBP deposit factors are mapped to GBP CP A1/P1 factors.)

For spread risk factors:

- Mapping to risk factors of a similar or related asset class (e.g. Jumbo ARM factors are mapped to agency ARM factors)
- Mapping to local-currency ARR/LIBOR spread factors (e.g. 1/1 agency ARM factor is mapped to the 1Y USD SOFR spread-over-treasury factor)

There is a special type of mapped factors known as the ALL factors[61]. Securities with exposure to these factors are denominated in currencies without dedicated risk factors. These ALL factors are in turn mapped to factors of a major currency (mostly EUR or USD) within the same asset class. For example, NOK does not have its dedicated ARR spread factors. Thus, NOK interest rate swaps will have exposure to ALL LIBOR ARR spread factors (which are mapped to EUR SONIA spread factors.)
Mapped factors will be identified with an asterisk (*) next to their factor titles in the list of factors. Detailed mapping information is provided in table 82.

Table 82: Mapped Factors

Factor Type	Subtype	SubSubtype	Factor	Mapped To
IR	Rates		UYU 3M	BRL 3M
			UYU 1Y	BRL 1Y
			UYU 2Y	BRL 2Y
			UYU 3Y	BRL 3Y
			UYU 5Y	BRL 5Y
			UYU 7Y	BRL 7Y
			UYU 10Y	BRL 10Y
			UYU 15Y	BRL 15Y
			UYU 20Y	BRL 20Y
			UYU 25Y	BRL 25Y
			UYU 30Y	BRL 30Y
			LKR 7Y	LKR 5Y
			LKR 10Y	LKR 5Y
			LKR 15Y	LKR 5Y
			LKR 20Y	LKR 5Y
			LKR 25Y	LKR 5Y
			LKR 30Y	LKR 5Y
			MYR 15Y	MYR 10Y
			MYR 20Y	MYR 10Y
			MYR 25Y	MYR 10Y

[61] "All" in this context should be interpreted as "all [other currencies not specifically covered]."

Factor Type	Subtype	SubSubtype	Factor	Mapped To
			MYR 30Y	MYR 10Y
			RUB 15Y	RUB 10Y
			RUB 20Y	RUB 10Y
			RUB 25Y	RUB 10Y
			RUB 30Y	RUB 10Y
Swap	Vol		Normal volatility factors (USD_normvol) {1M,3M,6M} * {1Y,2Y,3Y,5Y,7Y,10Y,15Y} {1Y,2Y,3Y,5Y} * {1Y,2Y,3Y,5Y,7Y,10Y,15Y,20Y}	Normal volatility sofr factors (USD_normvol_sofr)
			Log-normal volatility (USD_impvol)	Log-normal sofr volatility (USD_impvol_sofr)
Spread	Leveraged Loans		Non-USD Non-Distressed Factor	USD Market Factor
			Non-USD Distressed Factor	USD Distressed Factor
	Munis	Taxable	Tax Muni 5Y	Tax Muni 30Y
			Tax Muni 10Y	Tax Muni 30Y
Spread	Securitized	Agency ARM	1/1 Jumbo Hyb ARM*	1/1 Agy Hyb ARM
			3/1 Jumbo Hyb ARM	3/1 Agy Hyb ARM
			5/1 Jumbo Hyb ARM	5/1 Agy Hyb ARM
			7/1 Jumbo Hyb ARM	7/1 Agy Hyb ARM
			10/1 Jumbo Hyb ARM	10/1 Agy Hyb ARM
			1/1 Agy Hyb ARM	USD libor 1Y
			10/1 Agy Hyb ARM	7/1 Agy Hyb ARM
		CMBS	CMBS IO pre2005	CMBS AJ pre05
			CMBS IO 2006-2008	CMBS AJ 06/07
			CMBS IO post09	CMBS AS post09
		CLO	USD CLO Generic	USD leveraged loan spread risk factor
	Corporates	EUR Corporates	Pan Euro France HY	EU Corp France
			Pan Euro Germany HY	EU Corp Germany

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